Biochemistry



Section K
Lipid Metabolism

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1g 葡萄糖释放约 16KJ 的能量
1g 脂肪可以放出约 39KJ 的能量。

K1 Structures and roles of fatty acids

K2 Fatty Acid Breakdown

K3 Fatty Acid Synthesis

K4 Metabolism of Triacylglycerols



K1

Structures and roles of fatty acids



1. Lipid

Biological lipids are a chemically diverse group of compounds, the common and defining feature of which is their insolubility in water.

Fats and oils (storage lipids)

Phospholipids and sterol (major elements of membranes)



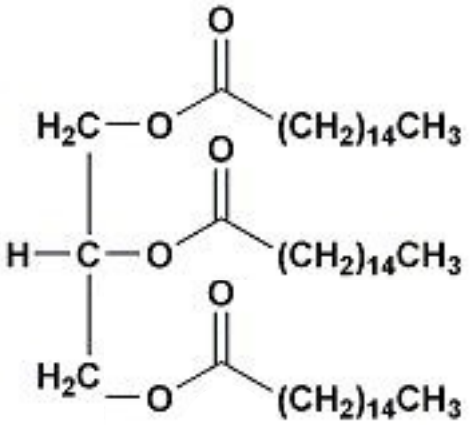
生物体内的脂类 (lipid)

（脂质，类脂）：脂肪酸和醇组成或其衍生物

- 皂化脂类
  - 单纯脂类
    - 三酰甘油
    - 蜡
  - 复合脂类
    - 磷脂
    - 糖脂
    - 硫脂
- 非皂化脂类
  - 萜　类
  - 类固醇

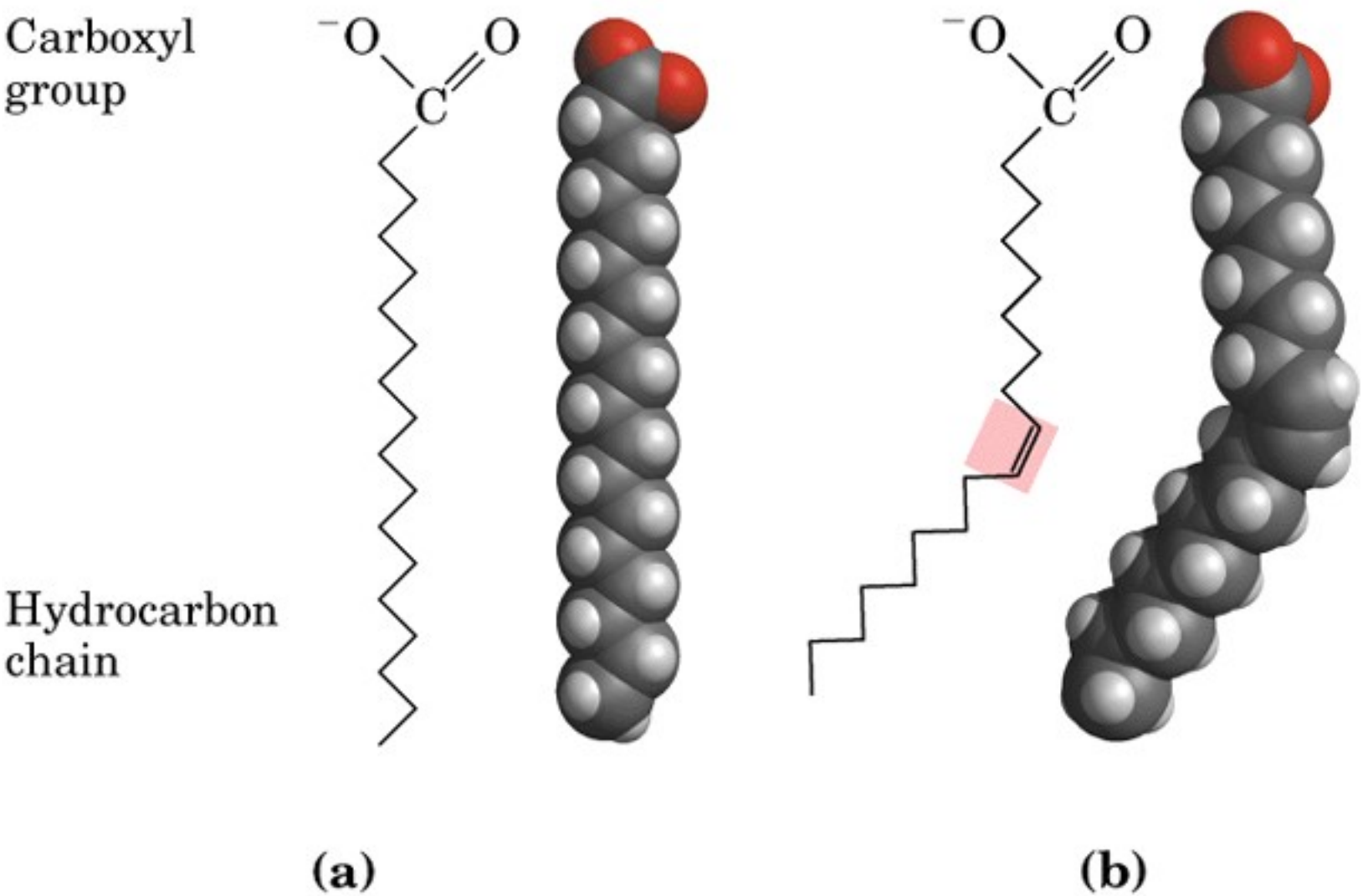
The fats and oils used almost universally as stored forms of energy in living organisms are derivatives of fatty acids.

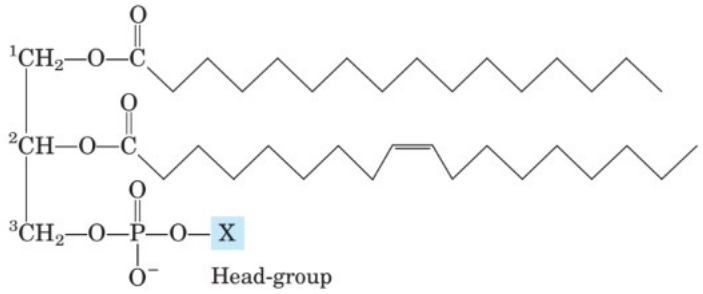
Typical type of fatty acid-containing compounds are triacylglycerols.

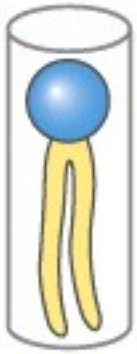


(C4 to C36)

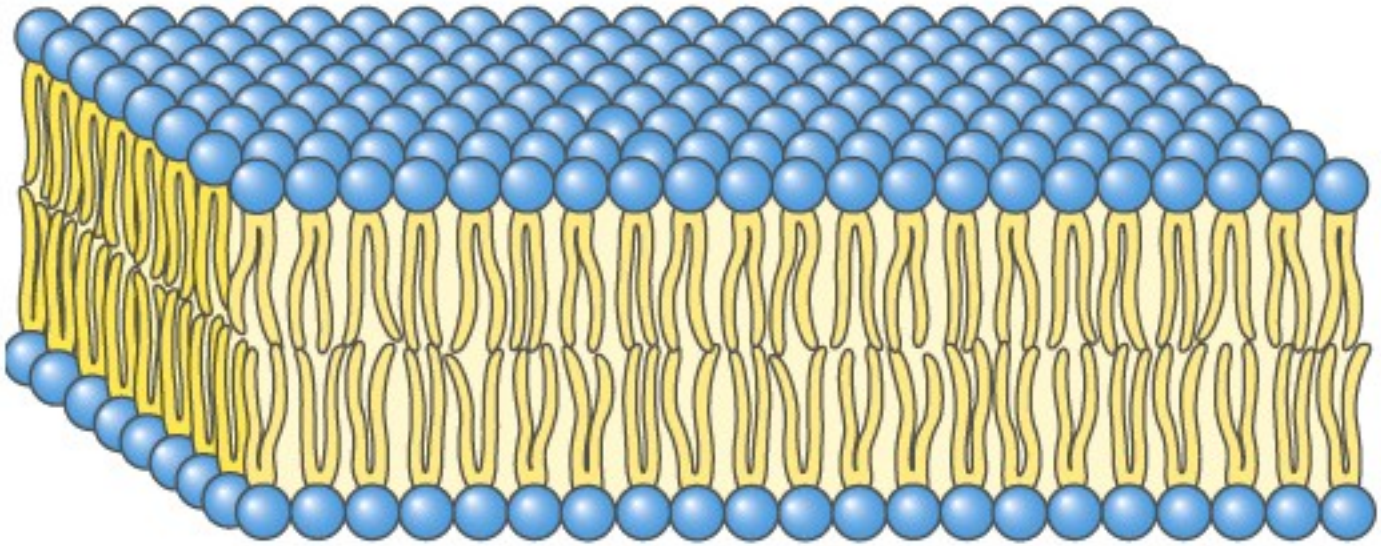
2. Structure and properties of Fatty acids







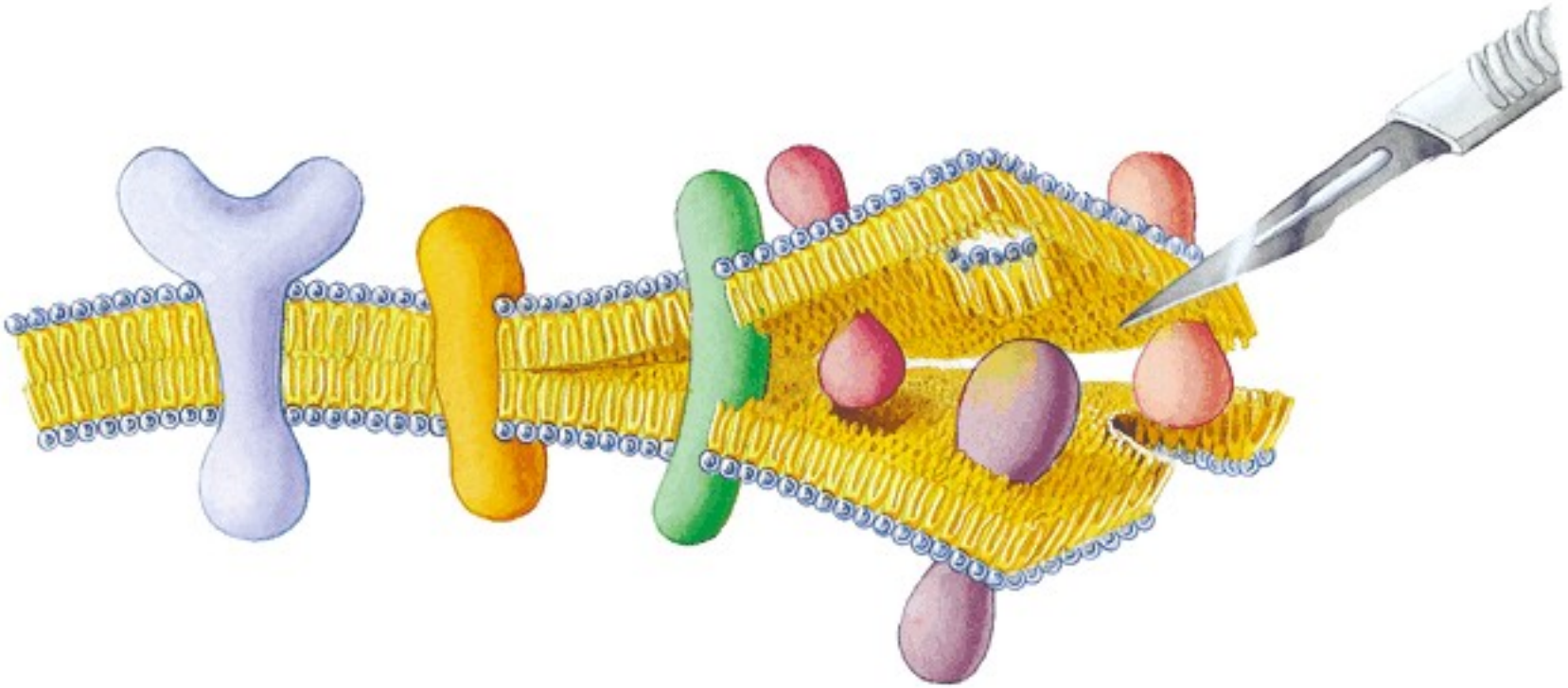
Individual units are cylindrical (cross-section of head equals that of side chain)

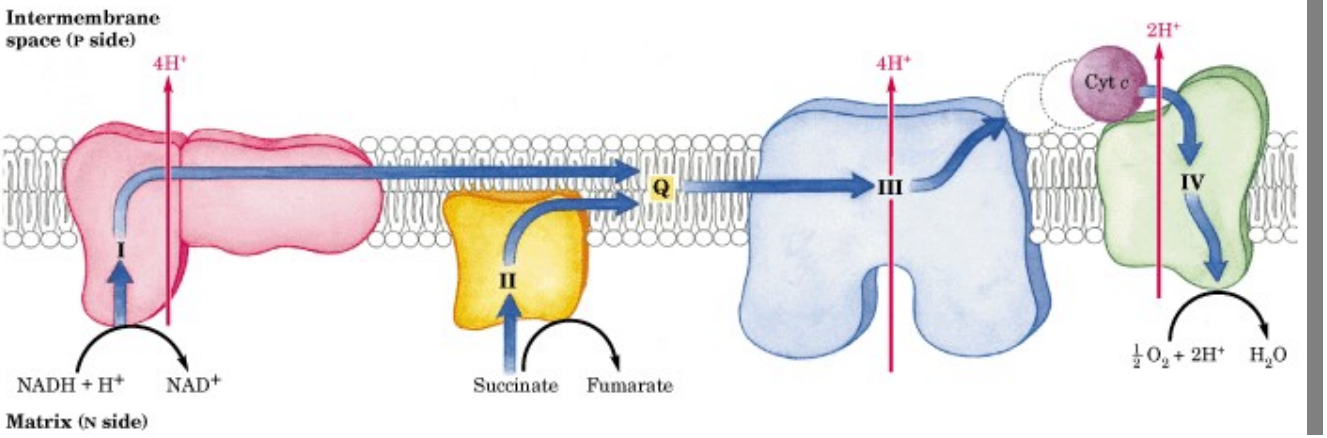


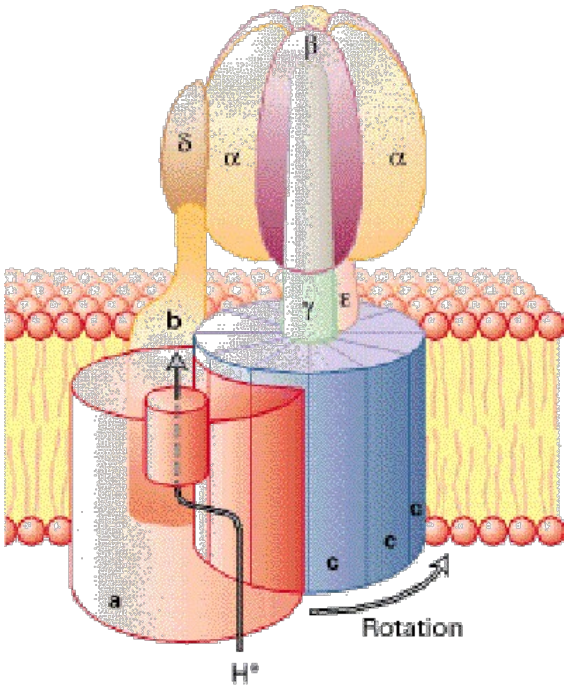
Bilayer

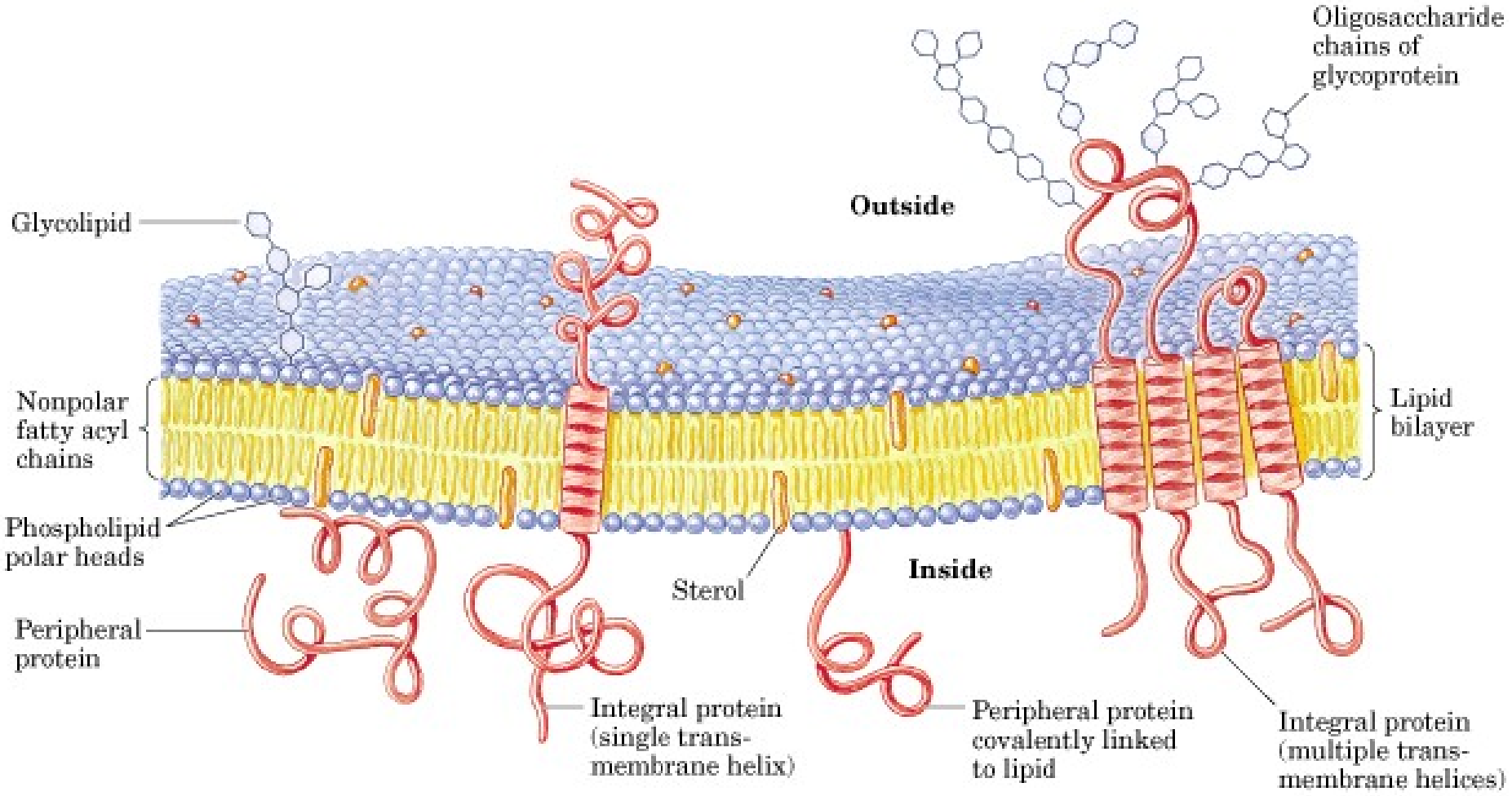
Cell membrane






Intermembrane space (P side)
4H+
4H+
2H+
Cyt c
Q
I
II
III
IV
NADH + H+
NAD+
Succinate
Fumarate
½ O2 + 2H+
H2O
Matrix (N side)


β
δ
α
α
γ
ε
b
c
c
c
c
a
Rotation
H+


Oligosaccharide chains of glycoprotein
Outside
Glycolipid
Nonpolar fatty acyl chains
Lipid bilayer
Phospholipid polar heads
Sterol
Inside
Peripheral protein
Integral protein (single trans-membrane helix)
Peripheral protein covalently linked to lipid
Integral protein (multiple trans-membrane helices)

Membranes consist mainly of lipids and proteins with mass ratio ranges between 1:4 to 4:1 (membranes with different functions have different proteins, some lipids and proteins have covalently linked carbohydrates).

1.2.1 In membranes the three major classes of lipids are the glycerophospholipids (甘油磷脂), the sphingolipids （鞘磷脂） and the sterols （固醇）.



(1)The glycerophospholipids (甘油磷脂) have a glycerol backbone that is attached to two fatty acid hydrocarbon chains and a phosphorylated head group.

These include：

phosphatidate（磷脂酸）

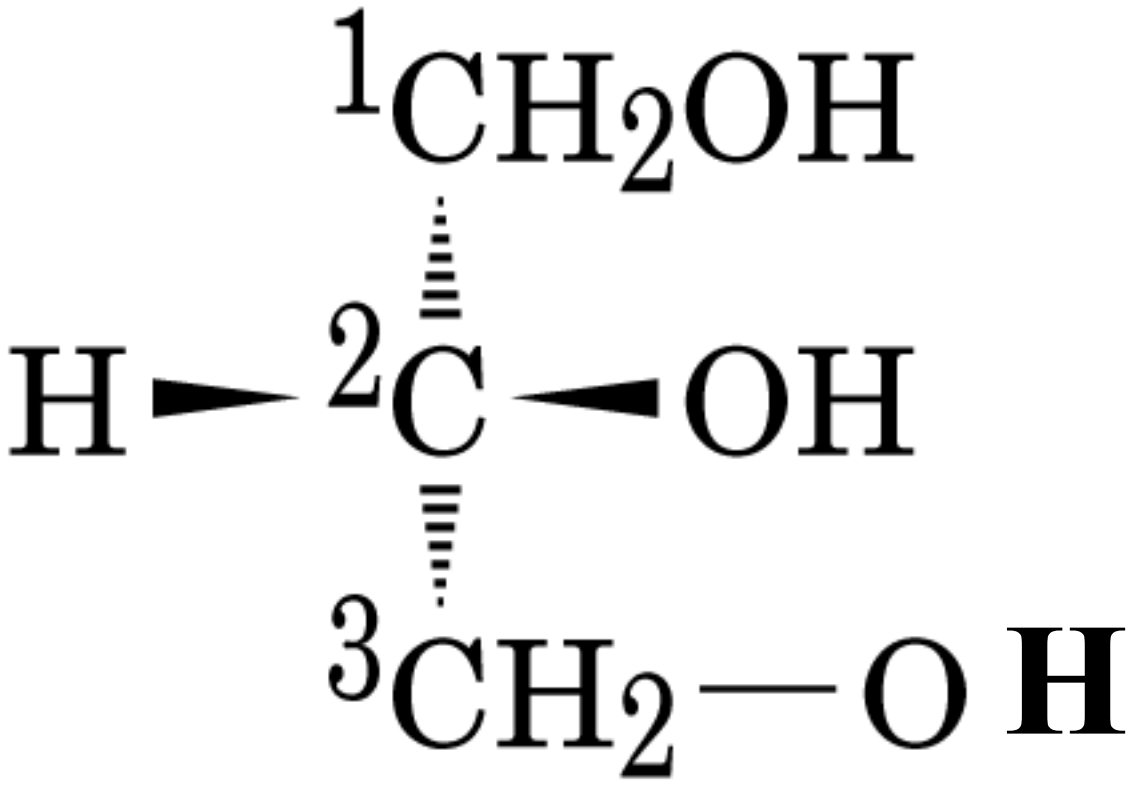
phosphatidylcholine （磷脂酰胆碱）

phosphatidylethanolamine （磷脂酰胆胺）

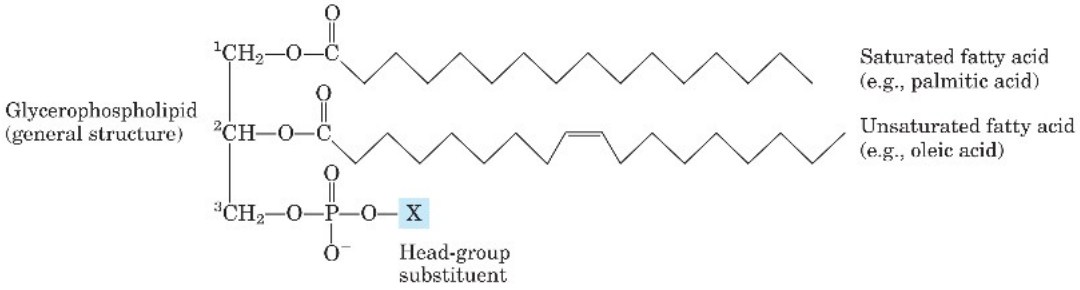
phosphatidylglycerol （磷脂酰甘油）

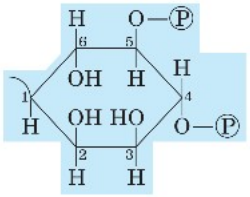
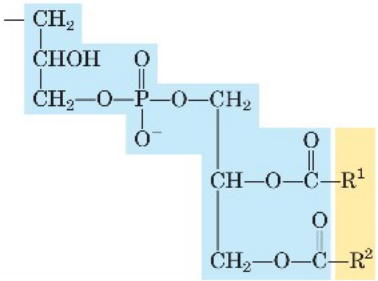
phosphatidylinositol （磷脂酰肌醇）

diphosphatidylglycerol （二磷脂酰甘油，心磷脂）



glycerol



Name of glycerophospholipid	Name of X	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—	— H	−1
Phosphatidylethanolamine	Ethanolamine	— CH_2—CH_2—$\overset{+}{N}H_3$	0
Phosphatidylcholine	Choline	— CH_2—CH_2—$\overset{+}{N}(CH_3)_3$	0
Phosphatidylserine	Serine	— CH_2—$CH(\overset{+}{N}H_3)$—COO^-	−1
Phosphatidylglycerol	Glycerol	— CH_2—$CH(OH)$—CH_2—OH	−1
Phosphatidylinositol 4,5-bisphosphate	*myo*-Inositol 4,5-bisphosphate		−4
Cardiolipin	Phosphatidyl-glycerol		−2

phosphatidate（磷脂酸）

phosphatidylethanolamine （磷脂酰胆胺）

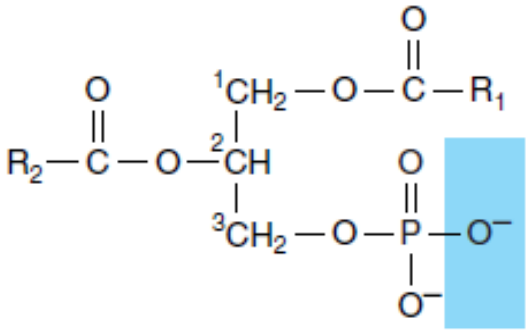
phosphatidylcholine （磷脂酰胆碱）

phosphatidylglycerol （磷脂酰甘油）

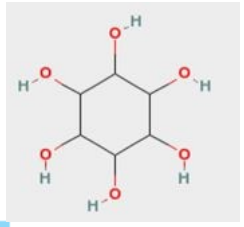
phosphatidylinositol （磷脂酰肌醇）

diphosphatidylglycerl （二磷脂酰甘油）心磷脂

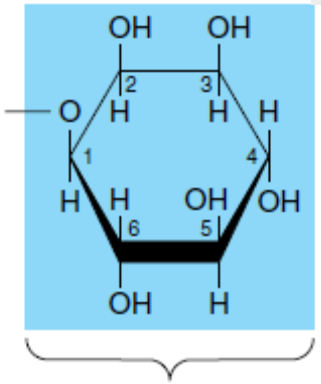




Phosphatidic acid

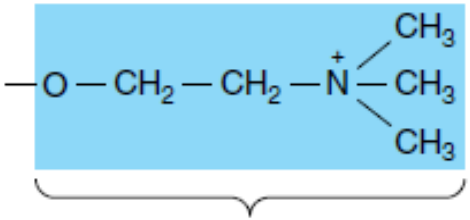


D



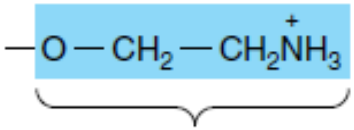
Myoinositol

A



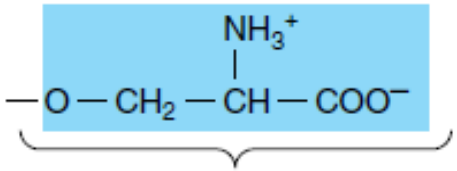
Choline

B



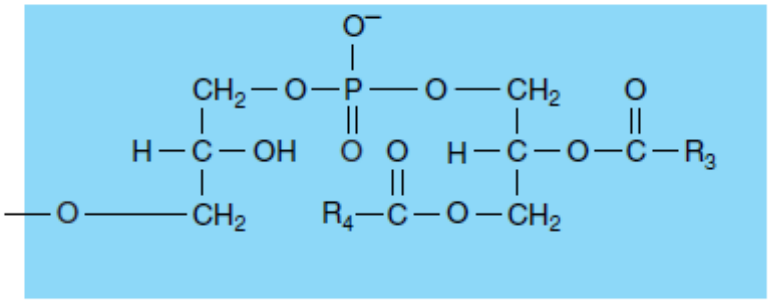
Ethanolamine

C



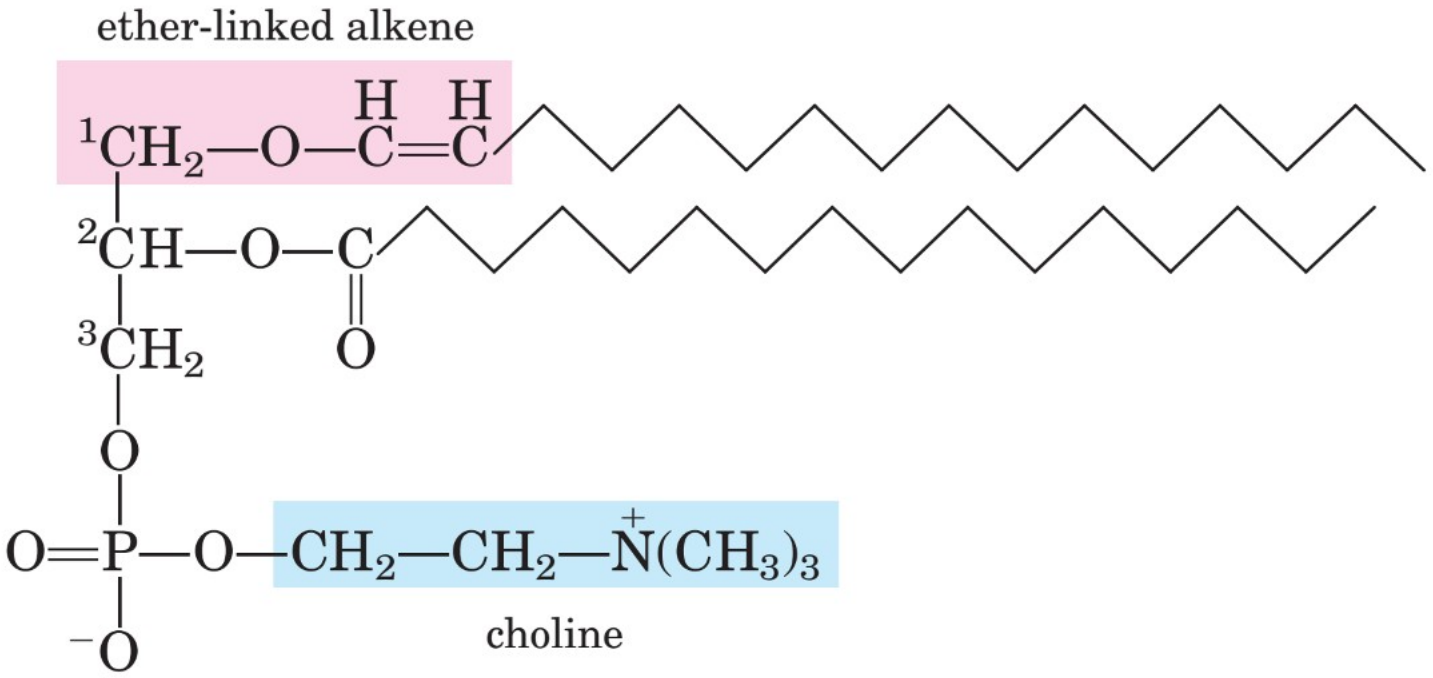
Serine

E

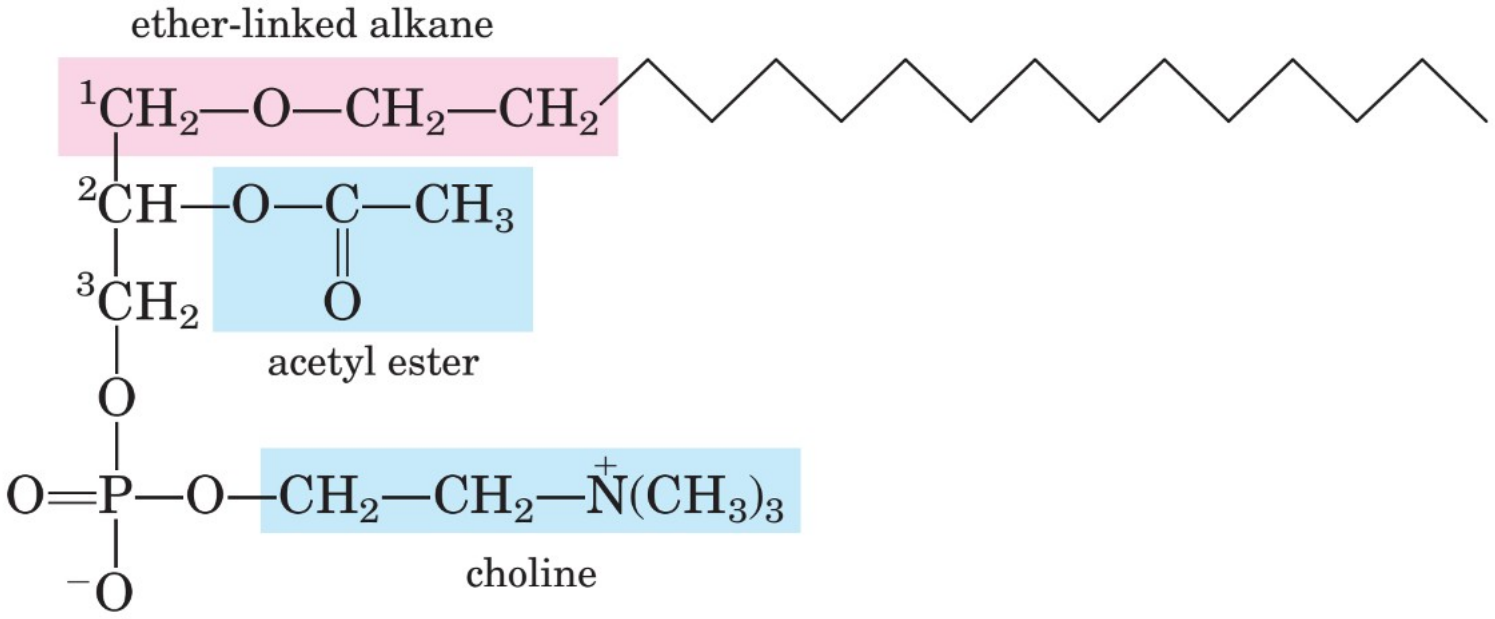


Phosphatidylglycerol





Plasmalogen



Platelet-activating factor

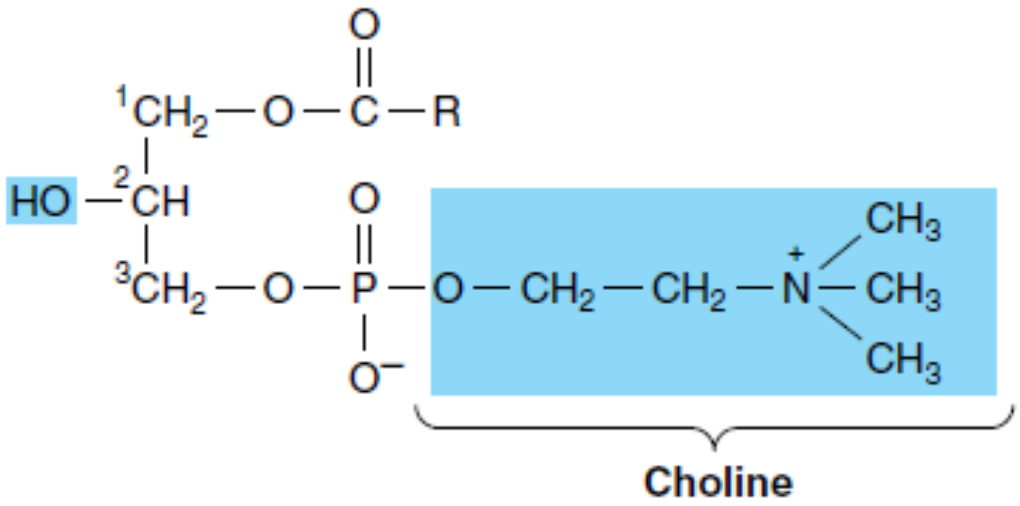


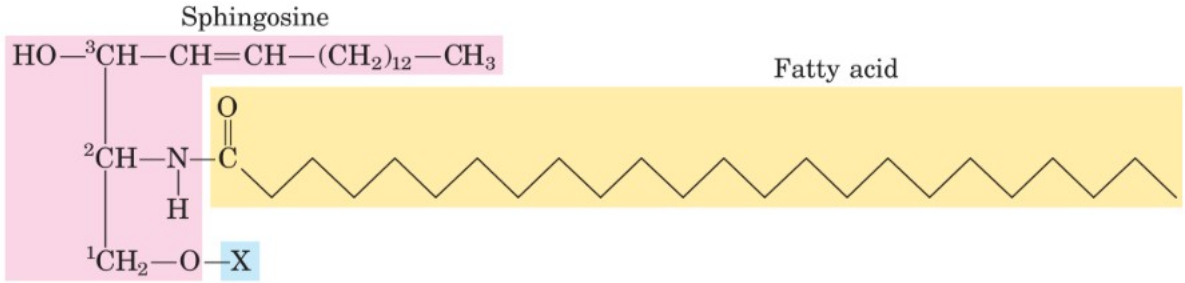
Figure 14–9. Lysophosphatidylcholine (lysolecithin).

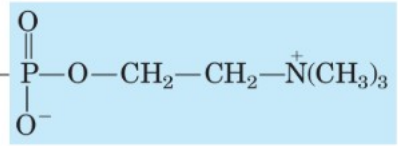
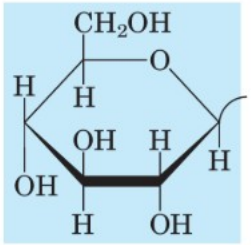
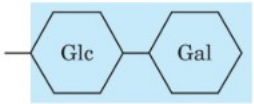
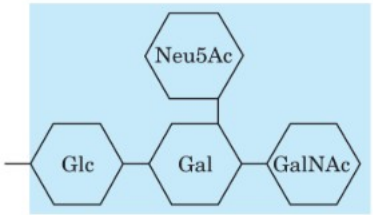
(2) The sphingolipids （鞘氨醇磷脂类） are base on sphingosine （鞘氨醇） to which a single fatty acid chain is attached and either a phosphorylated head group (sphingomyelin) or one or more sugar residues.



Sphinx

Sphingolipid (general structure)

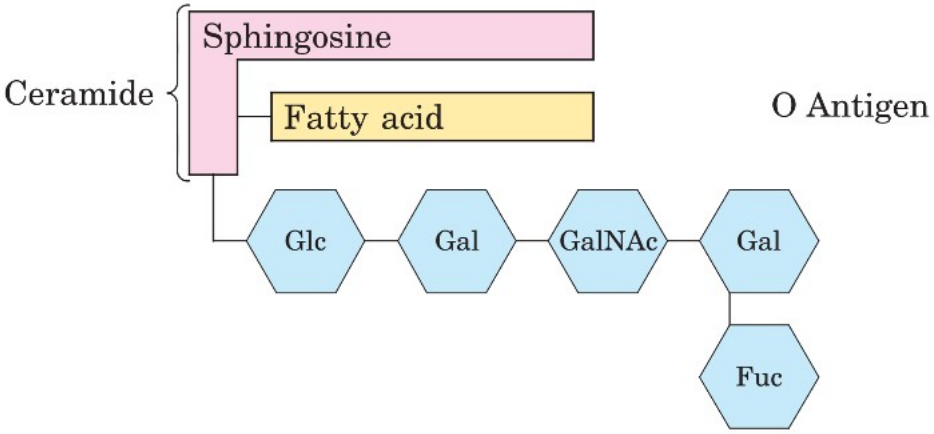


Name of sphingolipid	Name of X	Formula of X
Ceramide	—	—H
Sphingomyelin	Phosphocholine	
Neutral glycolipids Glucosylcerebroside	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

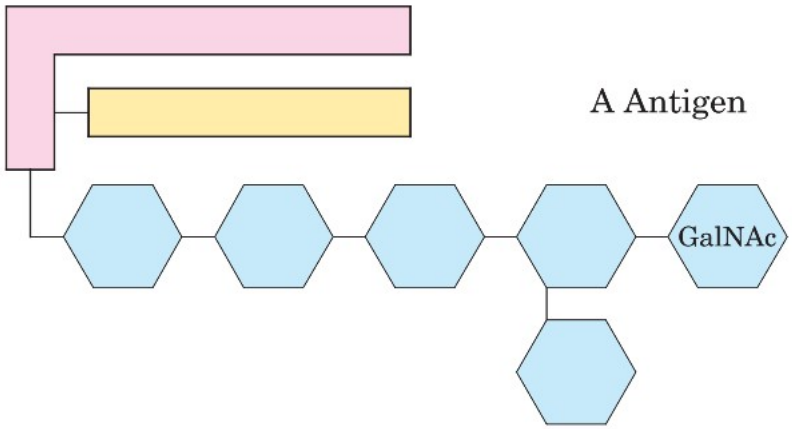
Biochemist

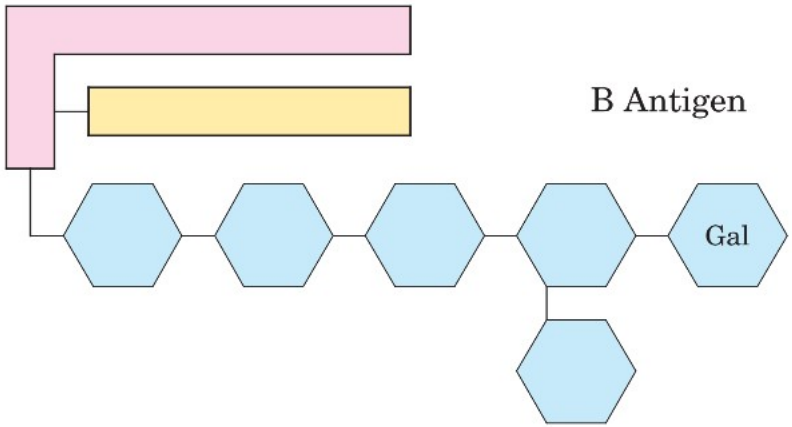
Ceramide

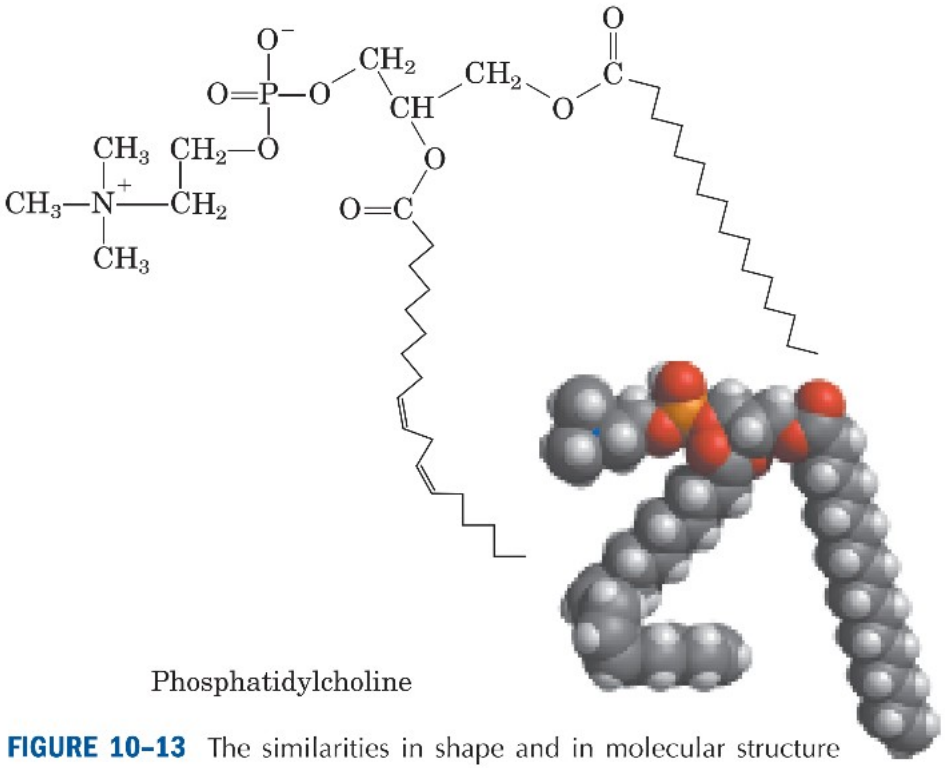
神经酰氨



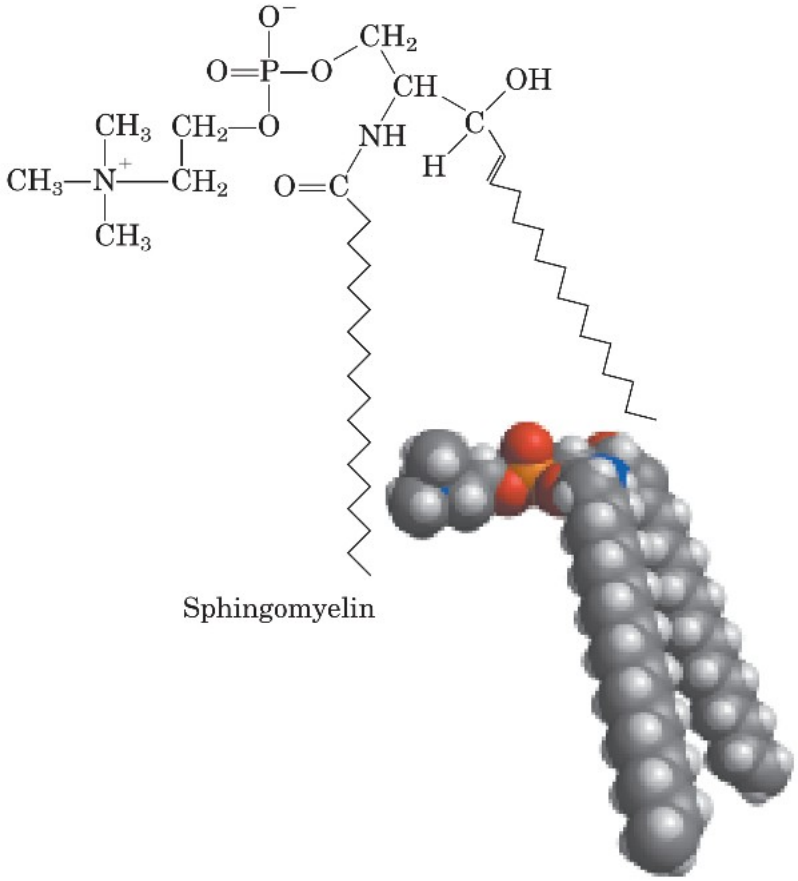
Γλψχοσπηινγολιπιδσ ασ δετερμιναντσ οφ βλοοδ γρουπσ.







Phosphatidylcholine



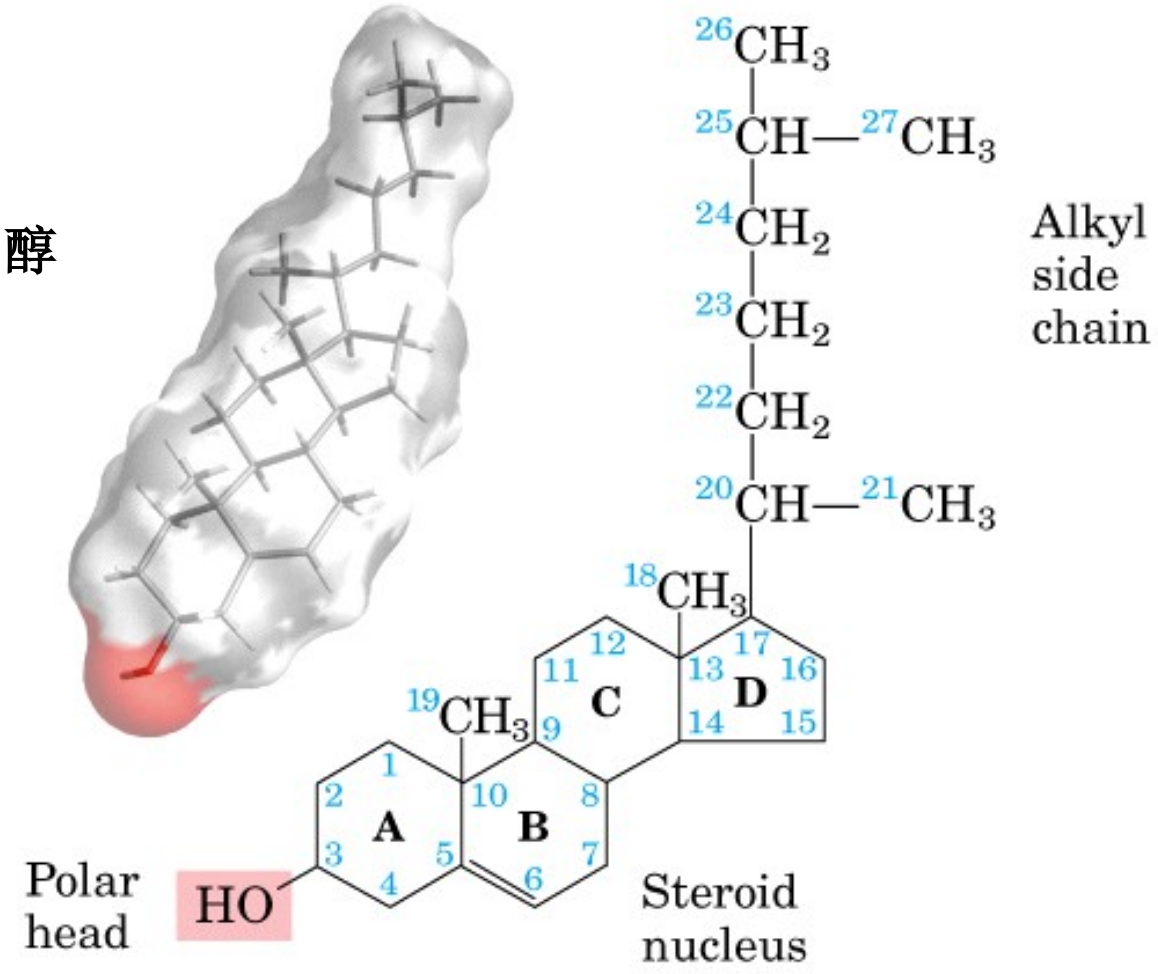
Sphingomyelin

FIGURE 10–13 The similarities in shape and in molecular structure of phosphatidylcholine (a glycerophospholipid) and sphingomyelin (a sphingolipid) are clear when their space-filling and structural formulas are drawn as here.

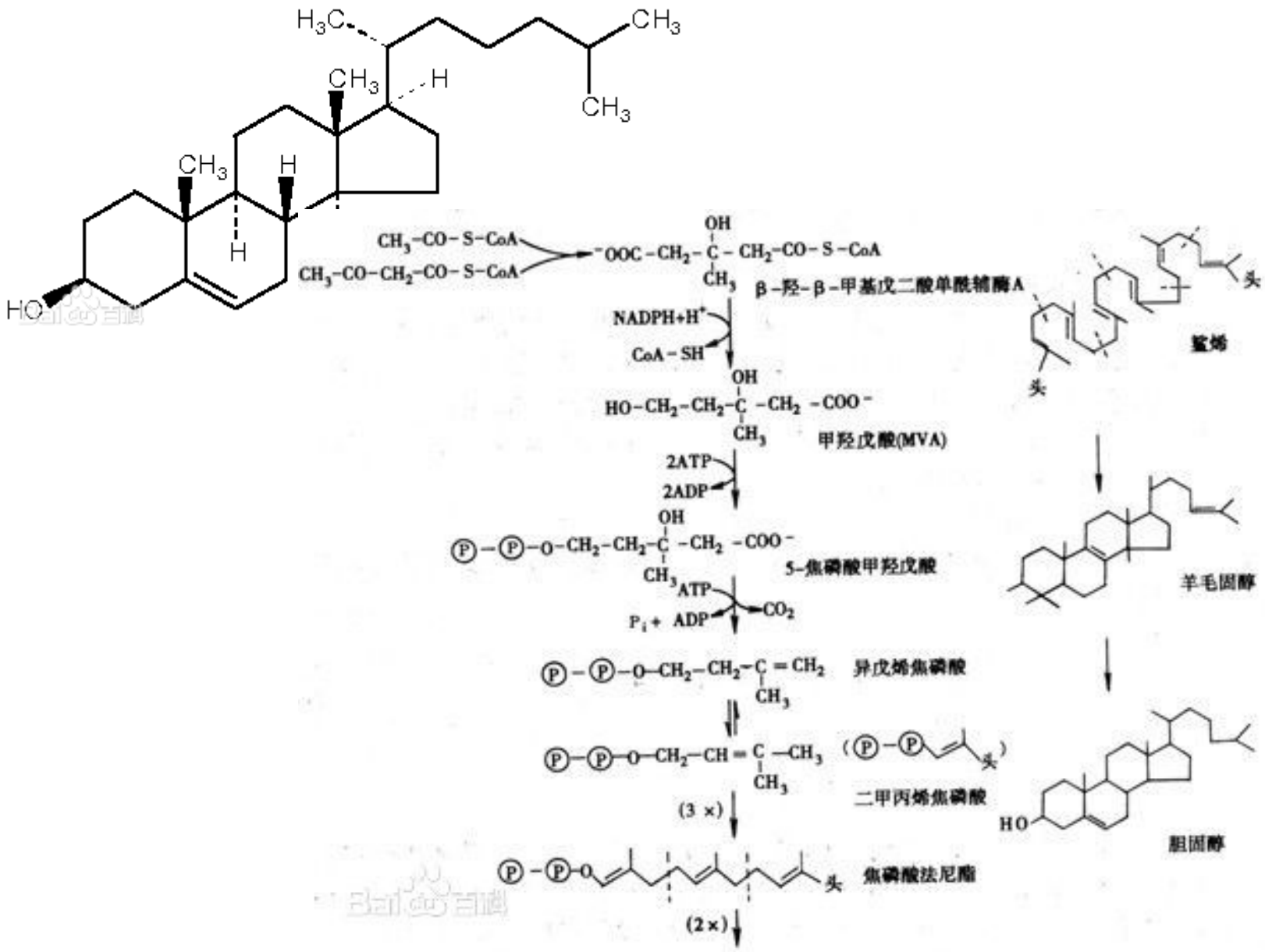
(3) The major sterol (固醇） in animal plasma membranes is cholesterol （胆固醇）, while the structurally related stigmasterol (豆甾醇） and β–sitosterol （谷甾醇） are found in plants.



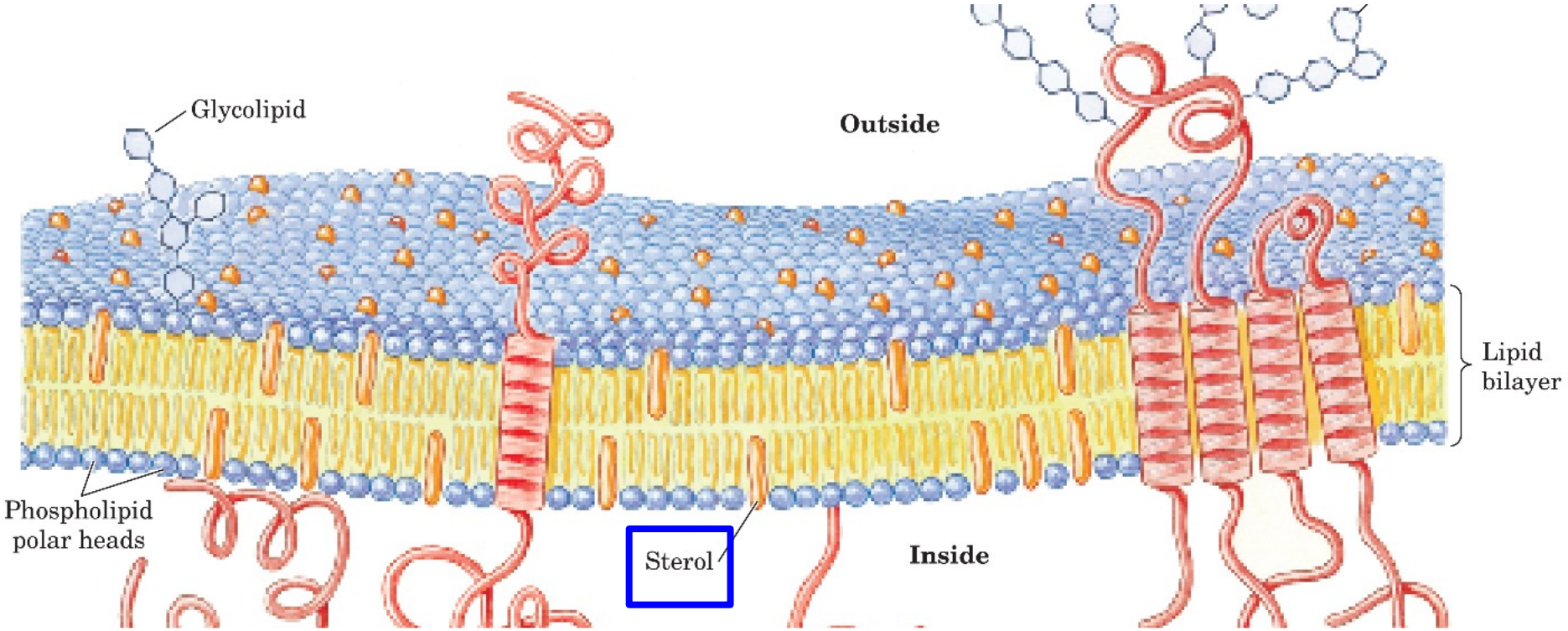
胆固醇

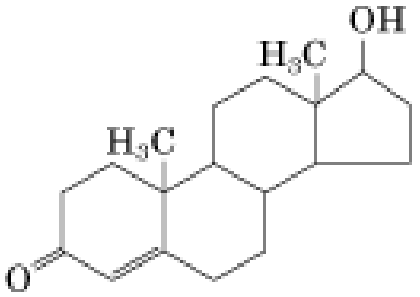


Sterols Have Four Fused Carbon Rings


CH3-CO-S-CoA
CH3-CO-CH2-CO-S-CoA
β-羟-β-甲基戊二酸单酰辅酶A
NADPH+H+
CoA-SH
甲羟戊酸(MVA)
2ATP
2ADP
5-焦磷酸甲羟戊酸
ATP
Pi + ADP
CO2
异戊烯焦磷酸
二甲丙烯焦磷酸
(3×)
焦磷酸法尼酯
(2×)
鲨烯
羊毛固醇
胆固醇

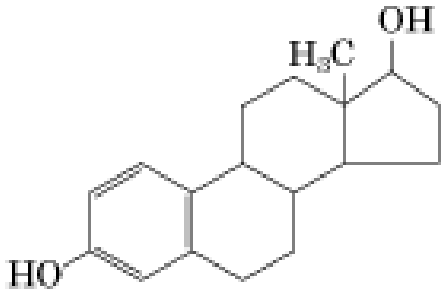
胆固醇


Glycolipid
Outside
Lipid bilayer
Phospholipid polar heads
Sterol
Inside

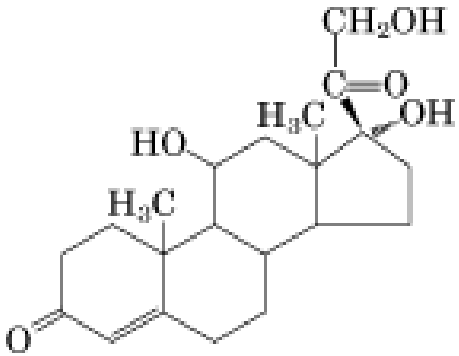


Testosterone

睾丸素

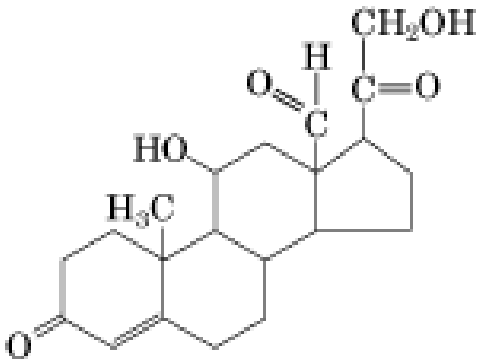


Estradiol 雌二醇

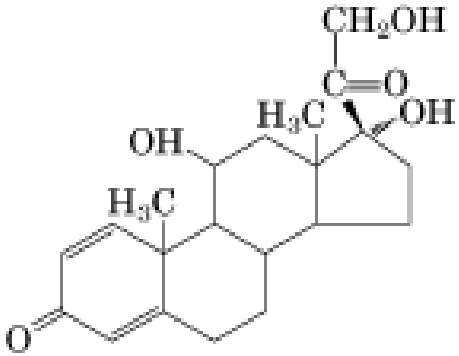


Cortisol

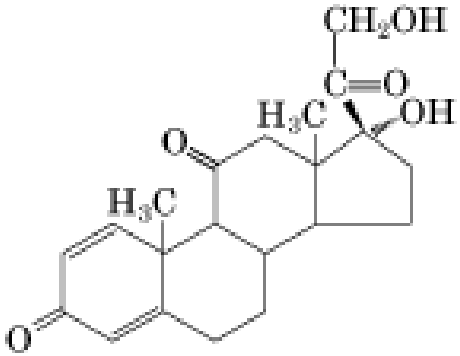
皮质（甾）醇



Aldosterone 醛甾酮



Prednisolone



Prednisone



Prostaglandins (前列腺素)

前列腺素（ PG ）实际上是一类具有生物活性的物质的总称，目前已发现的有几十种，它的基本结构为含一个环戊烷及两个脂肪侧链的二十碳脂肪酸，即前列腺烷酸。

由于环戊烷上的不饱和键位置或取代基不同，可将已知的前列素分为 E 、 F 、 A 、 B 四大类。



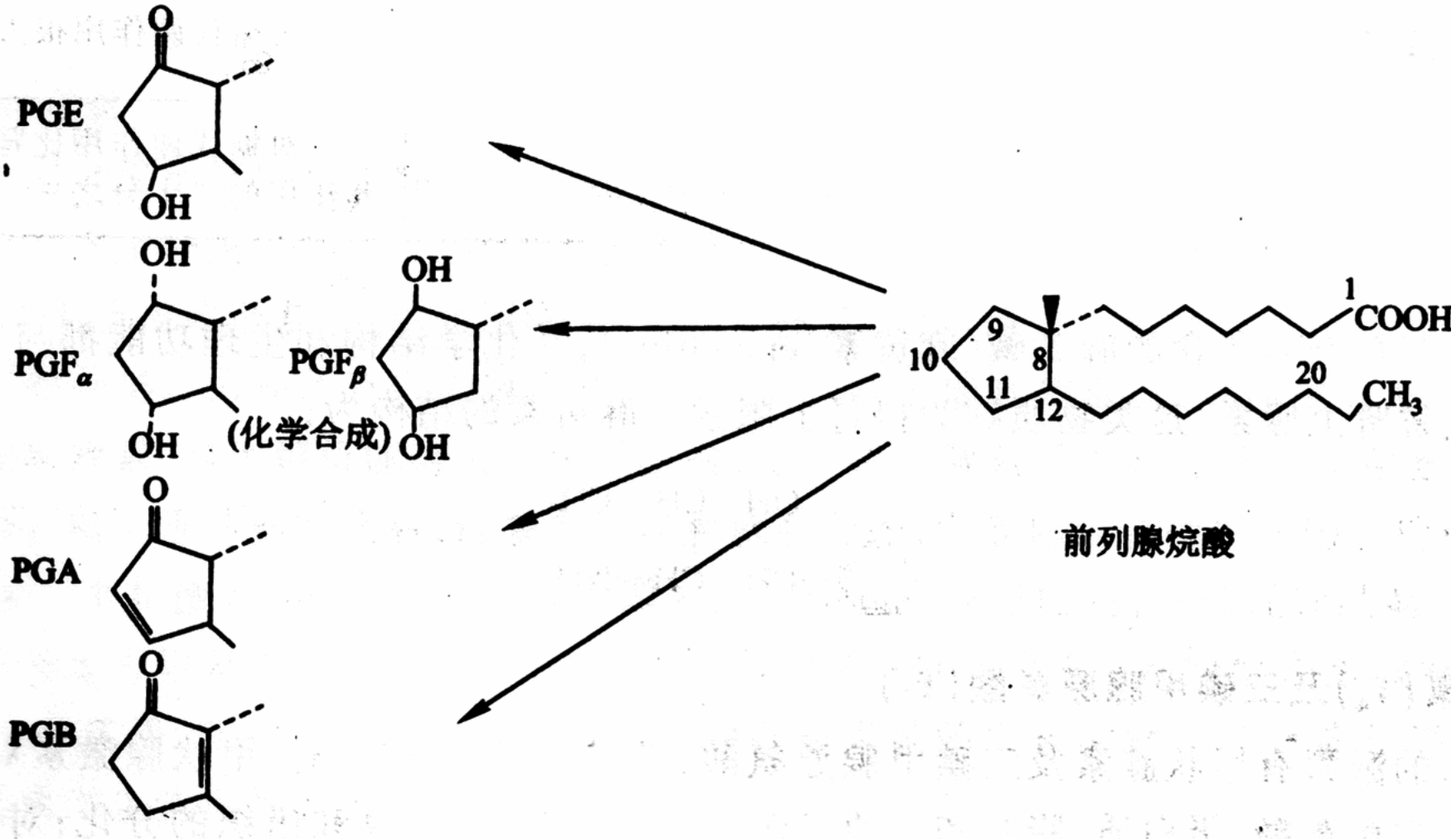
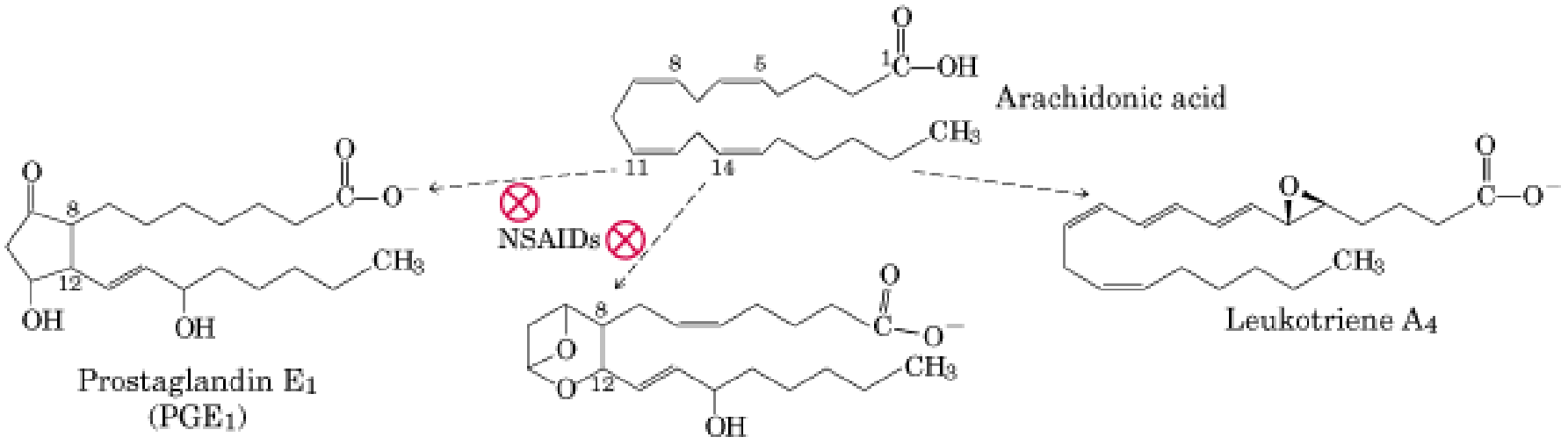


图 17－6　前列腺烷酸及各类前列腺素的结构



凝血噁烷

Eicosanoids

(b)

3. Nomenclature

Fatty acids are named according to the number of carbon atoms in the chain and the number and position of any double bonds.

Palmitate	**(C16:0)**	**棕榈酸（软脂酸）**
Stearate	**(C18:0)**	**硬脂酸**
Oleate	**(C18:1 （ Δ^9 ））**	**油酸**
Linoleate	**(C18:2 （ $\Delta^{9,12}$ ））**	**亚油酸**
Linolenate	**(C18:3)**	**亚麻酸**
Arachidonate	**(C20:4)**	**花生四稀酸**



4. Roles

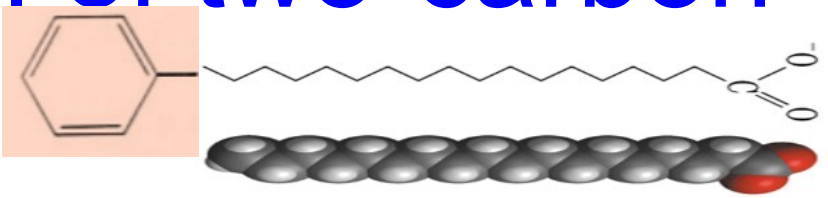
- Energy stores (triacylglycerols) and fuel molecule
- Components of membranes (glycerophospholipids and sphingolipids)
- As hormones and intracellular second messengers (DAG, diacylglycerol)
- Covalently joined with some proteins



Fatty Acid Breakdown
(K2)

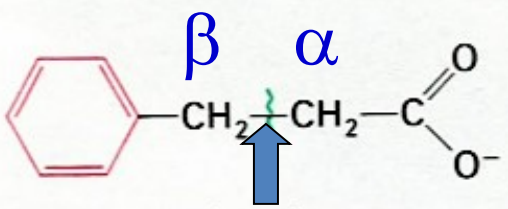


1. Early labeling experiments (1904): fatty acids are degraded by sequential removal of two-carbon units

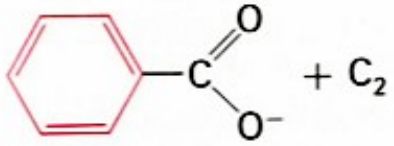


When dogs were fed with odd-numbered fatty acids attached to a phenyl group, benzoate was excreted; and when fed with even-numbered, phenylacetate was excreted. Hypothesis: the β-carbon is oxidized,with two-carbon units released by each round of oxidation.

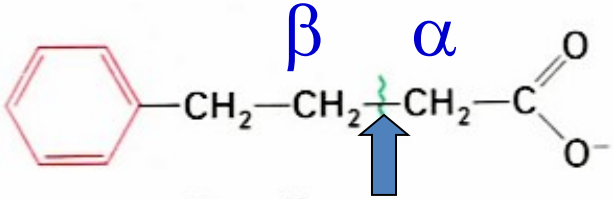




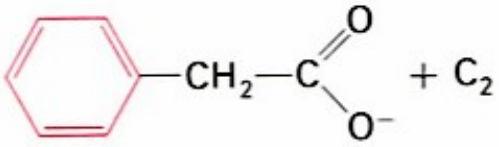
Phenylpropionate
(Fatty acid with odd number of carbon atoms)



Benzoate



Phenylbutyrate
(Fatty acid with even number of carbon atoms)



Phenylacetate

Franz Knoop's labeling Experiments (1904):
fatty acids are degraded by oxidation at the β carbon, i.e., **β oxidation**.

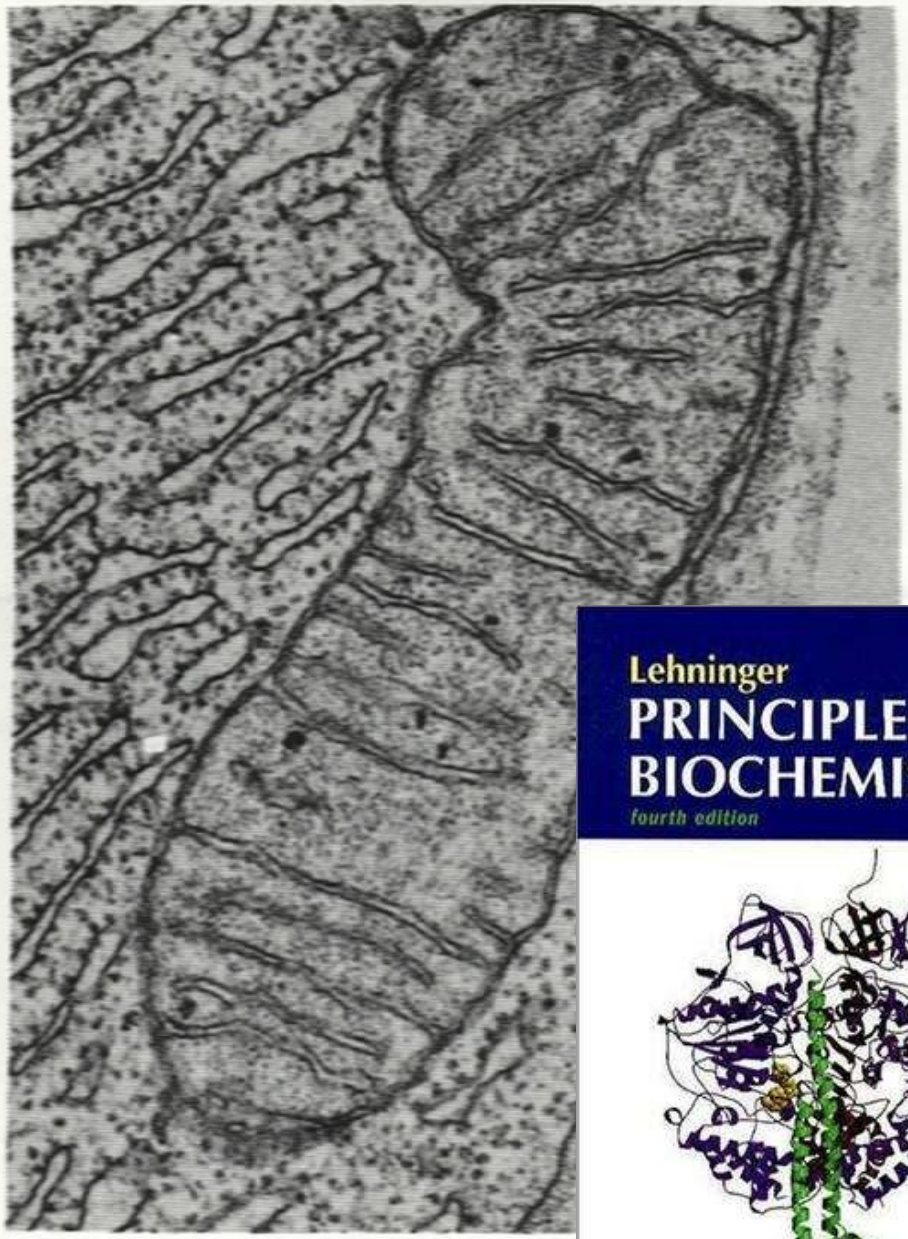


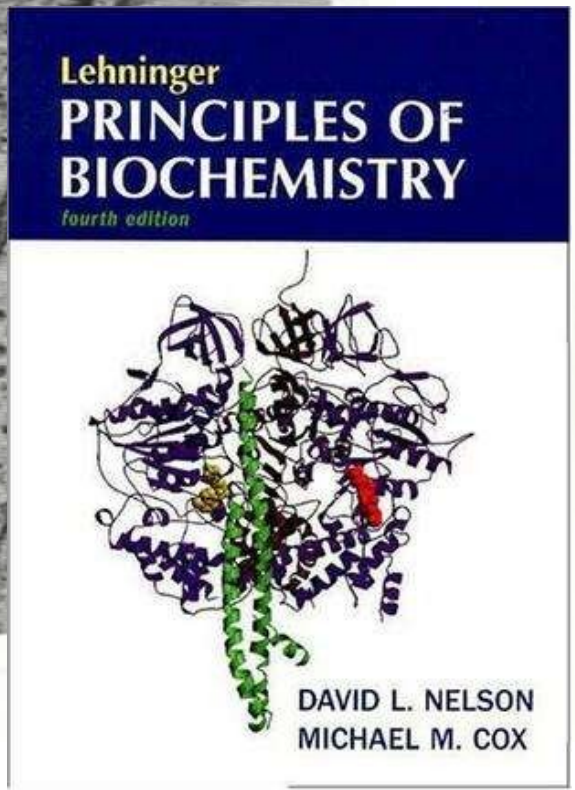
- **These experiments are a landmark in biochemistry, in using synthetic label (the phenyl group here) to elucidate reaction mechanism, and was done long before radioisotopes was used in biochemistry!**



2. Fatty acid oxidation was found to occur in mitochondria









Albert L. Lehninger
1917–1986

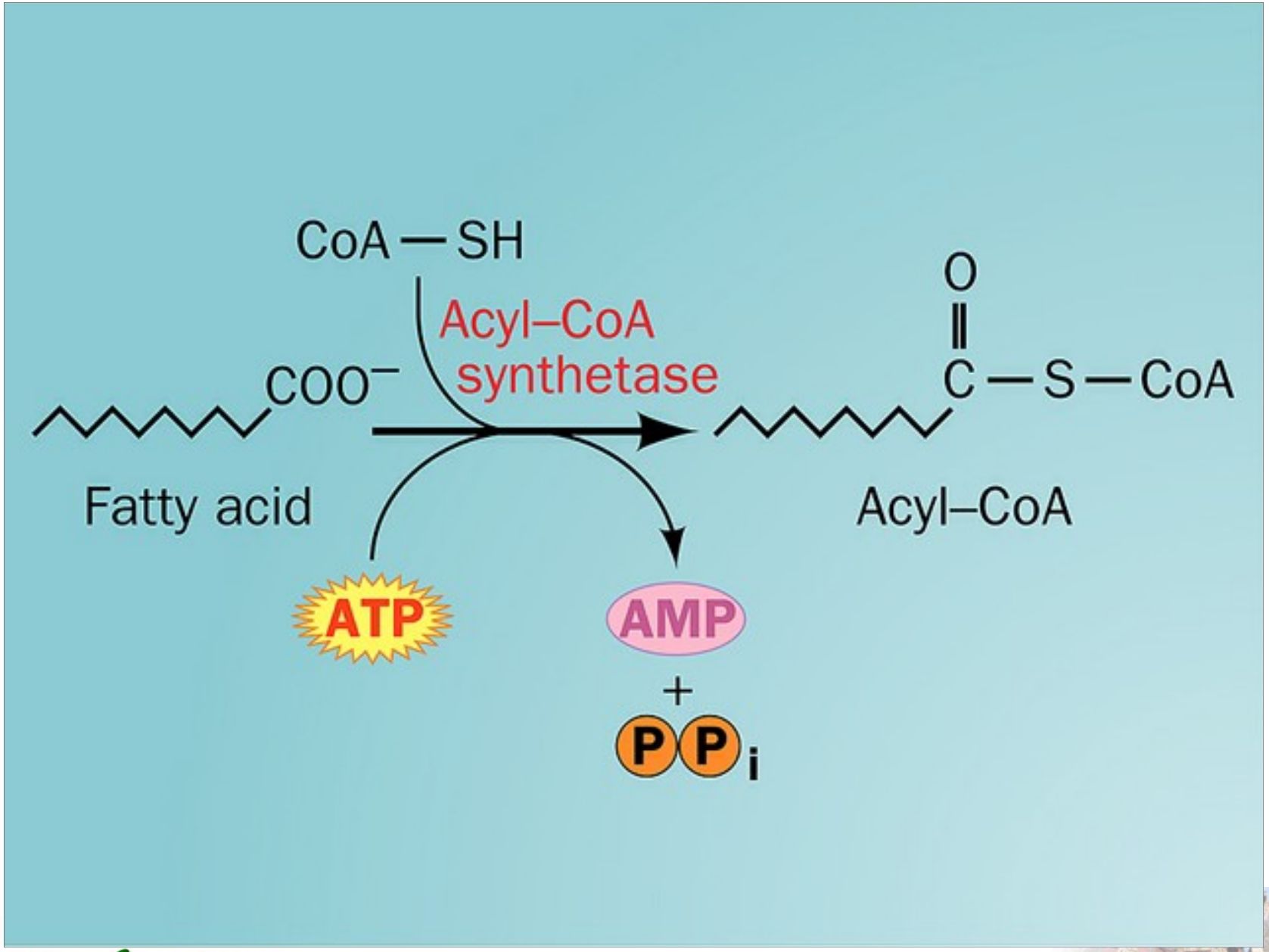
1. Activation of Fatty acid （outer membrane of mito）
2. Transport of fatty acyl-COA
3. β-oxidation： (mitochondria matrix)
   Oxidation by FAD
   Hydration
   Oxidation by NAD^+
   Thiolysis by CoA.



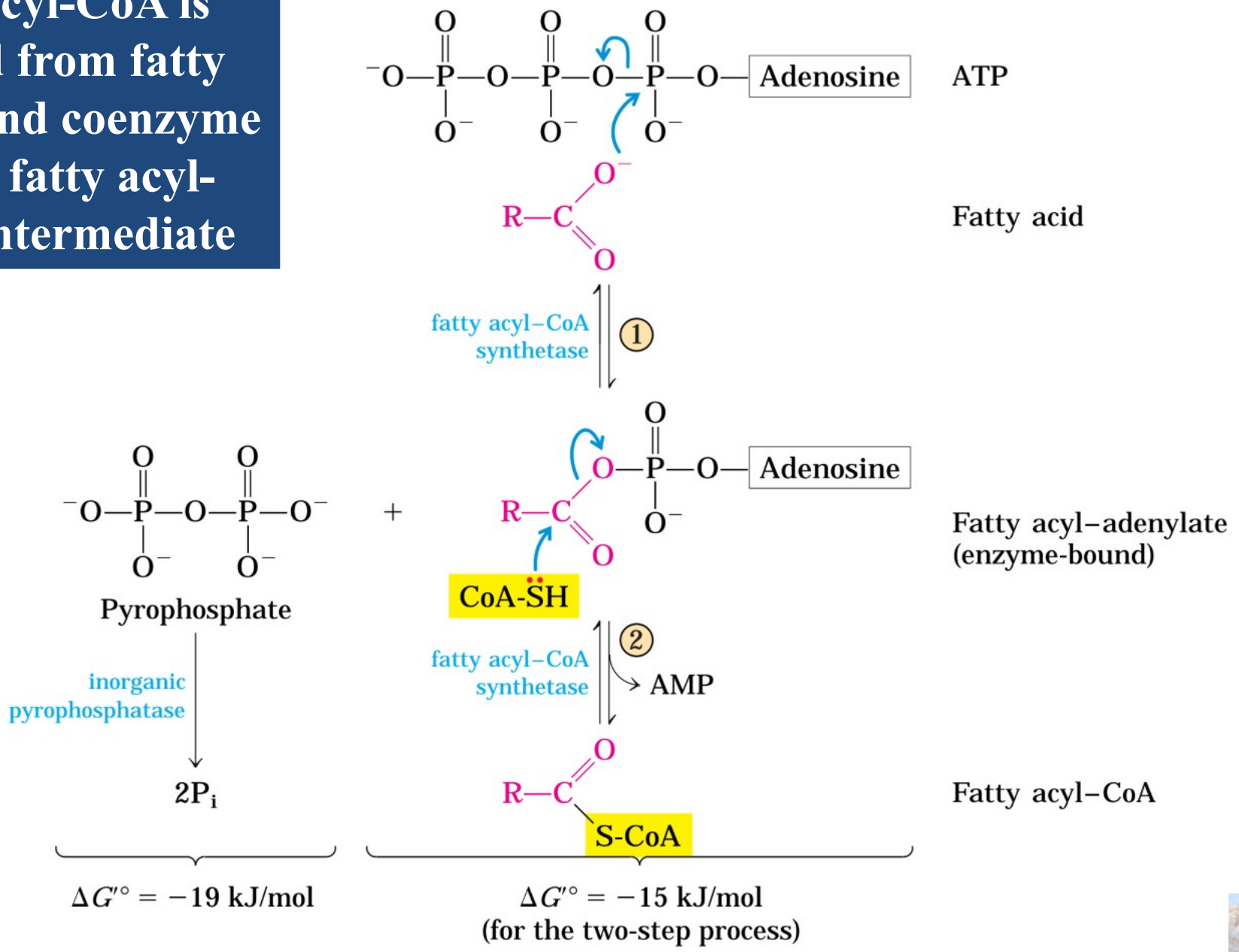
3. Fatty acids are activated on the outer membrane of mitochondria

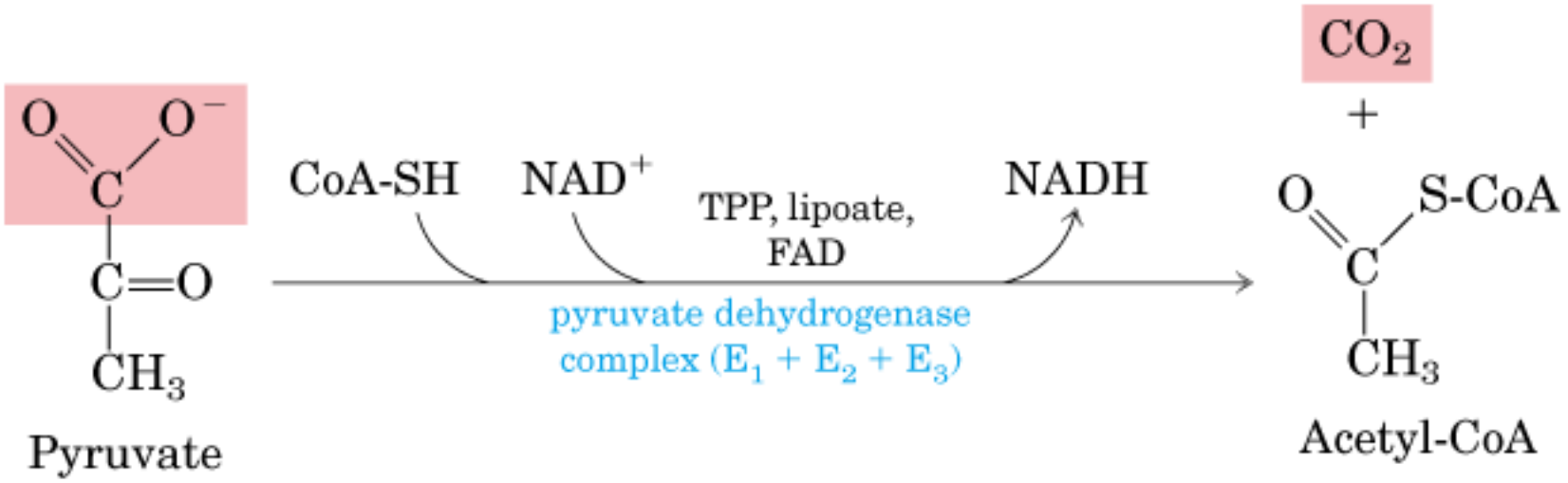
Fatty acids are converted to fatty acyl-CoA (a high energy compound) via a fatty-acyl-adenylate intermediate (enzyme-bound) by the action of fatty acyl-CoA synthetases (also called fatty acid thiokinase，硫激酶).




CoA — SH
Acyl–CoA
synthetase
COO⁻
O
C — S — CoA
Fatty acid
Acyl–CoA
ATP
AMP
+
PPi

Fatty acyl-CoA is formed from fatty acids and coenzyme A via a fatty acyl-AMP intermediate

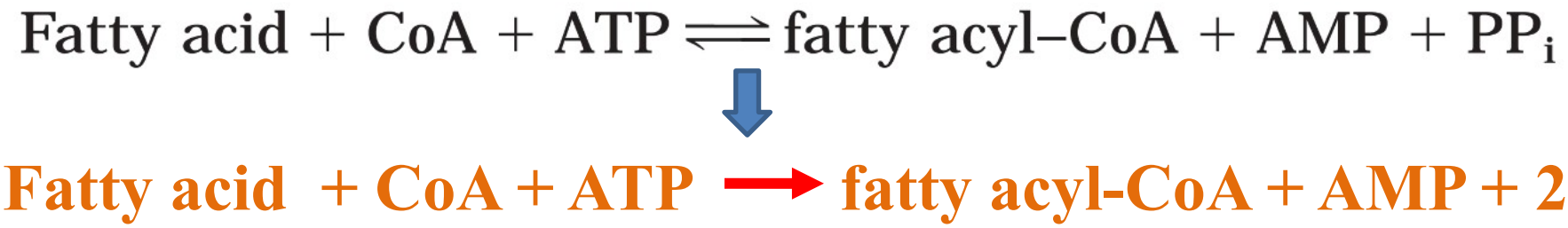




$\Delta G'^{\circ} = -33.4$ kJ/mol

The oxidative decarboxylation of pyruvate in mitochondria: producing acetyl-CoA and CO_2.

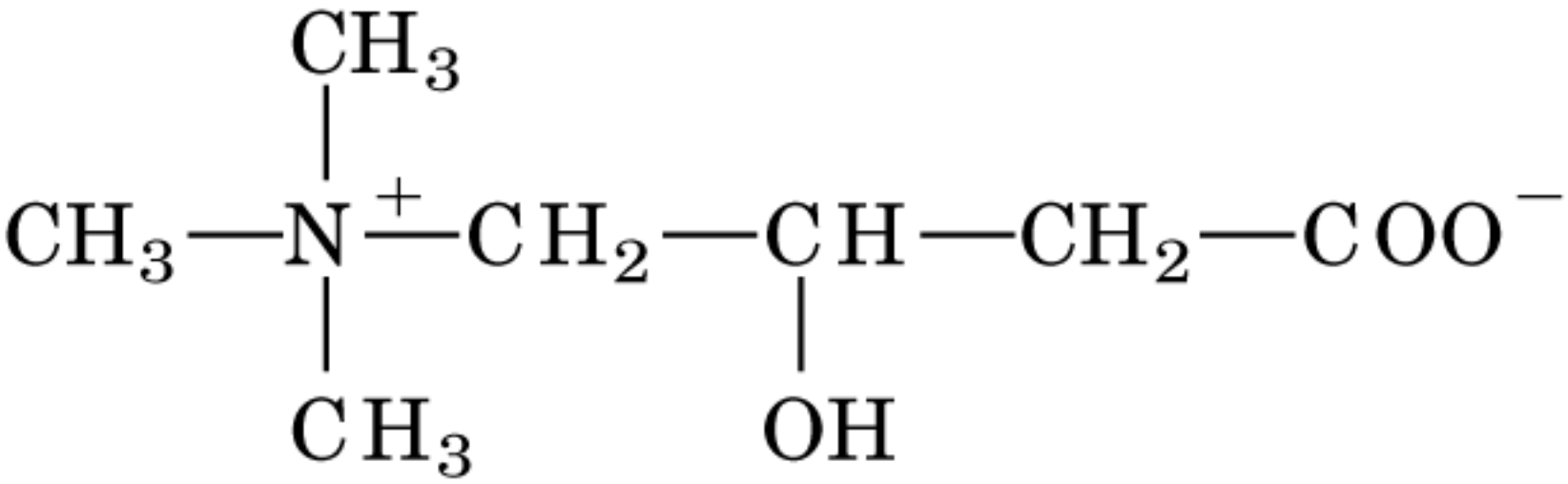




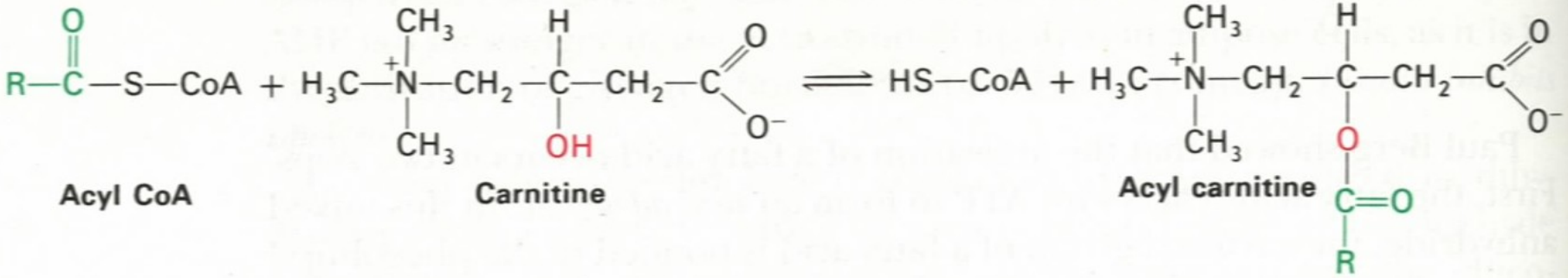
The fatty acid is activated by forming a thioester link with CoA before entering the mitochondria.

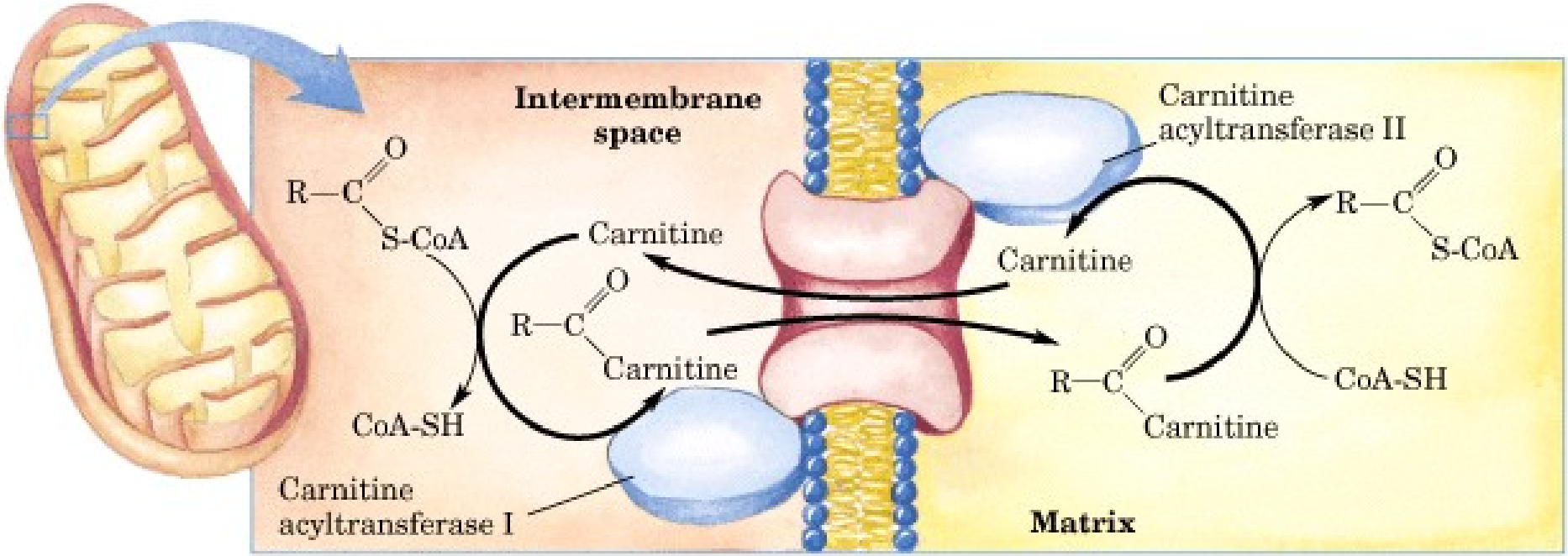
The inner mitochondrial membrane is not permeable to long-chain acyl CoA derivatives.

4. Activated (long chain) fatty acids are carried into the matrix by carnitine



Carnitine 肉碱（肉毒碱）


Acyl CoA
Carnitine
Acyl carnitine


Intermembrane space
Carnitine acyltransferase II
R—C(=O)—S-CoA
Carnitine
Carnitine
R—C(=O)—Carnitine
R—C(=O)—S-CoA
CoA-SH
R—C(=O)—Carnitine
CoA-SH
Carnitine acyltransferase I
Matrix

- The fatty acyl group is attached to **carnitine**（肉碱） by the action of **carnitine acyltransferase I** located on the outer face of the inner membrane, forming **fatty acyl-carnitine**, leaving the CoA in the cytosol.
- The **acyl carnitine/carnitine transporter** moves acyl-carnitine across the inner membrane of mitochondria.
- The acyl group is then transferred back to CoA to form fatty acyl-CoA by the action of **carnitine acyltransferase II** located on the inner face of the inner membrane.



5. Fatty acyl-CoA is oxidized to acetyl-CoA via multiple rounds of β oxidation

The β oxidation consists of four reactions:

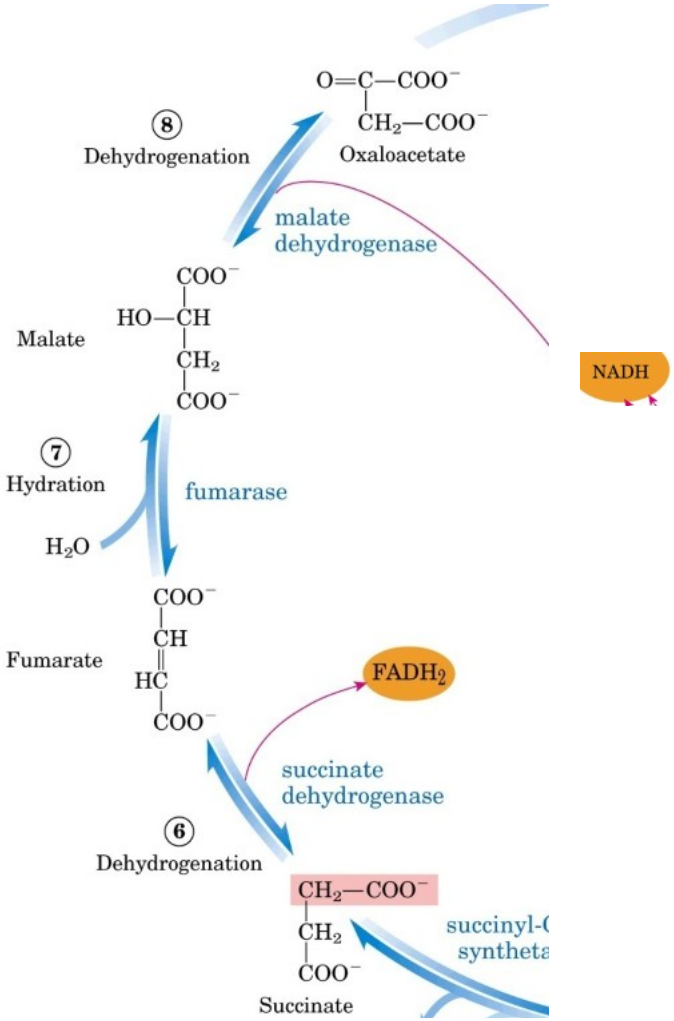
Oxidation by FAD

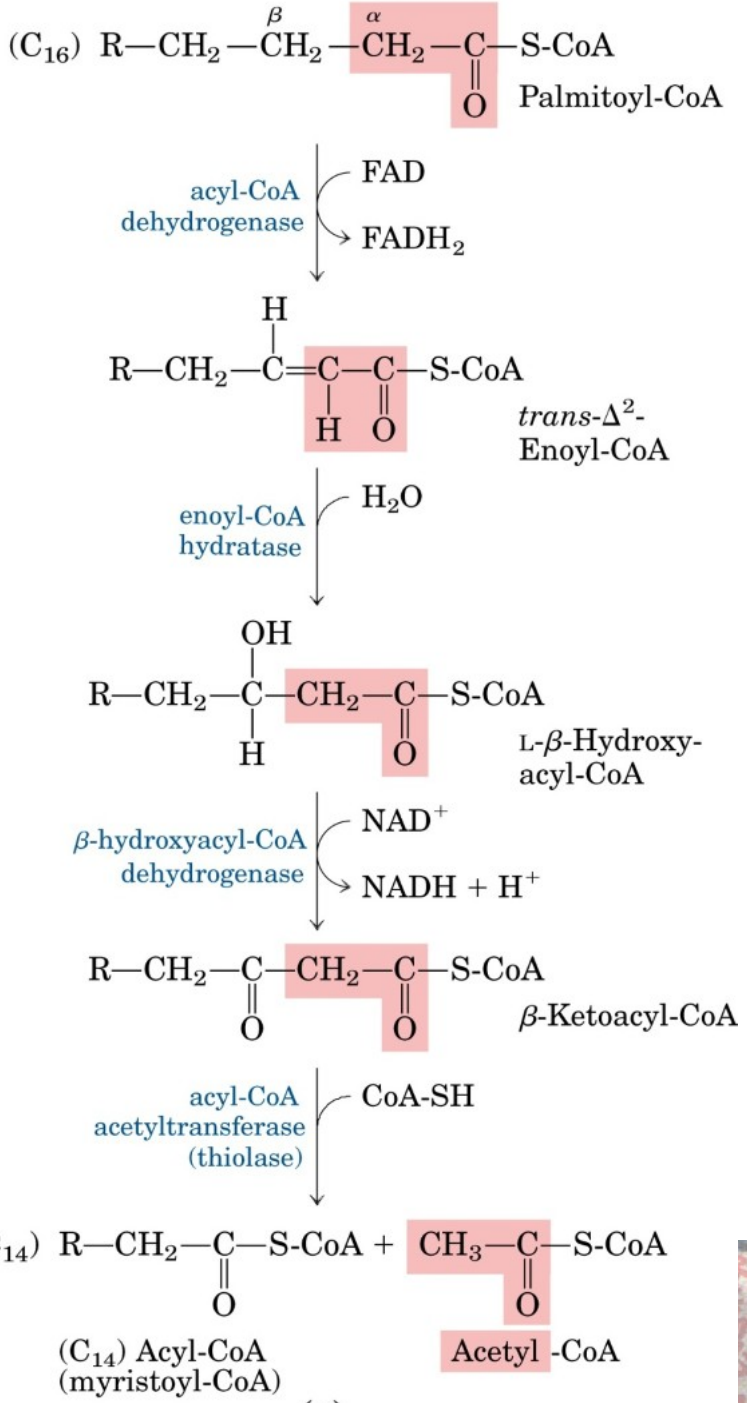
Hydration

Oxidation by NAD^+

Thiolysis by CoA.

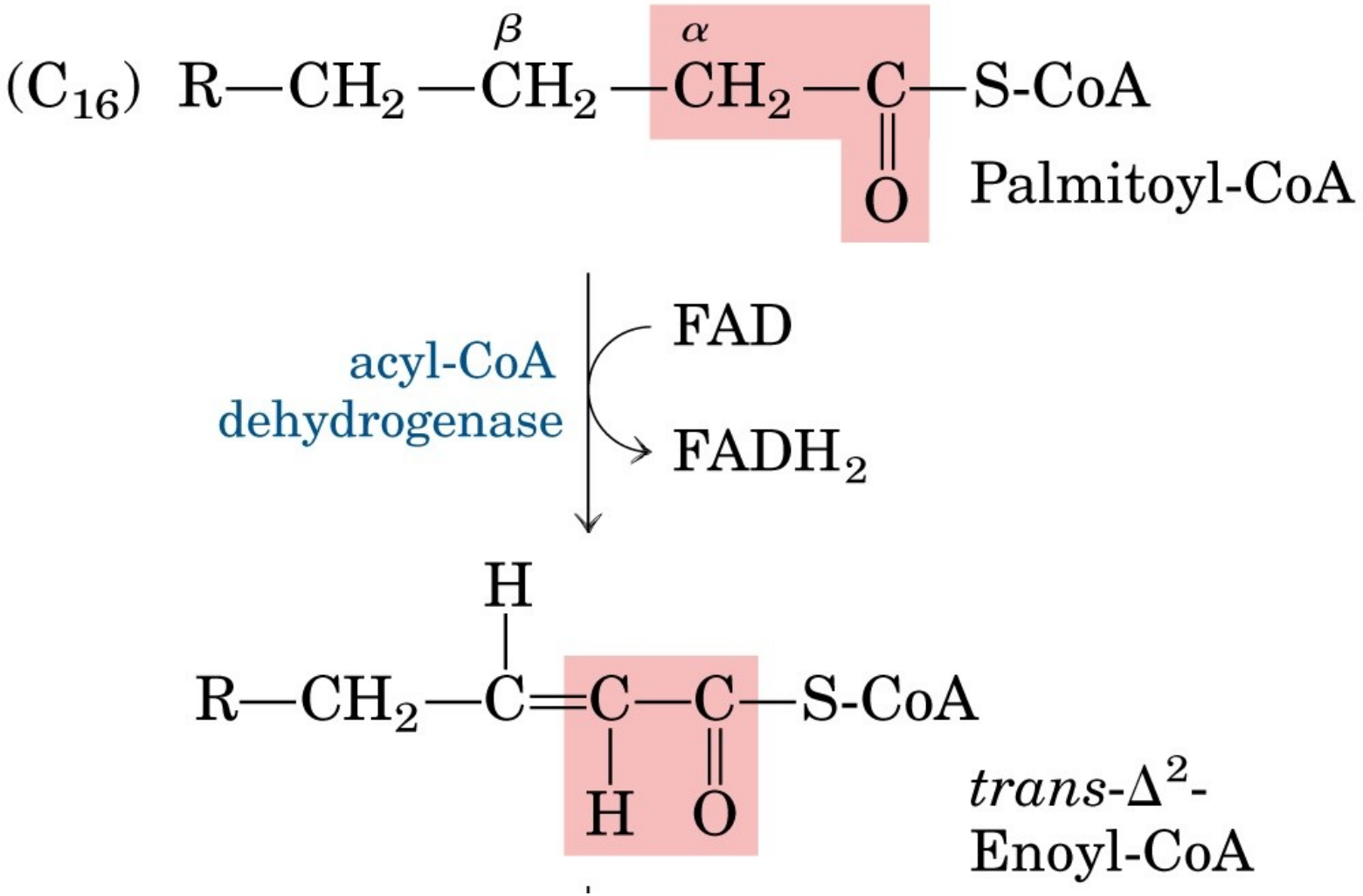


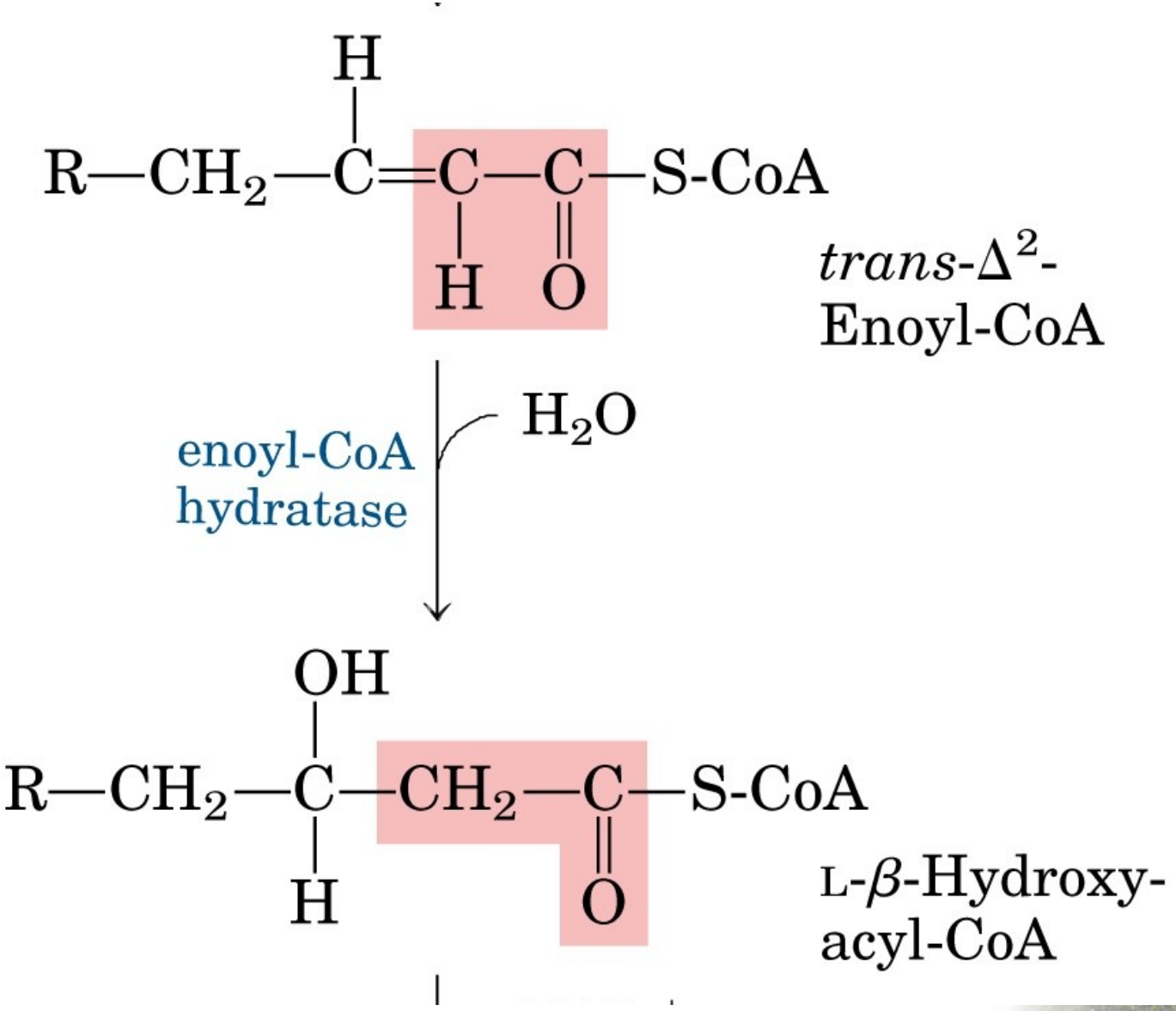


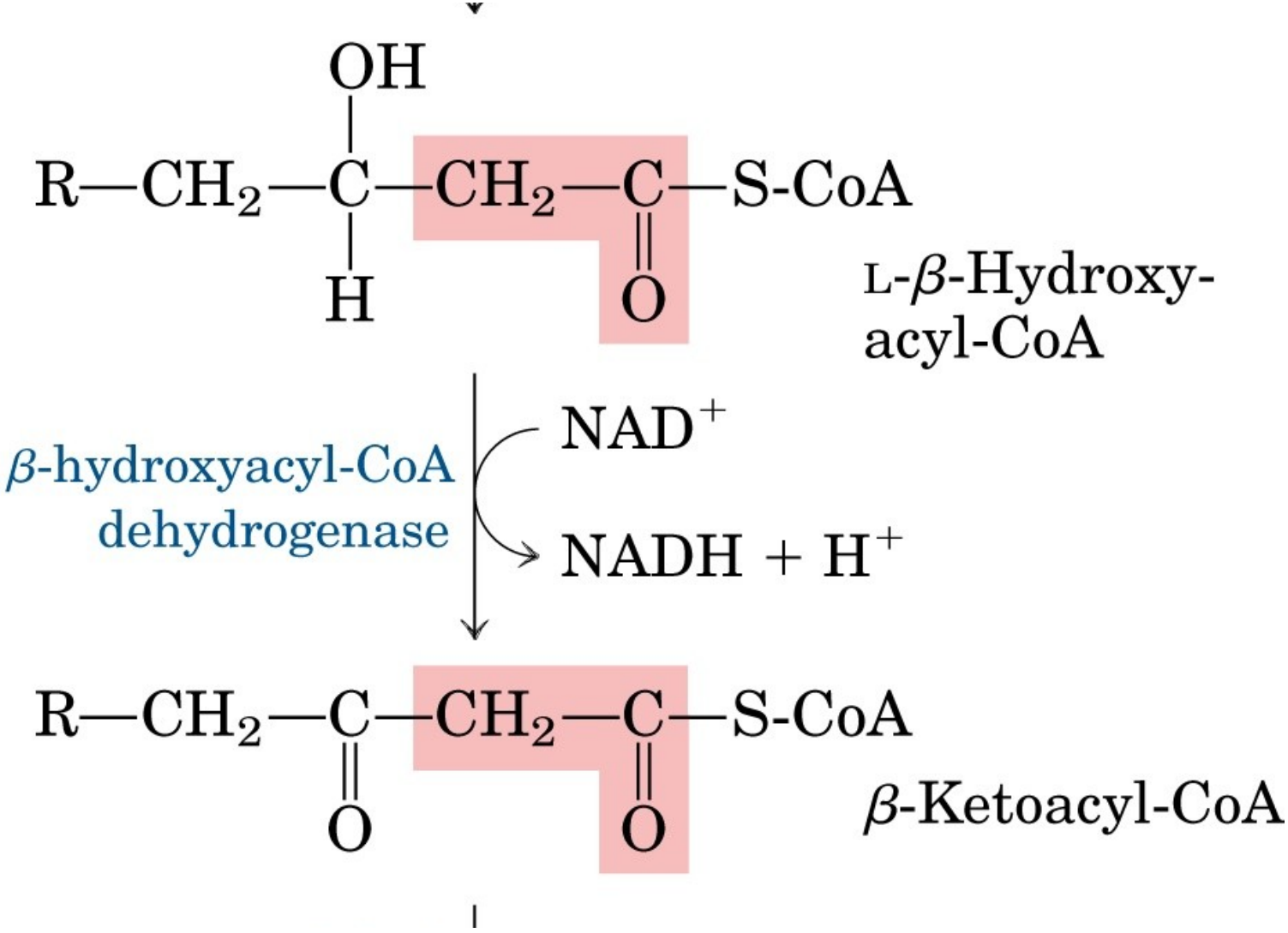


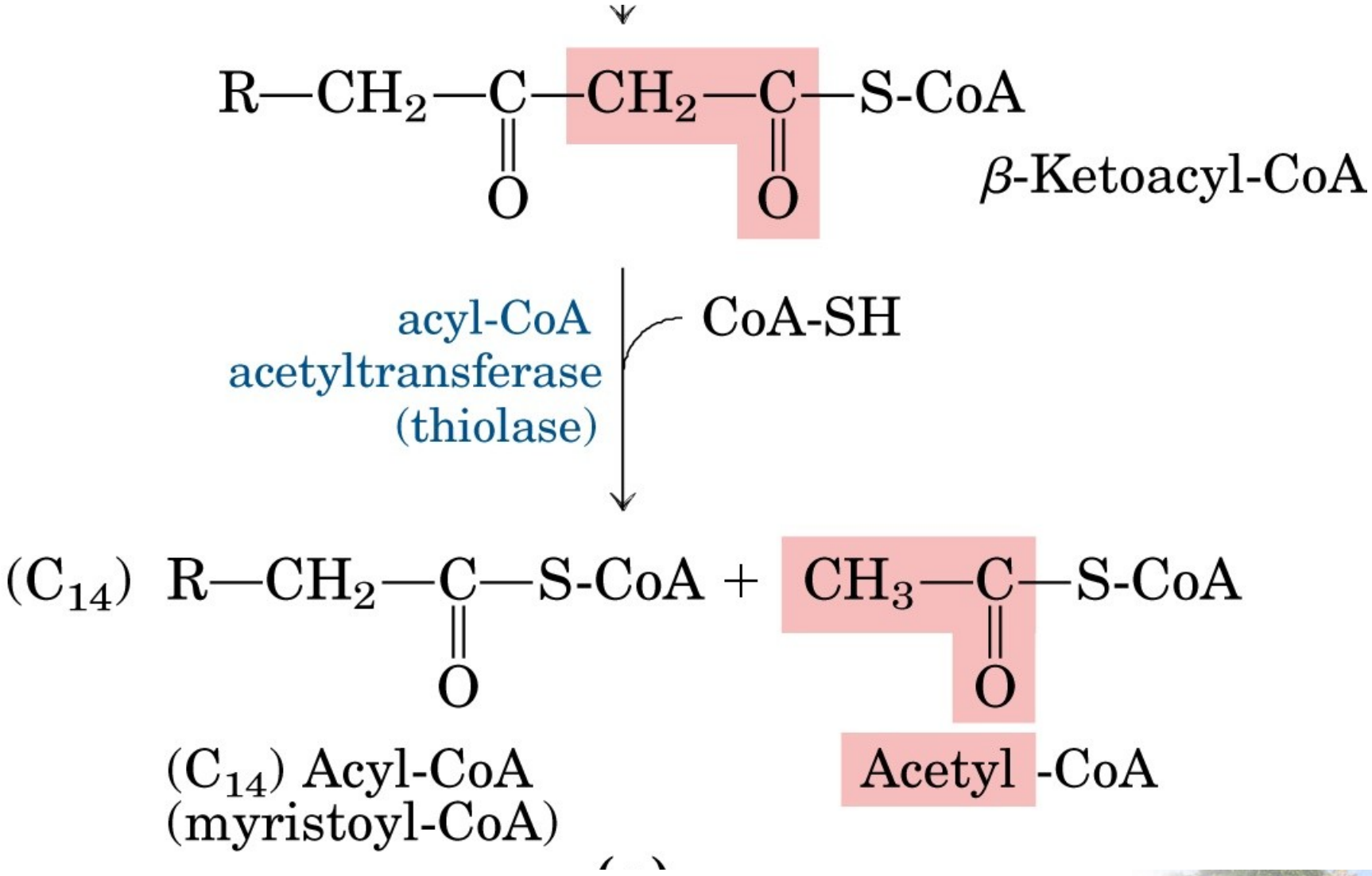
(a)

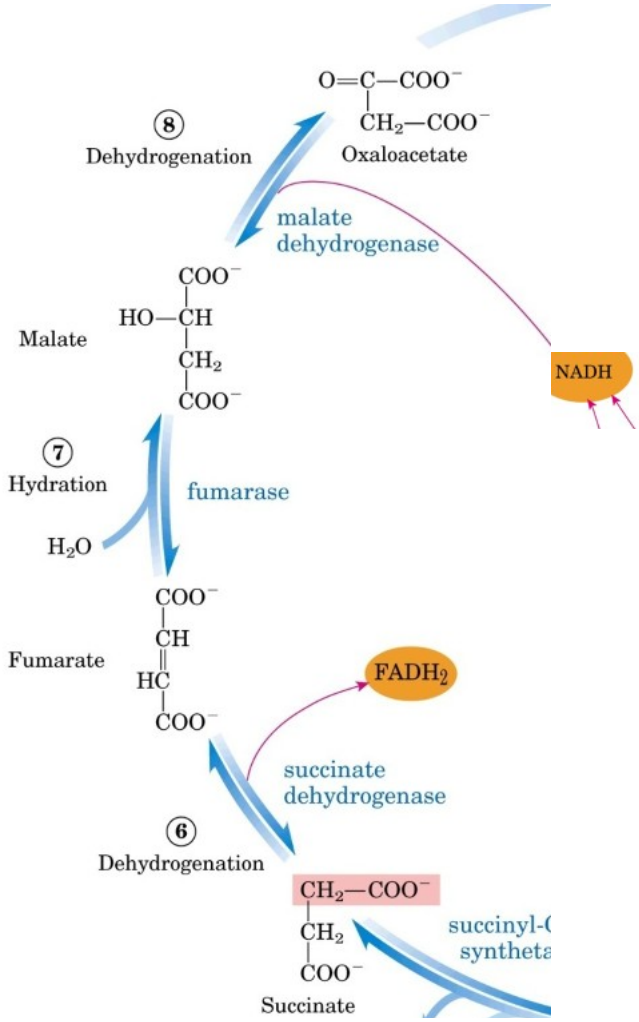
Biochemistry

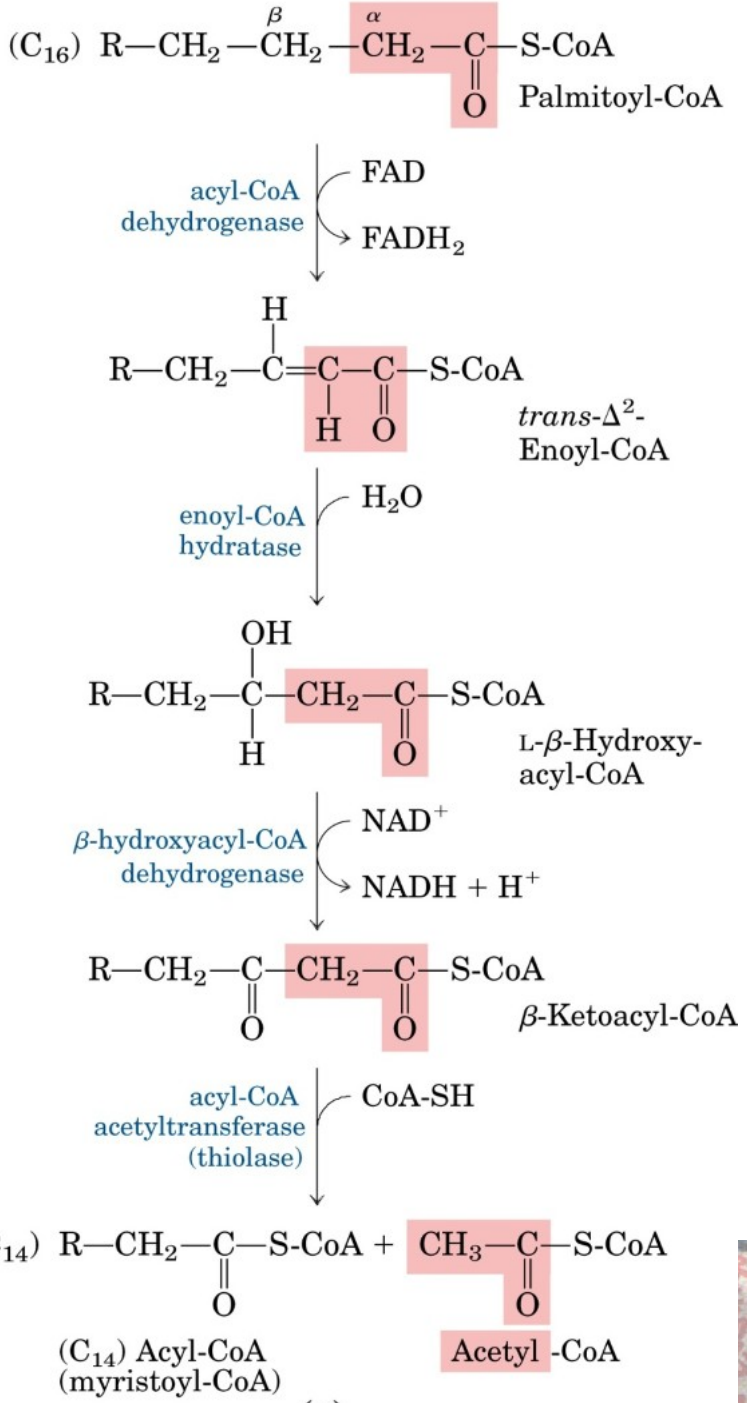

β
α
(C16) R—CH2—CH2—CH2—C—S-CoA
O
Palmitoyl-CoA
acyl-CoA
dehydrogenase
FAD
FADH2
H
R—CH2—C=C—C—S-CoA
H O
trans-Δ2-
Enoyl-CoA


H
R—CH2—C=C—C—S-CoA
H O
trans-Δ2-
Enoyl-CoA
enoyl-CoA
hydratase
H2O
OH
R—CH2—C—CH2—C—S-CoA
H O
L-β-Hydroxy-
acyl-CoA


OH
R—CH2—C—CH2—C—S-CoA
H
O
L-β-Hydroxy-acyl-CoA
β-hydroxyacyl-CoA dehydrogenase
NAD+
NADH + H+
R—CH2—C—CH2—C—S-CoA
O
O
β-Ketoacyl-CoA

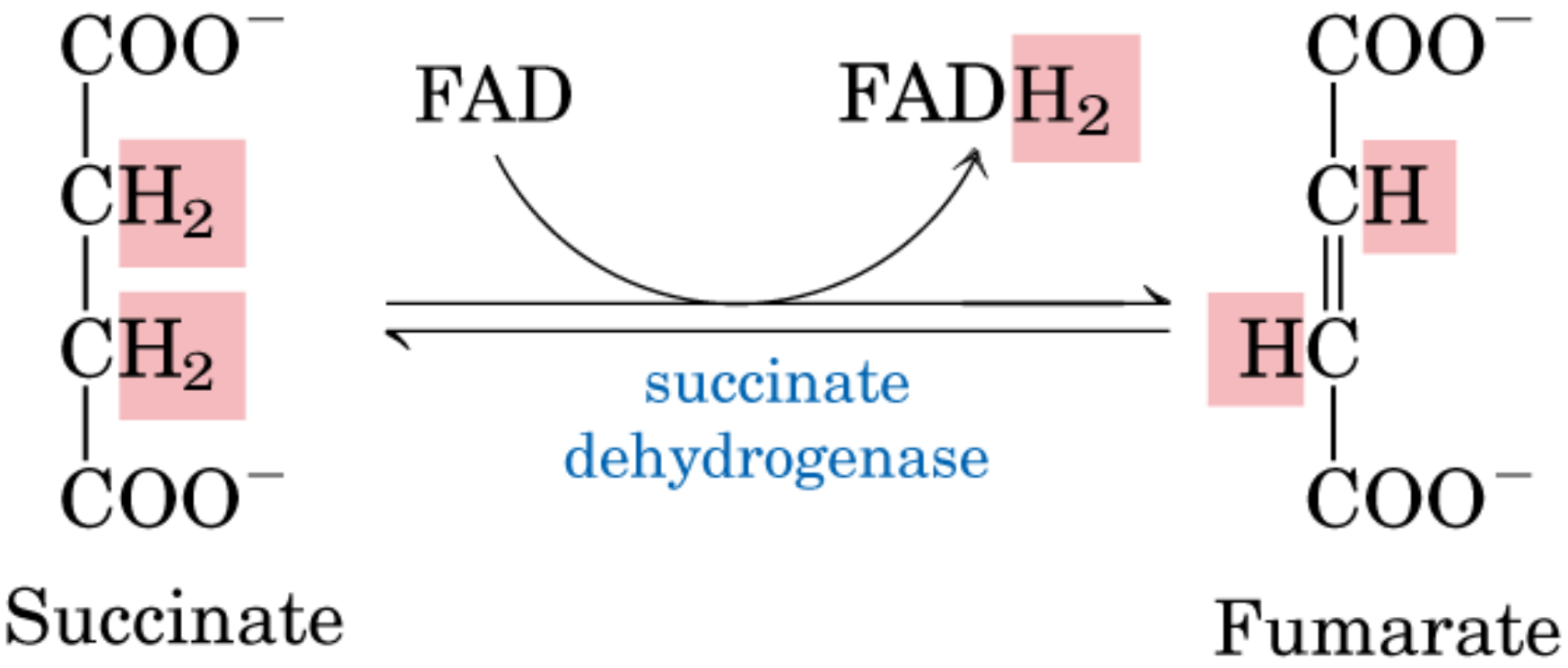

R—CH2—C—CH2—C—S-CoA
β-Ketoacyl-CoA
acyl-CoA acetyltransferase (thiolase)
CoA-SH
(C14) R—CH2—C—S-CoA + CH3—C—S-CoA
(C14) Acyl-CoA (myristoyl-CoA)
Acetyl -CoA

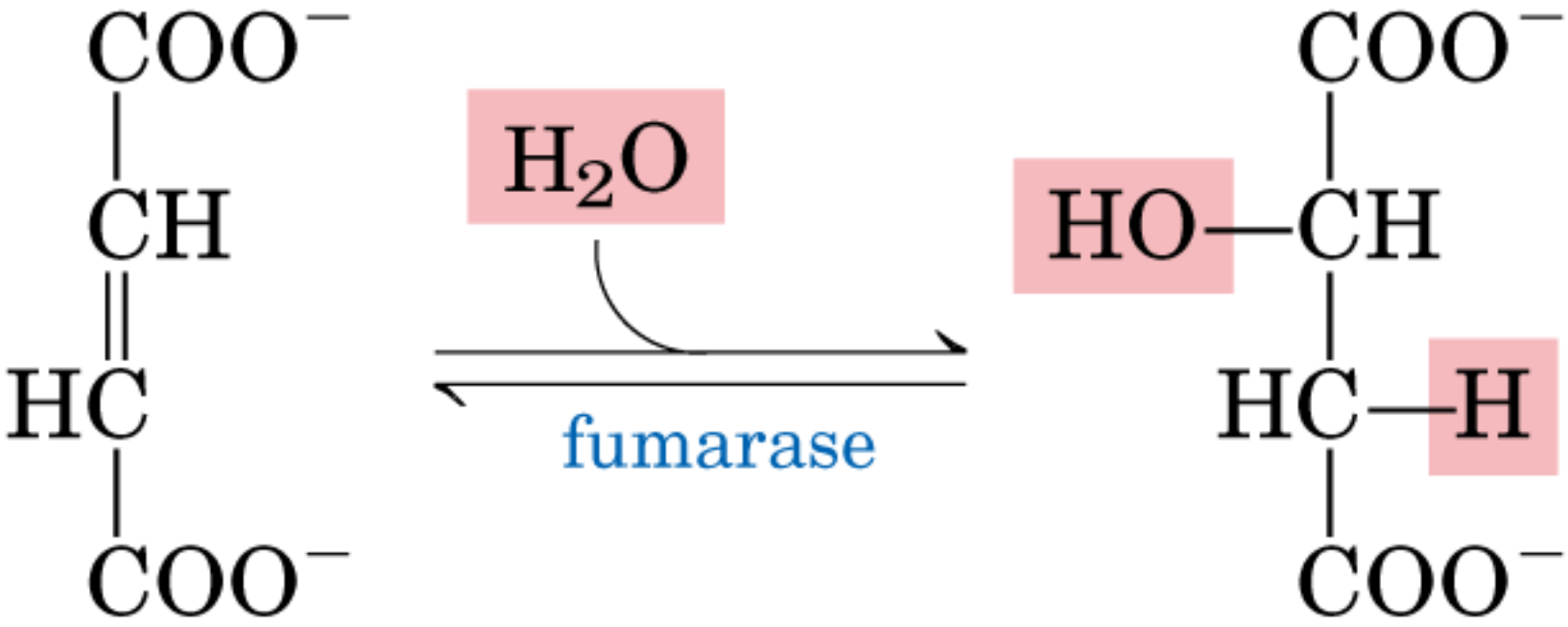




Biochemistry

Three similar reactions between the β-oxidation and citric acid cycle

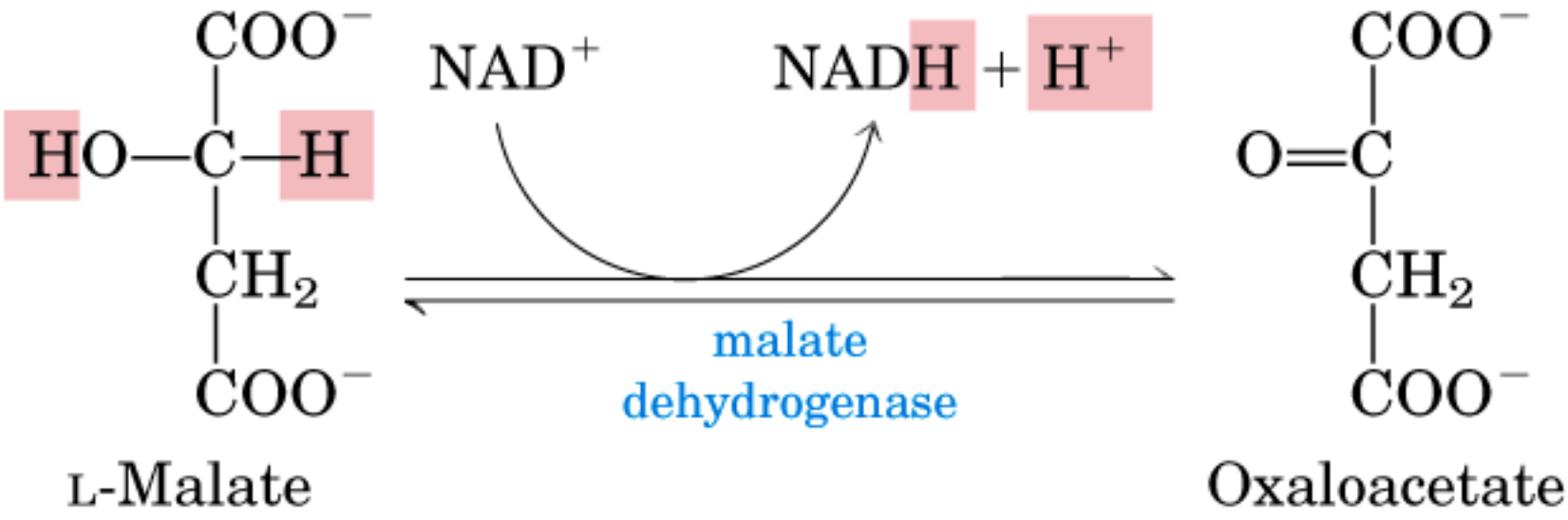




Fumarate

L-Malate

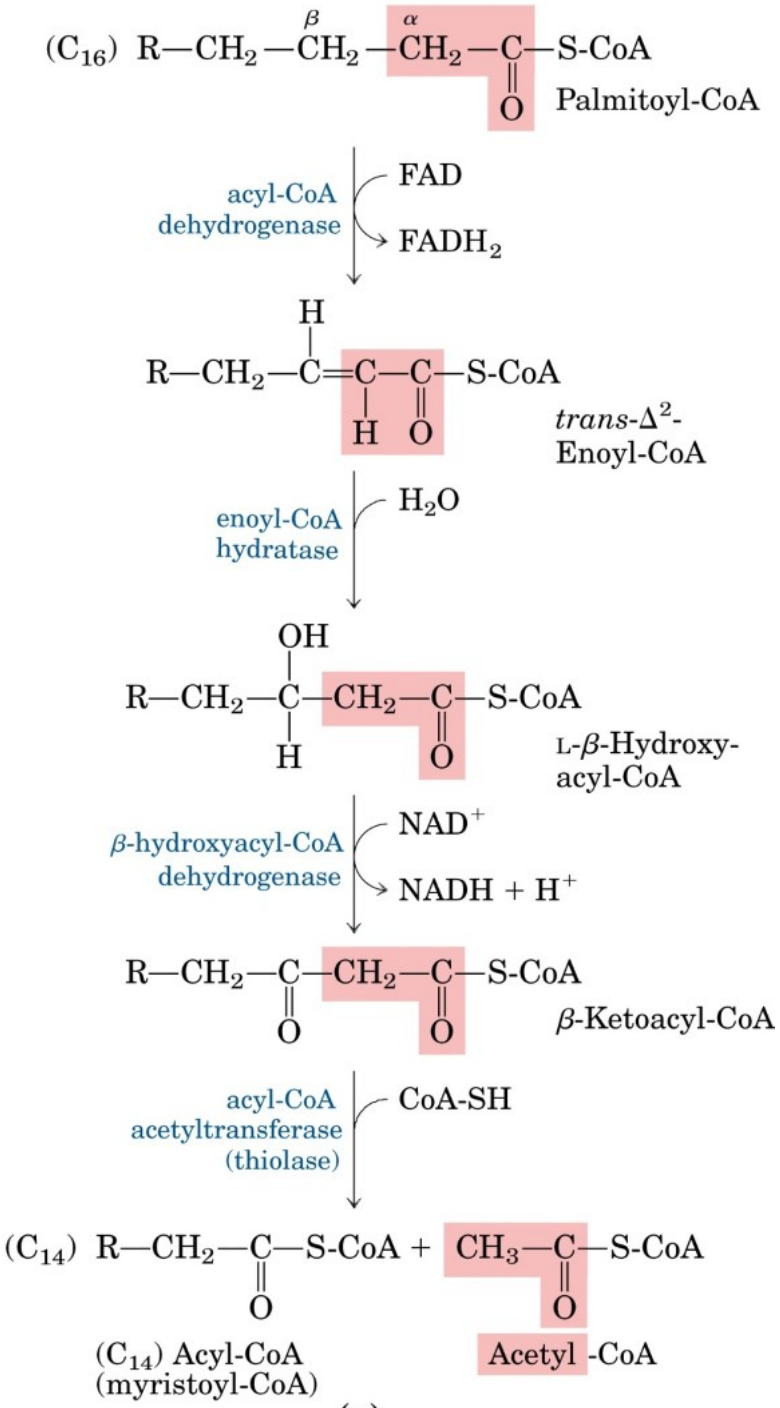
$$\Delta G'^{\circ} = -3.8 \text{ kJ/mol}$$



L-Malate

Oxaloacetate

$$\Delta G'^{\circ} = 29.7 \text{ kJ/mol}$$


(C16) R—CH2—CH2—CH2—C—S-CoA
β
α
O
Palmitoyl-CoA
acyl-CoA dehydrogenase
FAD
FADH2
H
R—CH2—C=C—C—S-CoA
H O
trans-Δ2-Enoyl-CoA
enoyl-CoA hydratase
H2O
OH
R—CH2—C—CH2—C—S-CoA
H O
L-β-Hydroxy-acyl-CoA
β-hydroxyacyl-CoA dehydrogenase
NAD+
NADH + H+
R—CH2—C—CH2—C—S-CoA
O O
β-Ketoacyl-CoA
acyl-CoA acetyltransferase (thiolase)
CoA-SH
(C14) R—CH2—C—S-CoA + CH3—C—S-CoA
O O
(C14) Acyl-CoA (myristoyl-CoA)
Acetyl -CoA

(a)

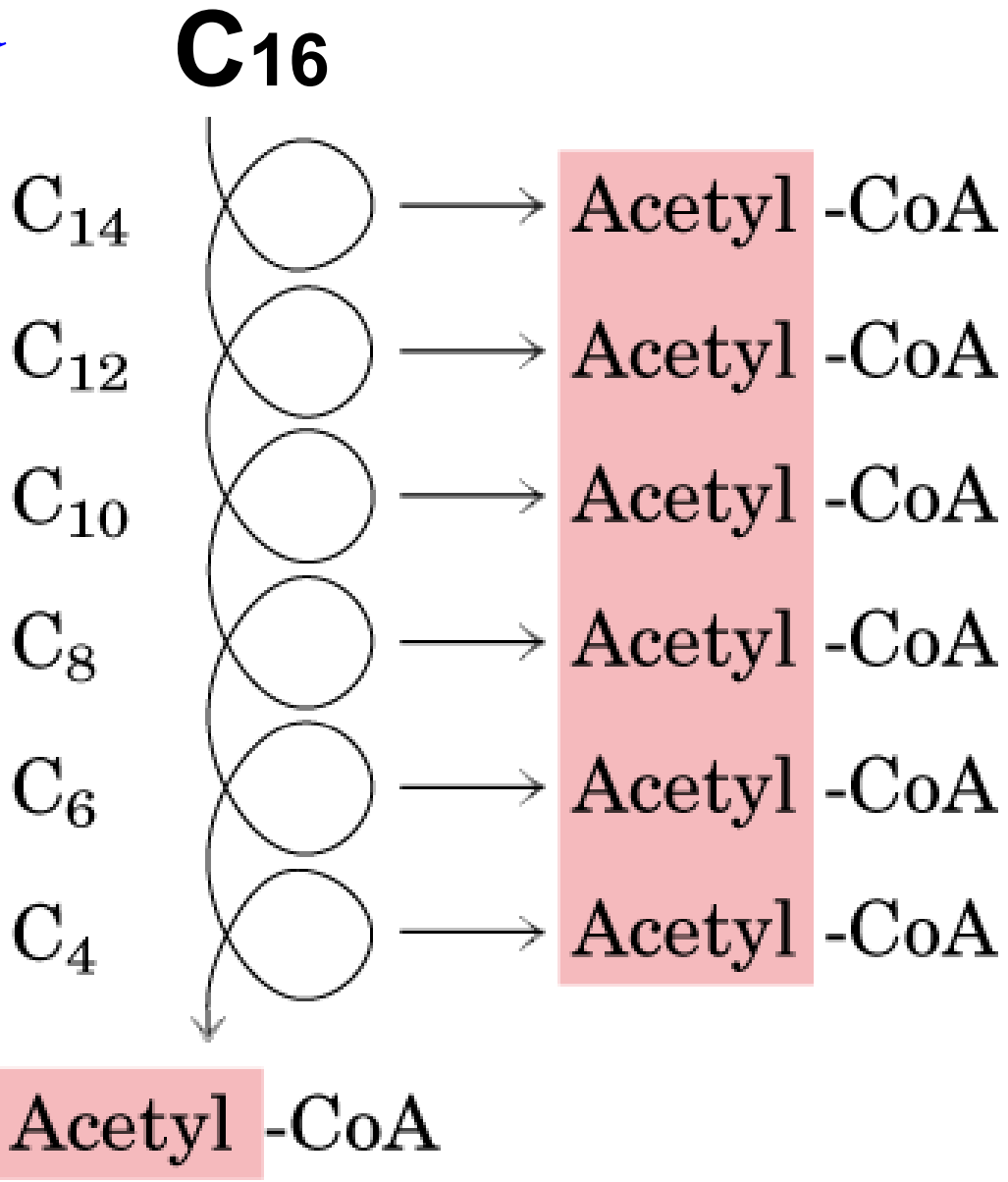
1. The 1st oxidation is catalyzed by the membrane-bound acyl-CoA dehydrogenase, converting acyl-CoA to *trans*-Δ^2-enoyl-CoA with electrons collected by FAD.
2. The hydration step, catalyzed by enoyl-CoA hydratase, converts the *trans*-Δ^2-enoyl-CoA to L-β-hydroxyacyl-CoA.
3. The second oxidation is catalyzed by L-β-hydroxyacyl-CoA dehydrogenase, converting L-β-hydroxylacyl-CoA to β-ketoacyl-CoA, with electrons collected by NAD^+.

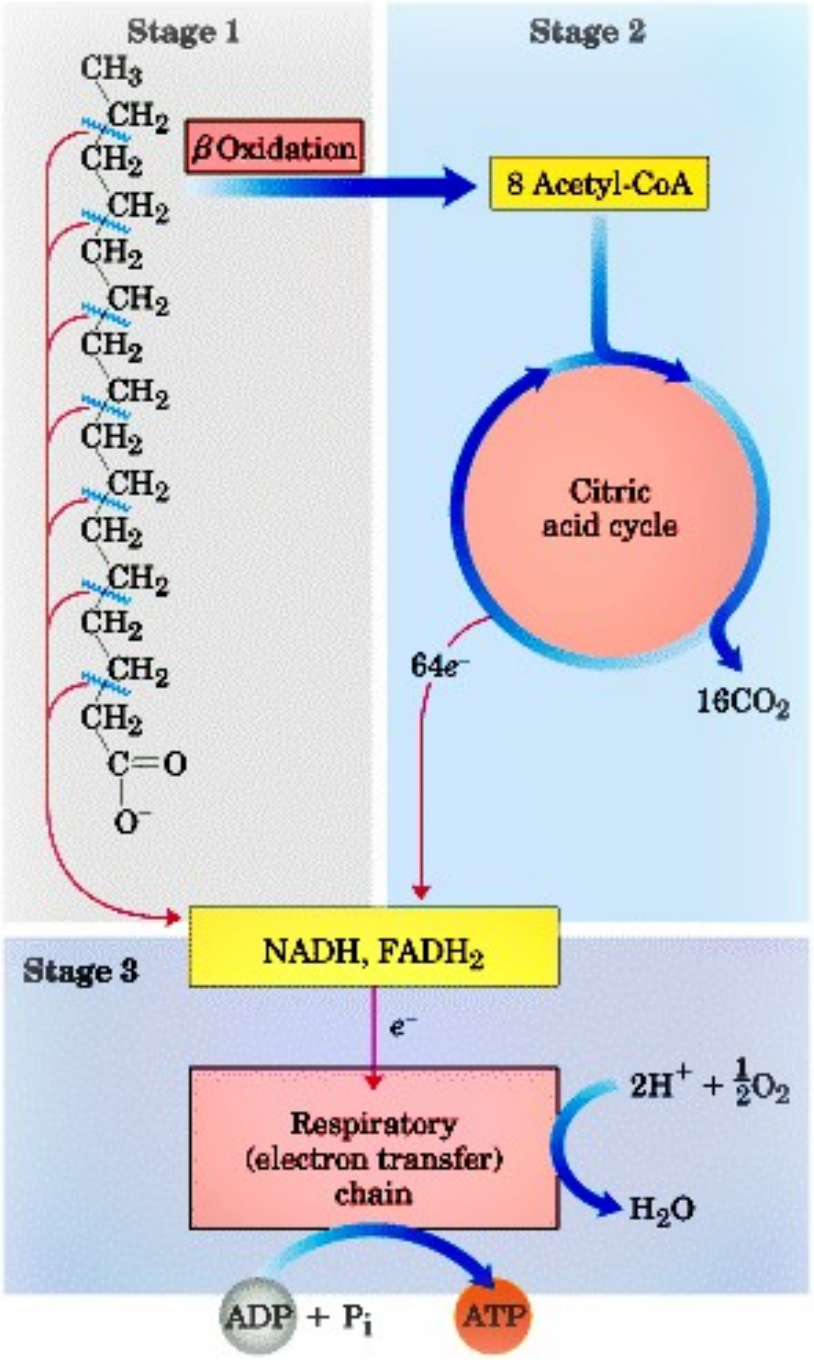


4. The acyl-CoA acetyltransferase (or commoly called thiolase) catalyzes the attack of CoA, cleaving β-ketoacyl-CoA between the α and β carbon (thiolysis), generating two acyl-CoA molecules with one entering the citric acid cycle and the other reenter the β oxidation pathway.



- Emergy yield




Stage 1
Stage 2
CH_3
CH_2
β Oxidation
8 Acetyl-CoA
C=O
O^-
Citric acid cycle
$64e^-$
$16CO_2$
NADH, $FADH_2$
Stage 3
e^-
Respiratory (electron transfer) chain
$2H^+ + \frac{1}{2}O_2$
H_2O
ADP + P_i
ATP

Biochemist

$$\text{Palmitoyl-CoA} + 7\text{CoA} + 7\text{FAD} + 7\text{NAD}^+ + 7\text{H}_2\text{O} \longrightarrow 8\ \text{acetyl-CoA} + 7\text{FADH}_2 + 7\text{NADH} + 7\text{H}^+$$

- Emergy yield
- The complete oxidization of each 16-carbon palmitate (to H_2O and CO_2) yields 106 ATP
- (32 ATP per glucose, both having about 60% of actual energy recovery).



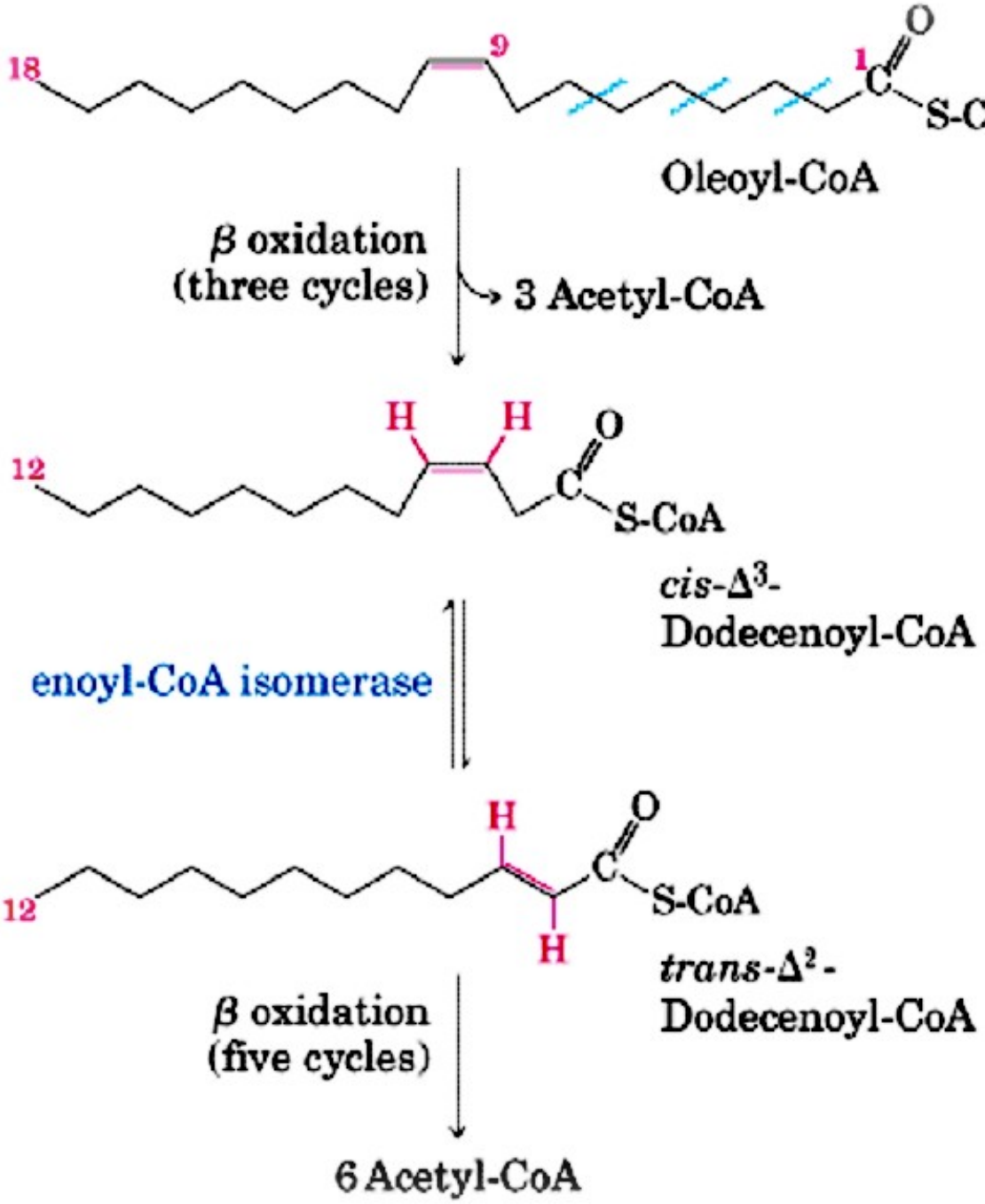
6. Oxidation of unsaturated fatty acids requires one or two auxiliary enzymes, an isomerase and a reductase

The isomerase converts a *cis*-Δ^3 double bond to a *trans*-Δ^2 double bond.

The reductase (2,4-dienoyl-CoA reductase) converts a *trans*-Δ^2, cis-Δ^4 structure to a *trans*-Δ^3 structure, which will be further converted to a *trans*-Δ^2 structure by the isomerase.

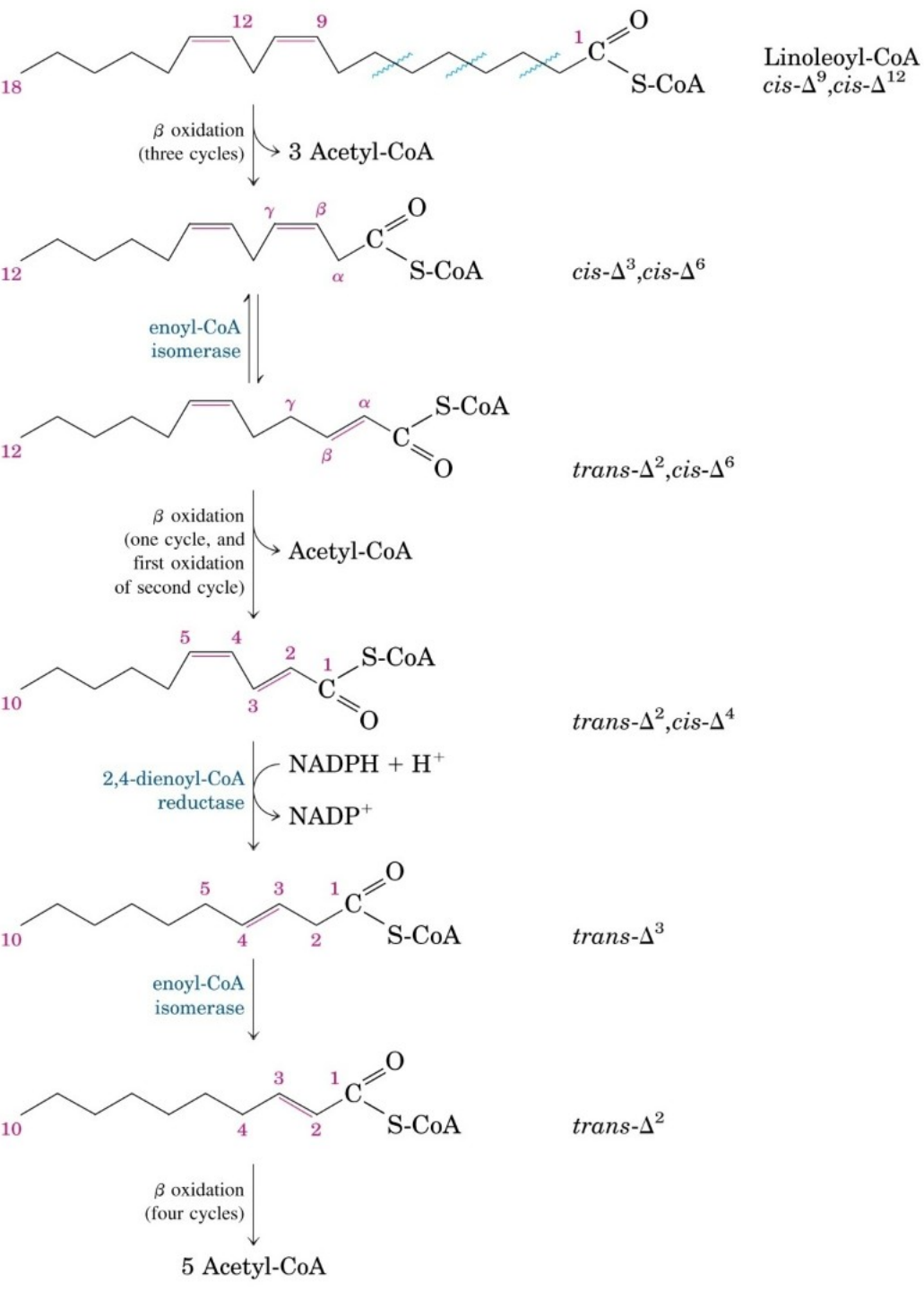
NADPH is needed for the reduction (from two double bonds to one).





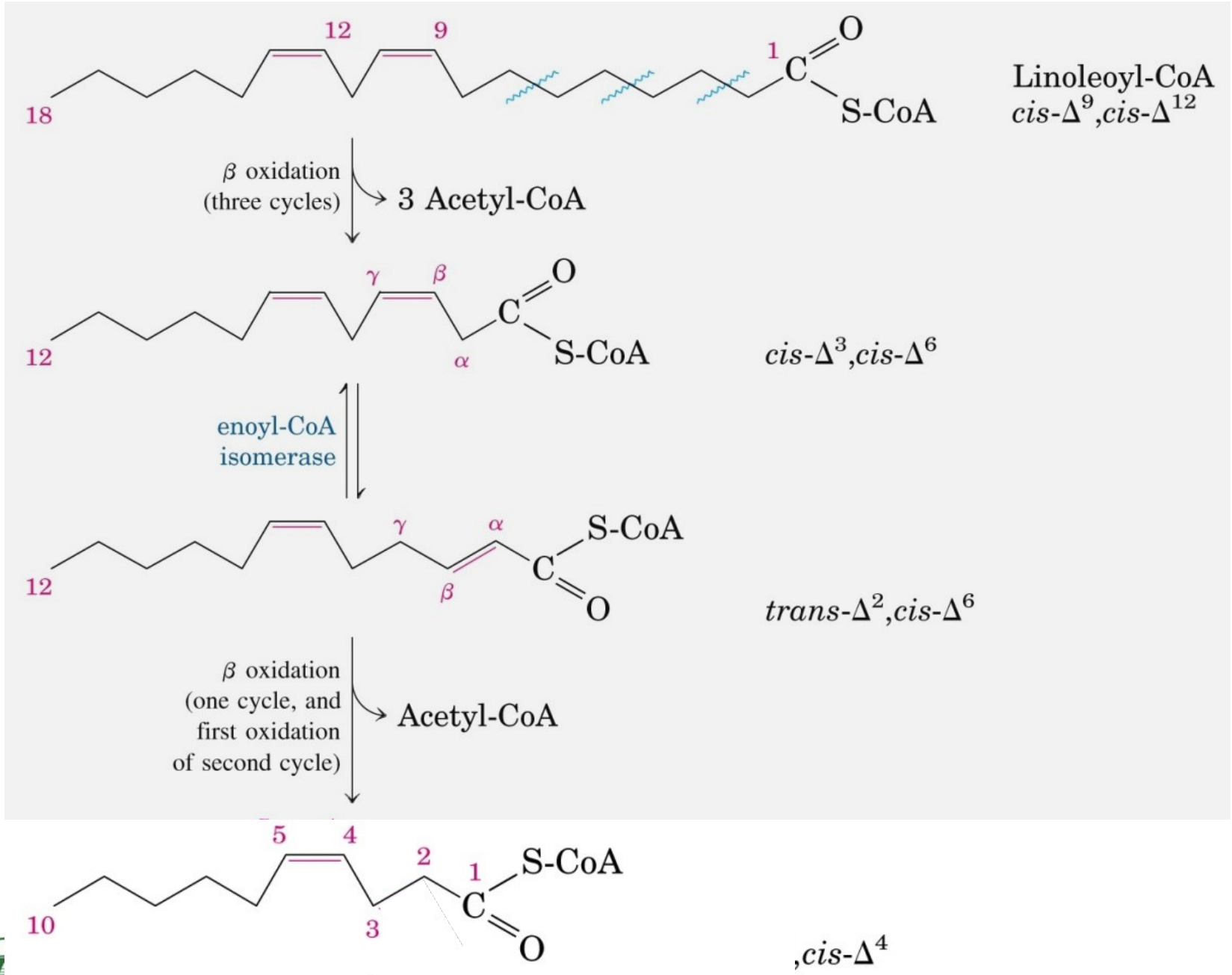
Oxidation of a monounsaturated fatty acid: the enoyl-CoA isomerase helps to reposition the double bond

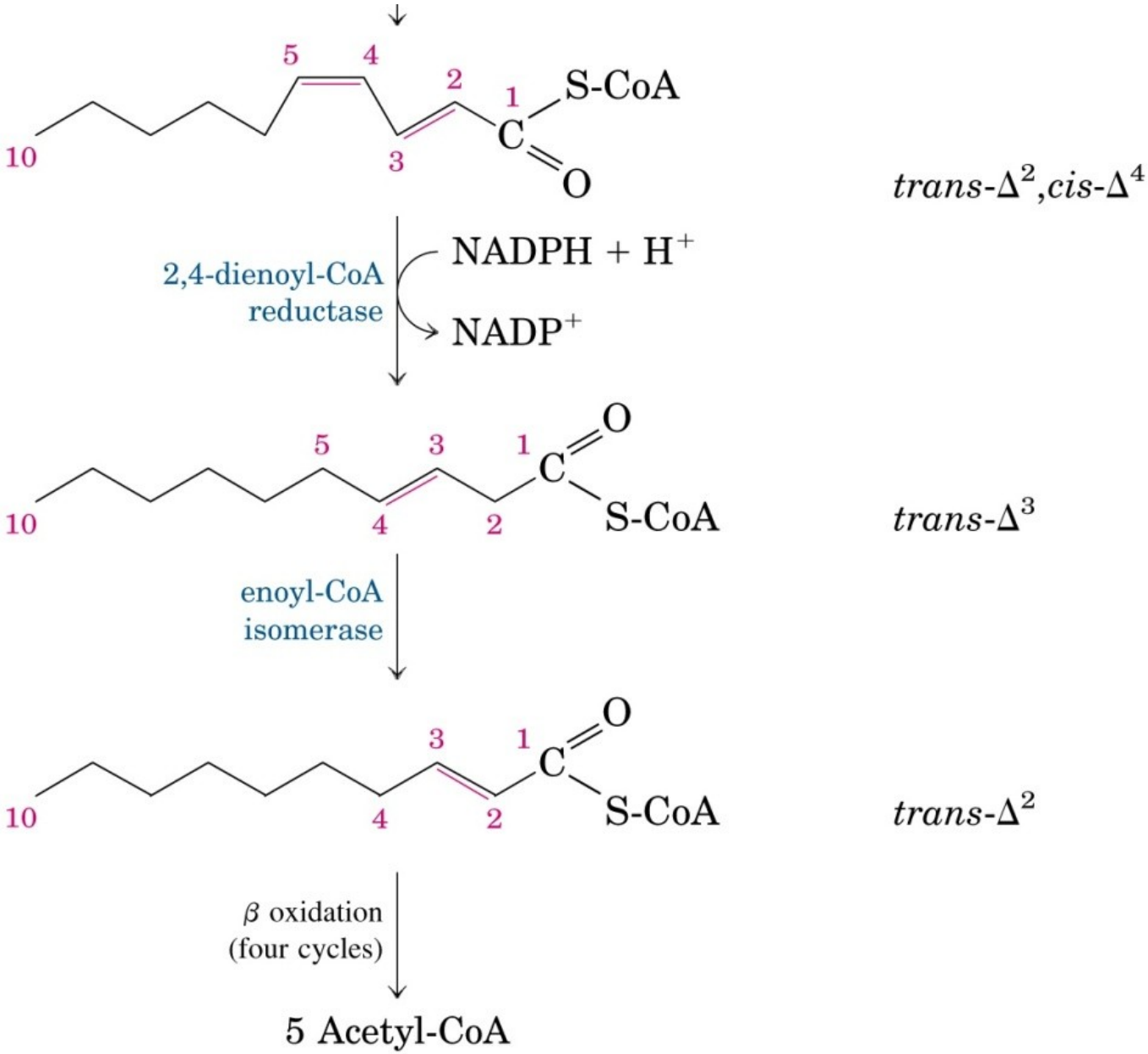


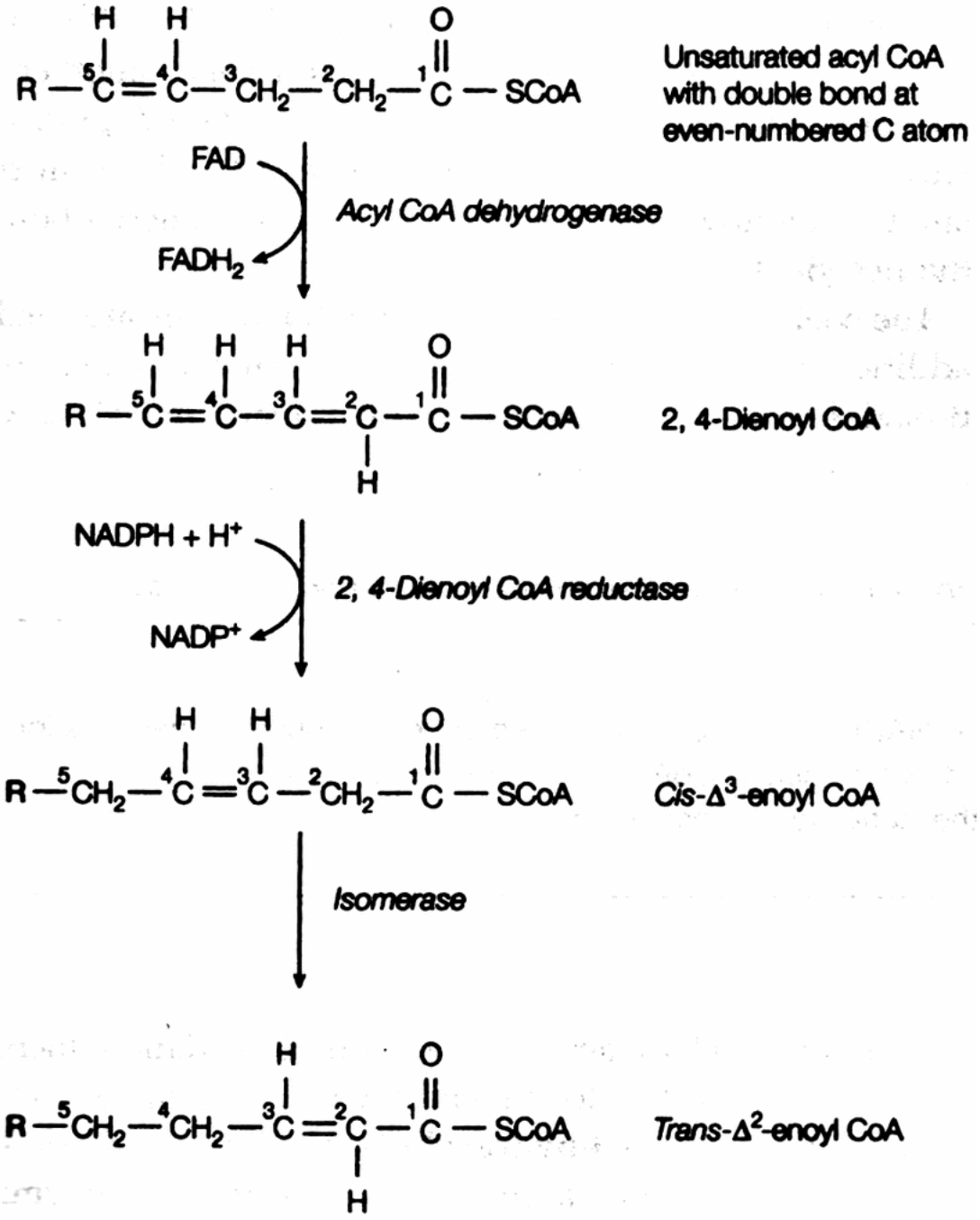


Both an isomerase and a reductase are needed for oxidizing polyunsaturated fatty acids.




12
9
1
O
C
S-CoA
18
Linoleoyl-CoA
cis-Δ9,cis-Δ12
β oxidation
(three cycles)
3 Acetyl-CoA
γ
β
O
C
12
S-CoA
α
cis-Δ3,cis-Δ6
enoyl-CoA
isomerase
γ
α
S-CoA
C
12
β
O
trans-Δ2,cis-Δ6
β oxidation
(one cycle, and
first oxidation
of second cycle)
Acetyl-CoA
5
4
2
1
S-CoA
C
10
3
O
,cis-Δ4


S-CoA
O
NADPH + H+
2,4-dienoyl-CoA reductase
NADP+
trans-Δ2,cis-Δ4
trans-Δ3
enoyl-CoA isomerase
trans-Δ2
β oxidation (four cycles)
5 Acetyl-CoA


Unsaturated acyl CoA with double bond at even-numbered C atom
FAD
FADH2
Acyl CoA dehydrogenase
2, 4-Dienoyl CoA
NADPH + H+
NADP+
2, 4-Dienoyl CoA reductase
Cis-Δ3-enoyl CoA
Isomerase
Trans-Δ2-enoyl CoA

7. The oxidation of odd-chain fatty acids forms C3 propionyl-CoA

Fatty acids having an odd number of carbon atoms are also degraded by the β–oxidation pathway in the same way as those with an even number of carbon atoms.

The only difference is that in the final round the five carbon acyl CoA intermediate is cleaved into C3 **propionyl CoA** and acetyl CoA.



8. Fatty acid oxidation also occurs in peroxisomes (glyoxysomes)

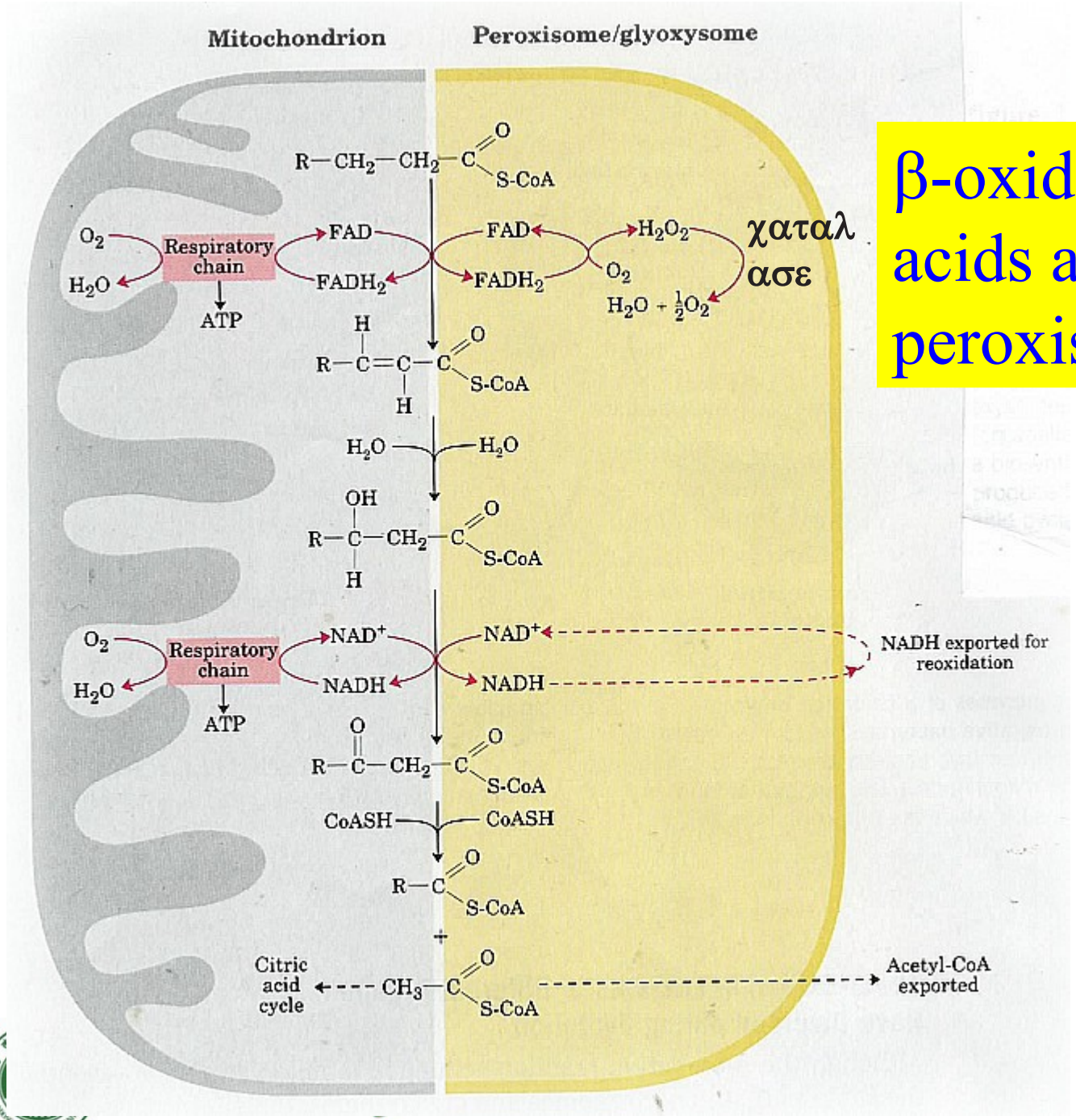
Peroxisome is now recognized as the principal organelle in which fatty acids are oxidized in most cell types.

The acetyl-CoA produced in animal peroxisomes is transported into cytosol, where it is used in the synthesis of cholesterol and other metabolites.



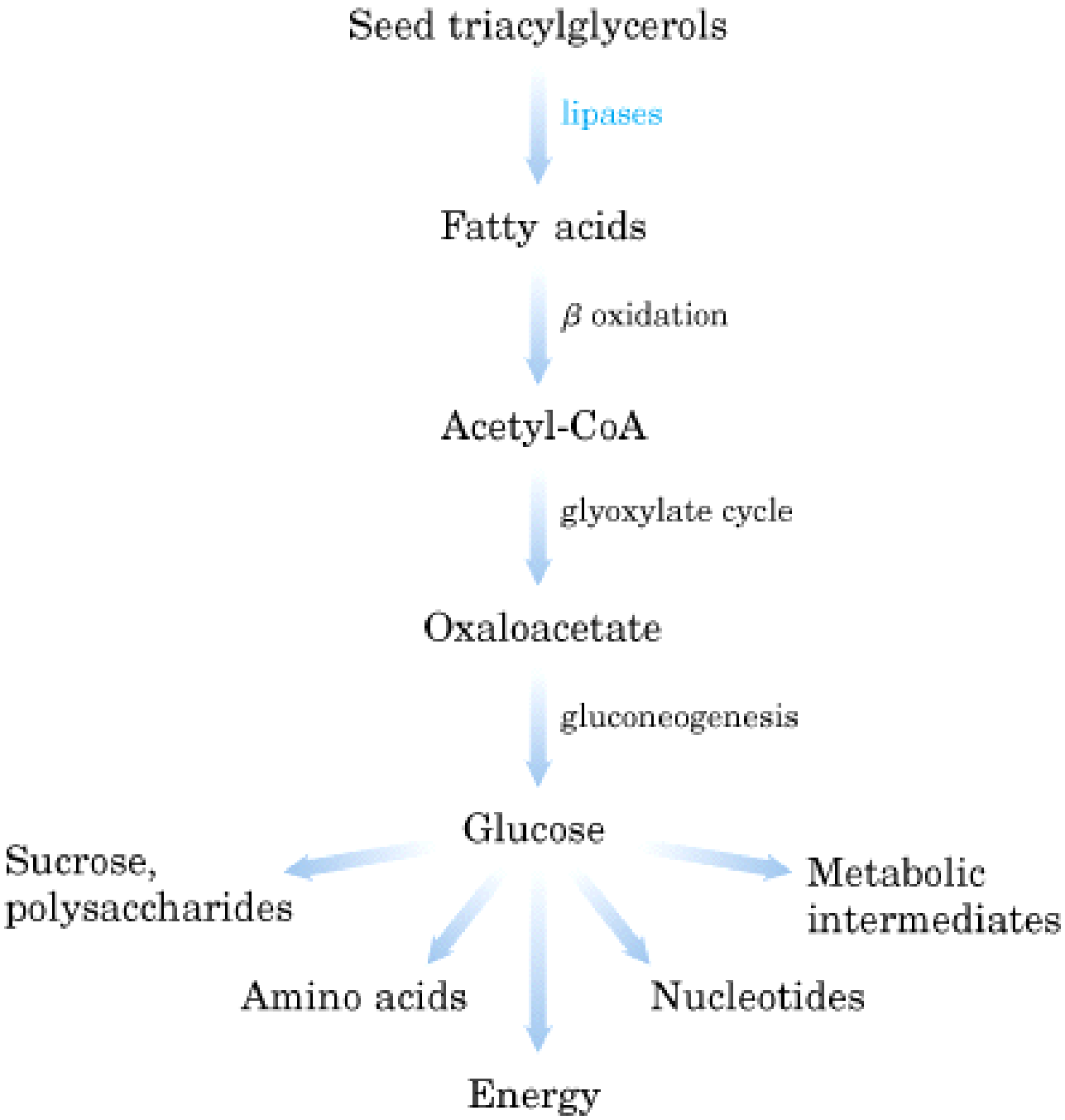
The acetyl-CoA produced in plant peroxisomes/ glyoxysomes (especially in germinating seeds) is converted to succinate via the glyoxylate cycle, and then to glucose via gluconeogenesis.





β-oxidation of fatty acids also occurs in peroxisomes




Seed triacylglycerols
lipases
Fatty acids
β oxidation
Acetyl-CoA
glyoxylate cycle
Oxaloacetate
gluconeogenesis
Glucose
Sucrose, polysaccharides
Metabolic intermediates
Amino acids
Nucleotides
Energy

9. Acetyl-CoA in liver can be converted to ketone bodies when carbohydrate supply is not optimal



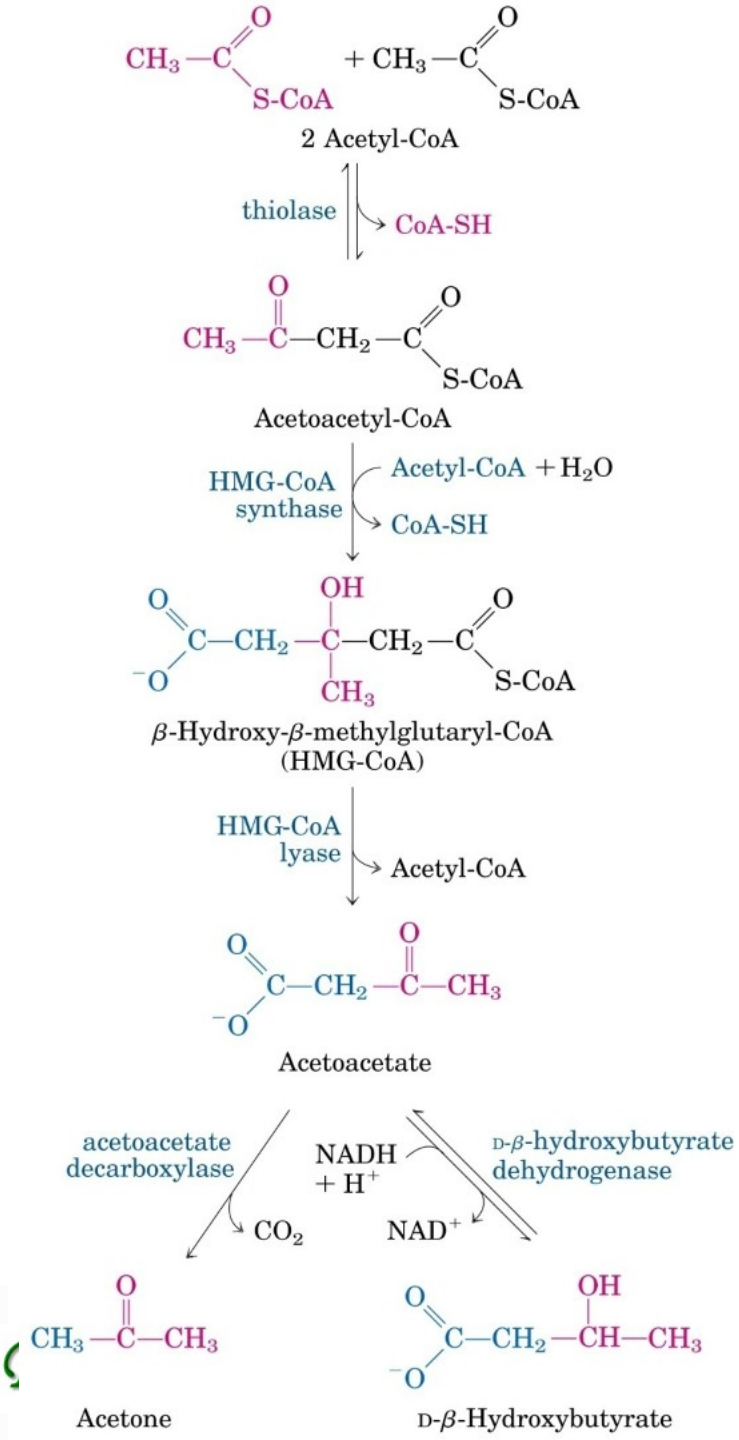
- In liver, The acetyl-CoA generated from active fatty acid oxidation can not be oxidized via the citric acid cycle and will be converted to acetoacetate （乙酰乙酸）, β-hydroxybutyrate （羟基丁酸）, and acetone （丙酮） (i.e., the ketone bodies) in mitochondria for export to other tissues.



For ketone body formation, first two acetyl-CoAs condense to form acetoacetyl-CoA (乙酰乙酰 **CoA)** catalyzed by thiolase; then addition of another acetyl-CoA forms β-hydroxyl-β-methylglutaryl-CoA **(**β– 羟－ β －甲基戊二酰 **CoA)**.

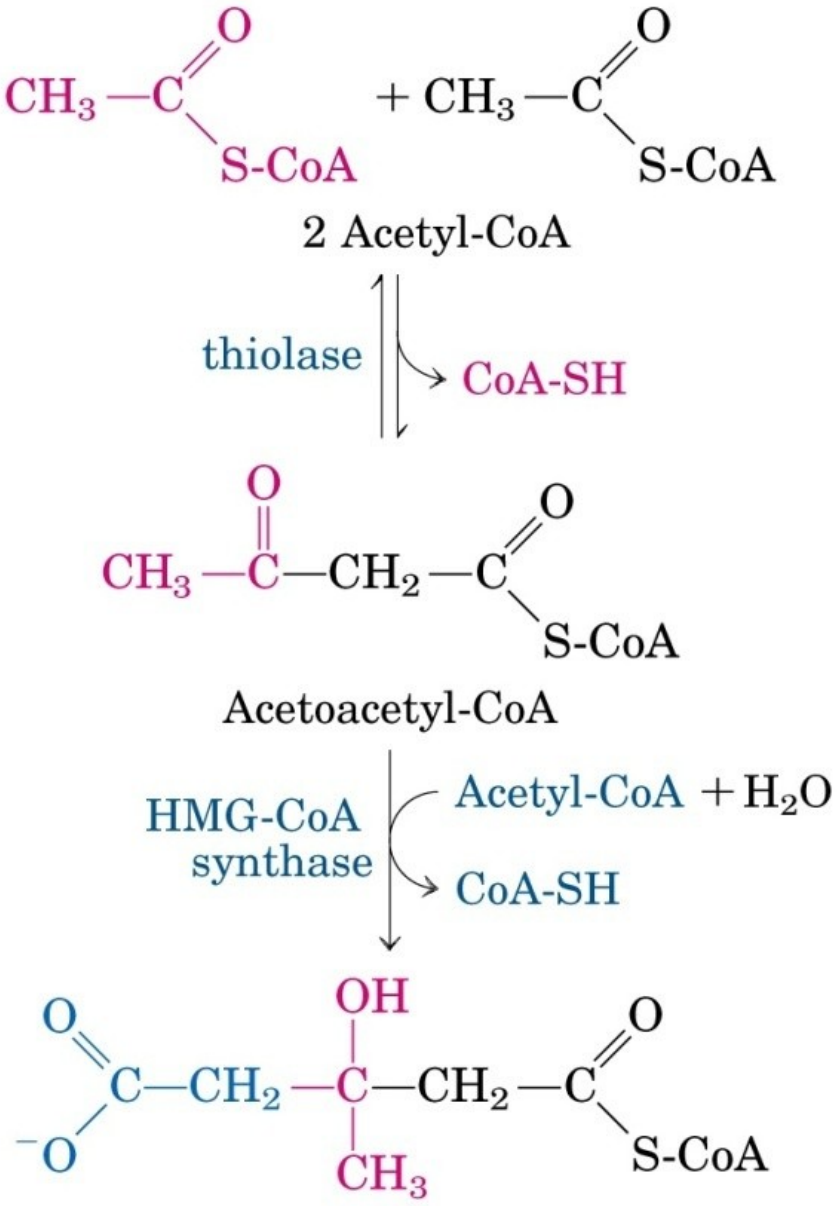
Acetoacetate can be decarboxylated to form acetone (decarboxylase) or reduced to D-β-hydroxylbutyrate **(**β– 羟丁酸 **)** (dehydrogenase).

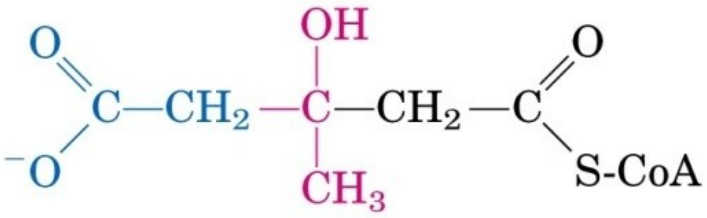




Acety-CoA can be converted to ketone bodies in liver.

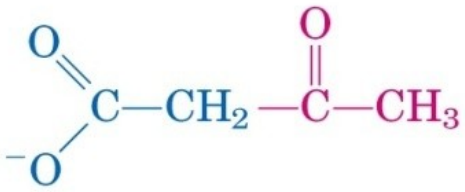



CH3—C(=O)—S-CoA + CH3—C(=O)—S-CoA
2 Acetyl-CoA
thiolase
CoA-SH
CH3—C(=O)—CH2—C(=O)—S-CoA
Acetoacetyl-CoA
HMG-CoA synthase
Acetyl-CoA + H2O
CoA-SH
⁻O—C(=O)—CH2—C(OH)(CH3)—CH2—C(=O)—S-CoA
β-Hydroxy-β-methylglutaryl-CoA
(HMG-CoA)



β-Hydroxy-β-methylglutaryl-CoA
(HMG-CoA)

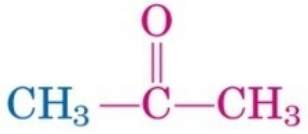
HMG-CoA lyase → Acetyl-CoA



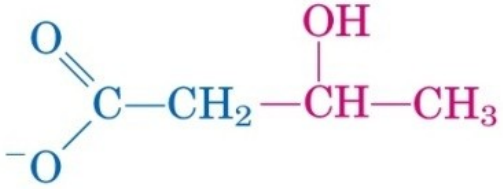
Acetoacetate

acetoacetate decarboxylase → CO_2

NADH + H^+ ⇌ NAD^+ (D-β-hydroxybutyrate dehydrogenase)



Acetone



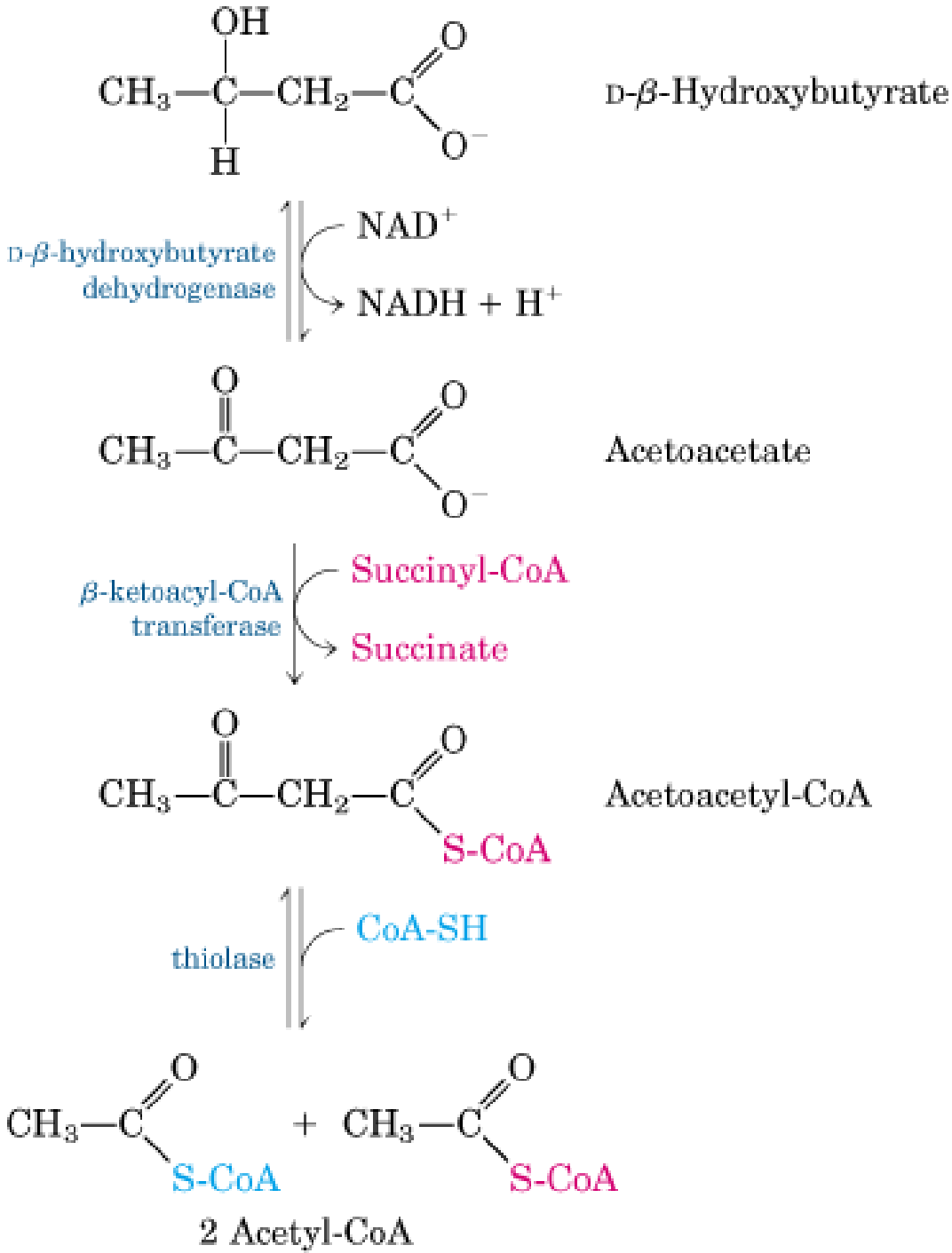
D-β-Hydroxybutyrate





10. Ketone bodies are converted back to acetyl-CoA in extrahepatic tissues





Ketone bodies are converted to acetyl-CoA in extrahepatic tissues.



Summary

Fatty acids are activated to the acyl-CoA form and is then carried into mitochondria by carnitine with the help of two carnitine acyltranseferase isozymes (I and II) located on the outside and inside of the inner membrane.
Acyl-CoA is converted to acetyl-CoA after going through multiple rounds of the four-step (dehydrogenation, hydration, dehydrogenation and thiolysis) β-oxidation pathway.



Oxidative degradation of unsaturated fatty acids need two extra enzymes: an isomerase and a reductase.

The rate of β-oxidation pathway is controlled by the rate at which acyl-CoA is transported into mitochondria.

Excess acetyl-CoA (under conditions when glucose metabolism is not optimal) can be converted to ketone bodies (acetoacetate, β-hydroxylbutyrate and acetone) in the liver cells and reconverted into acetyl-CoA in extrahepatic cells.



Fatty Acid Synthesis

(K3)



1. Fatty acid synthesis takes a different pathway from its degradation

Occurs in the cytosol (chloroplasts in plants).

Acetyl-CoA provides the first two carbons, which is elongated by sequential addition of two-carbon units donated from malonyl-CoA.

Intermediates are attached to the -SH groups of an acyl carrier protein (ACP).



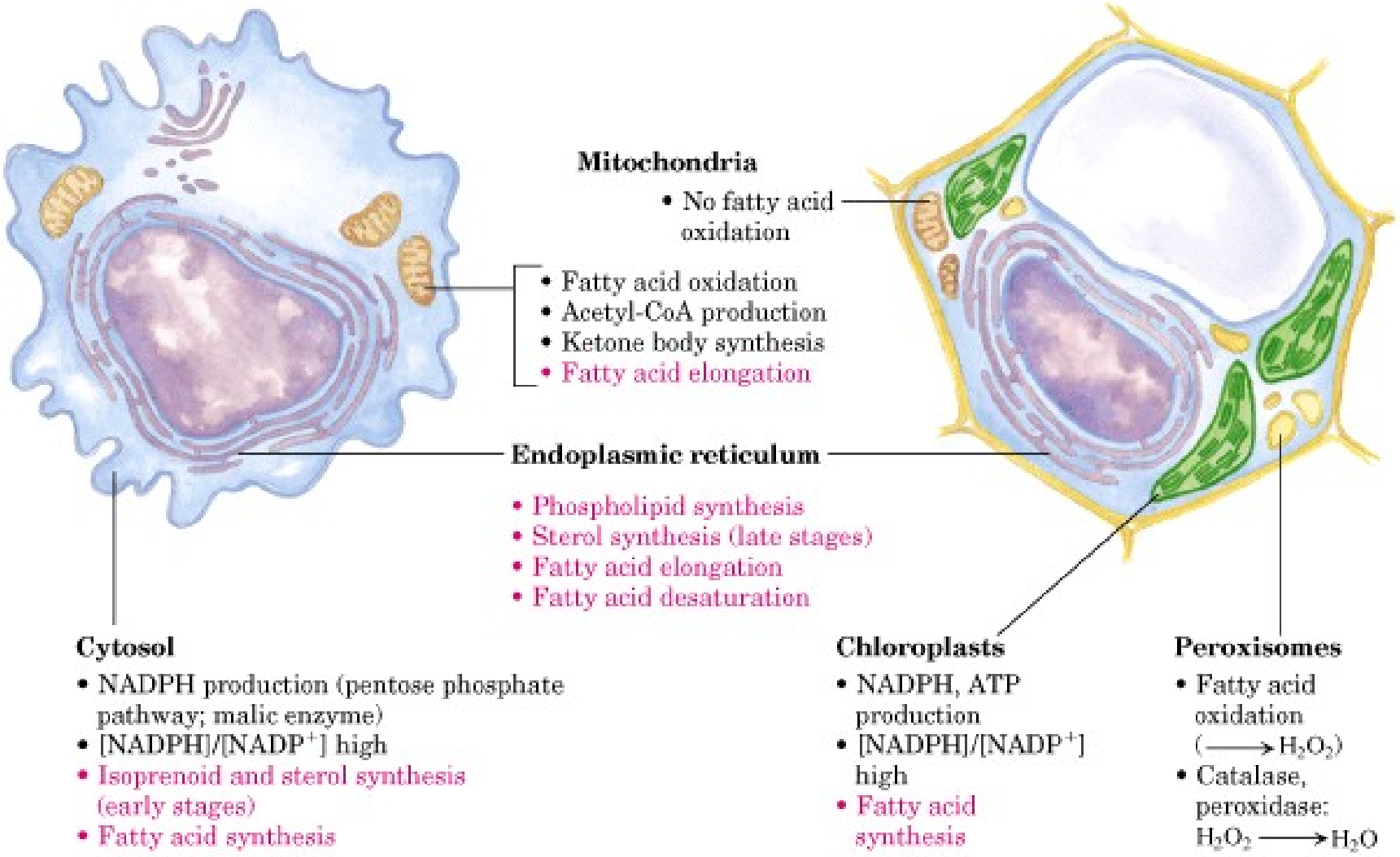
NADPH is the reductant.

The enzymes are associated as a multi-enzyme complex or even being in one polypeptide chain in higher organisms (fatty acid synthase).

Elongation by the fatty acid synthase complex stops upon formation of palmitate (C_{16}), further elongation and desaturation are carried out by other enzyme systems.

unsaturated fatty acids
saturated fatty acids
desatureation




Animal cells, yeast cells
Plant cells
Mitochondria
• No fatty acid oxidation
• Fatty acid oxidation
• Acetyl-CoA production
• Ketone body synthesis
• Fatty acid elongation
Endoplasmic reticulum
• Phospholipid synthesis
• Sterol synthesis (late stages)
• Fatty acid elongation
• Fatty acid desaturation
Cytosol
• NADPH production (pentose phosphate pathway; malic enzyme)
• [NADPH]/[NADP+] high
• Isoprenoid and sterol synthesis (early stages)
• Fatty acid synthesis
Chloroplasts
• NADPH, ATP production
• [NADPH]/[NADP+] high
• Fatty acid synthesis
Peroxisomes
• Fatty acid oxidation (⟶ H2O2)
• Catalase, peroxidase: H2O2 ⟶ H2O

1. Acetyl-COA 转运

2. Acetyl-COA 羧化为 malonyl-COA

3. Fatty acid synthesis：

Acetyl-COA 进位 KS 巯基，malonyl-COA 进位 ACP 巯基

Condensation

Reduction

Dehydration

Reduction

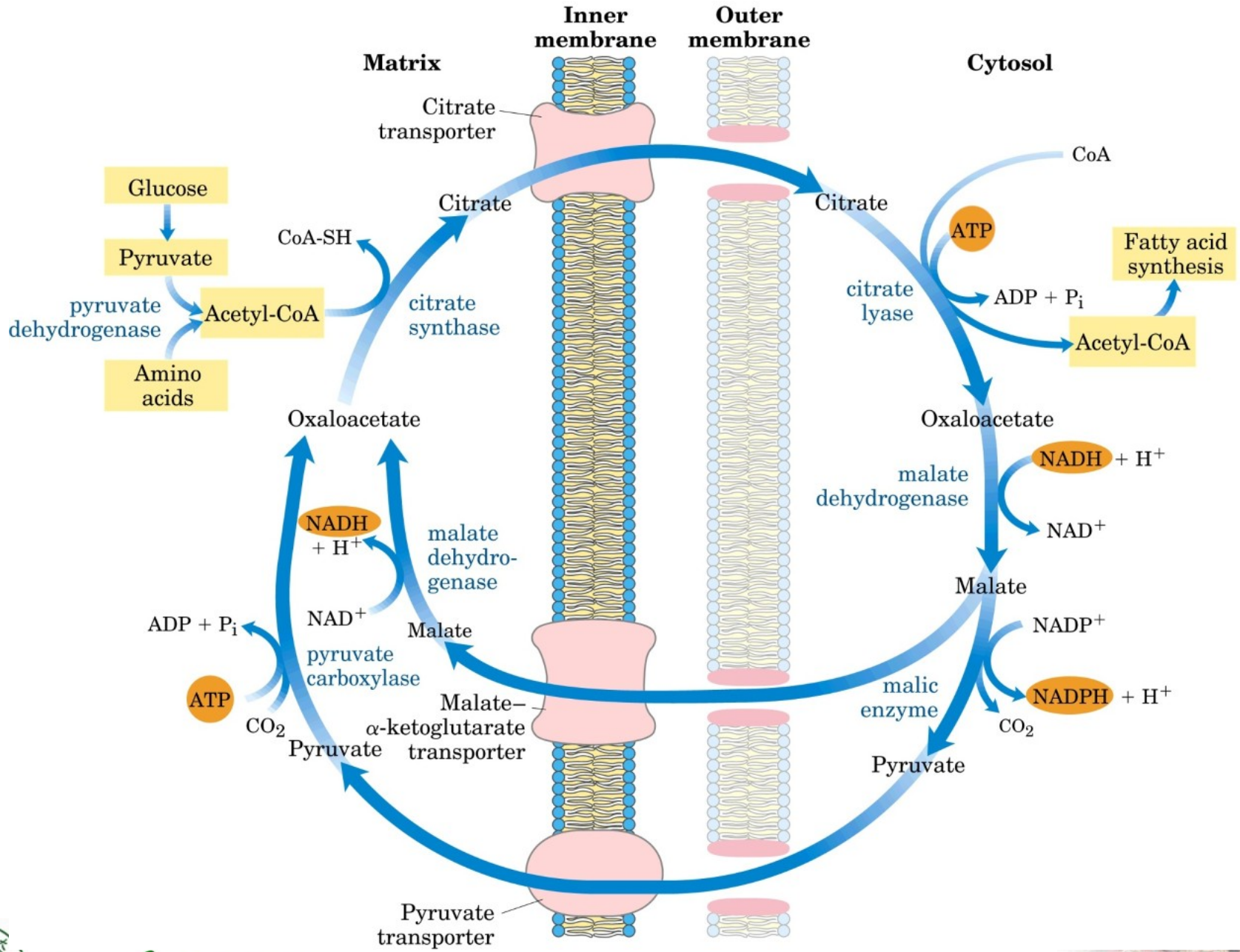
延长 2C fatty acyl 移位至外周巯基



2. The acetyl groups of the mitochondrion are transported into the cytosol in the form of citrate

The acetyl-CoA molecules are made from glucose and amino acids in mitochondria.
They are shuttled into the cytosol in the form of citrate via the citrate transporter of the inner membrane.
Acetyl-CoA is regenerated by the action of ATP-dependent citrate lyase in the cytosol.
Oxaloacetate is shuttled back into the mitochondria as malate or pyruvate.



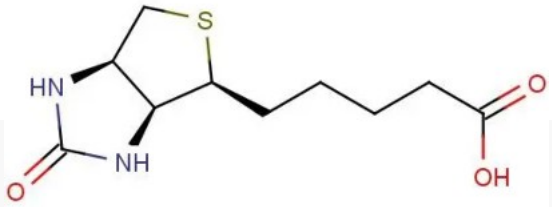


Biochemistry

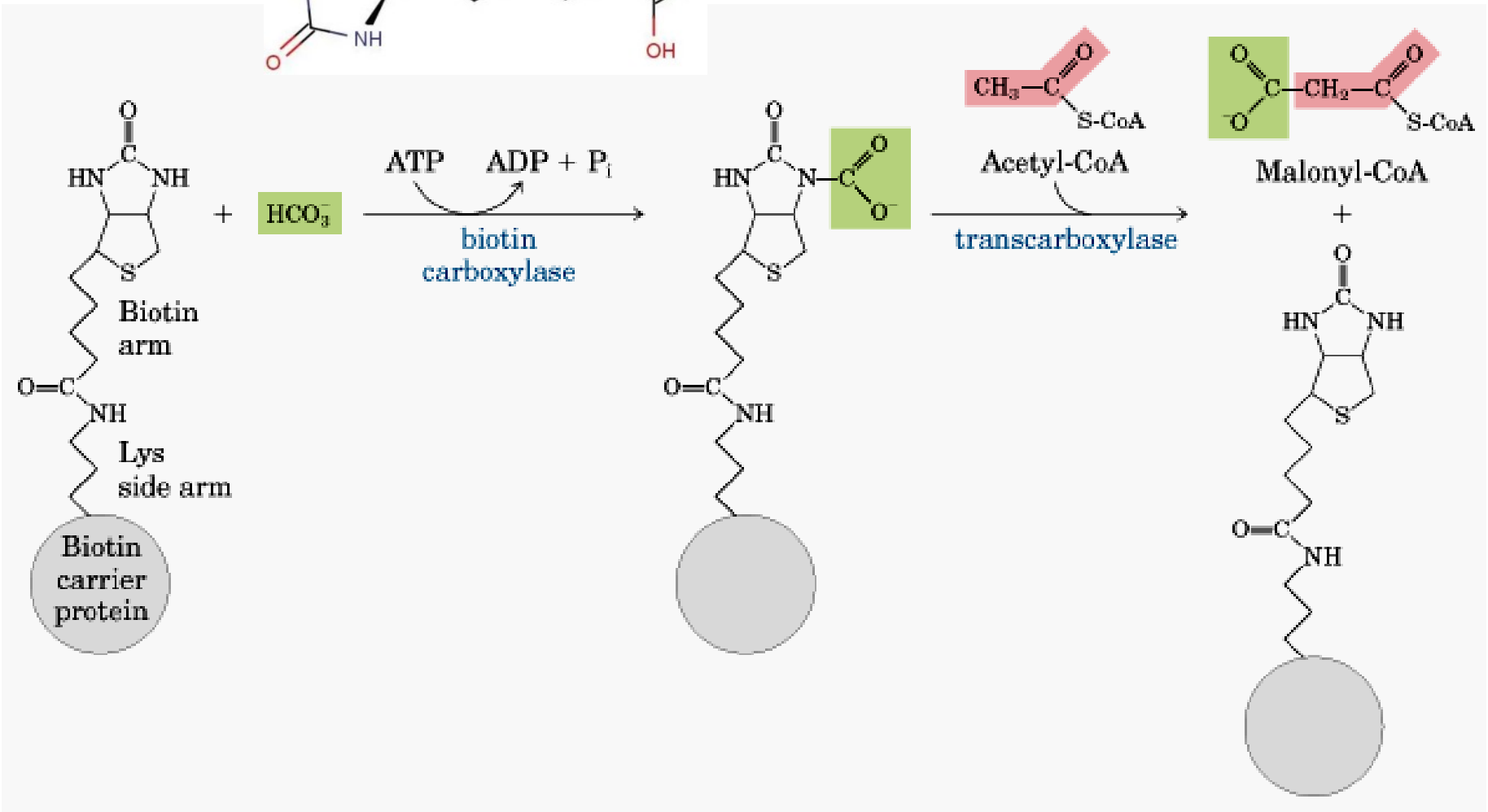
3. Malonyl-CoA is formed from acetyl-CoA and bicarbonate

Salih Wakil discovered that HCO_3^- is required for fatty acid synthesis.
Acetyl-CoA carboxylase catalyzes this carboxylation reaction.
The enzyme has three functional parts: a biotin carrier protein; an ATP-dependent biotin carboxylase; and a transcarboxylase.



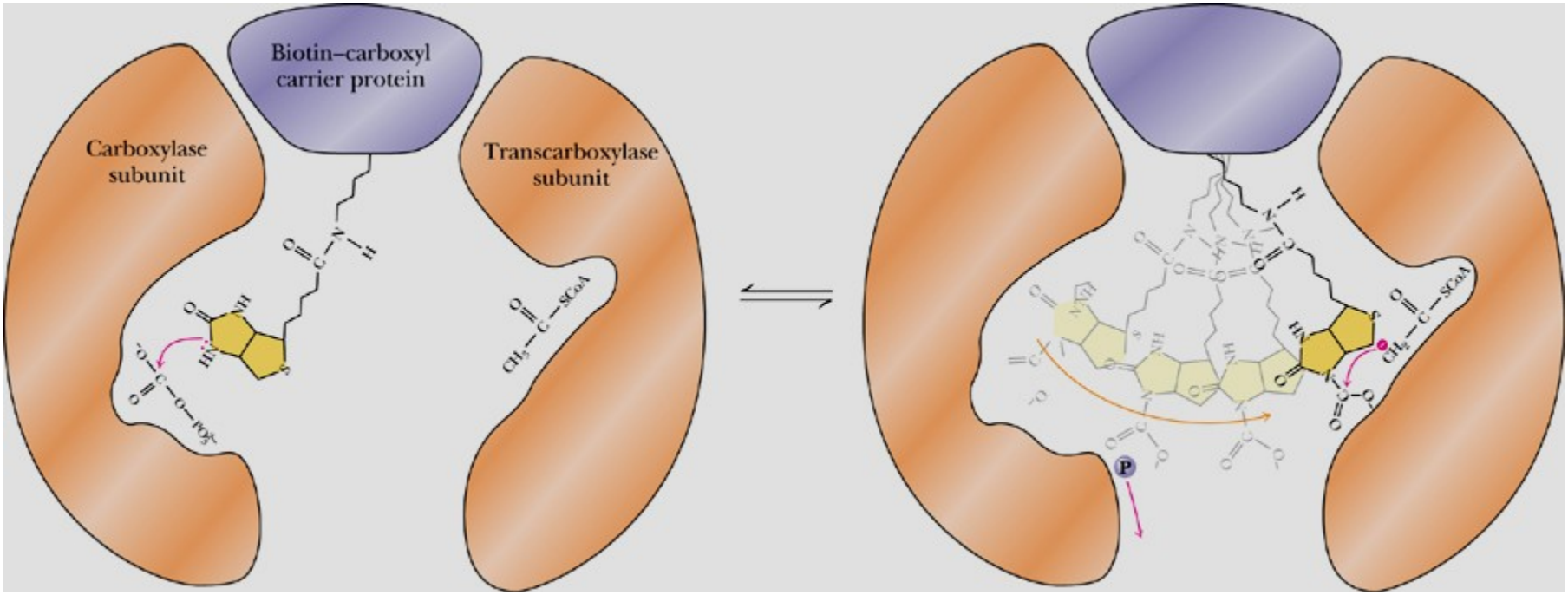


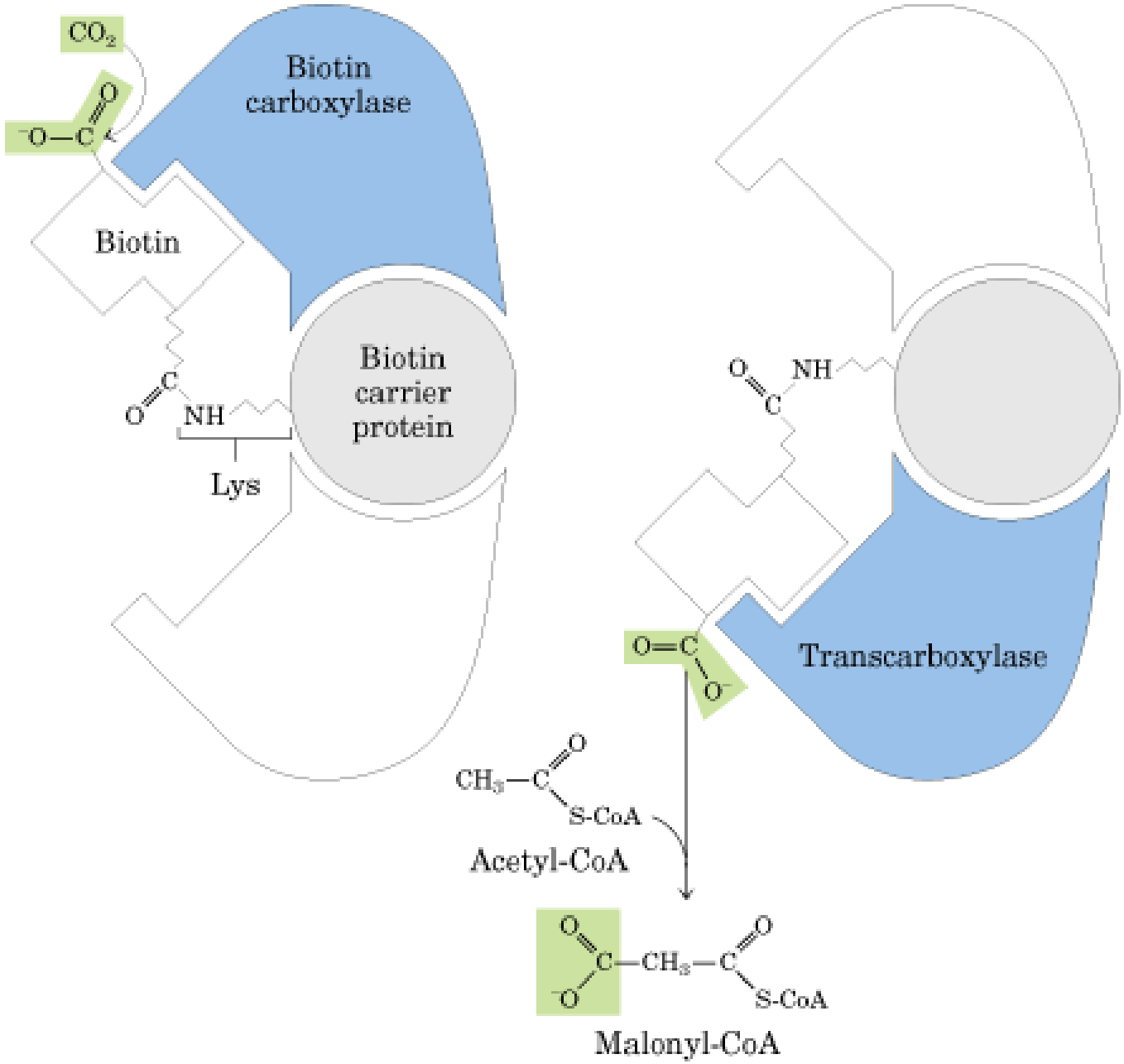
维生素B7 生物素



biotin carboxyl-carrier protein **BCCP**




Biotin–carboxyl carrier protein
Carboxylase subunit
Transcarboxylase subunit
P


CO_2
Biotin carboxylase
Biotin
Biotin carrier protein
NH
Lys
NH
Transcarboxylase
CH_3—C
S-CoA
Acetyl-CoA
C—CH_2—C
S-CoA
Malonyl-CoA

4. The acetyl and malonyl groups are first transferred to two –SH groups of the fatty acid synthase complex

The acetyl group of acetyl-CoA is first transferred to the –SH group of a Cys residue on the β-ketoacyl-ACP synthase (KS) in a reaction catalyzed by acetyl-CoA-ACP transacetylase (AT).

The malonyl group is transferred from malonyl-CoA to the –SH group of the acyl carrier protein (ACP) catalyzed by malonyl-CoA-ACP transacetylase (MT).

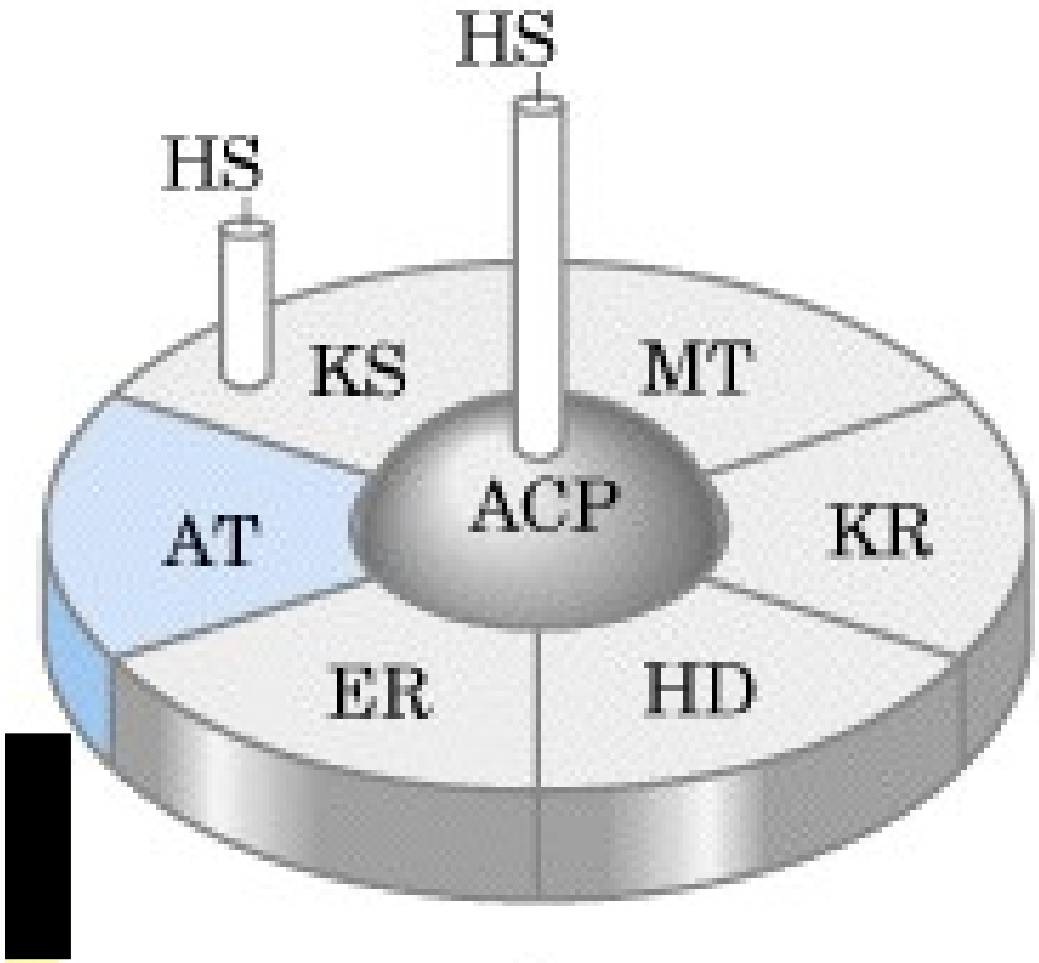


table 21–1

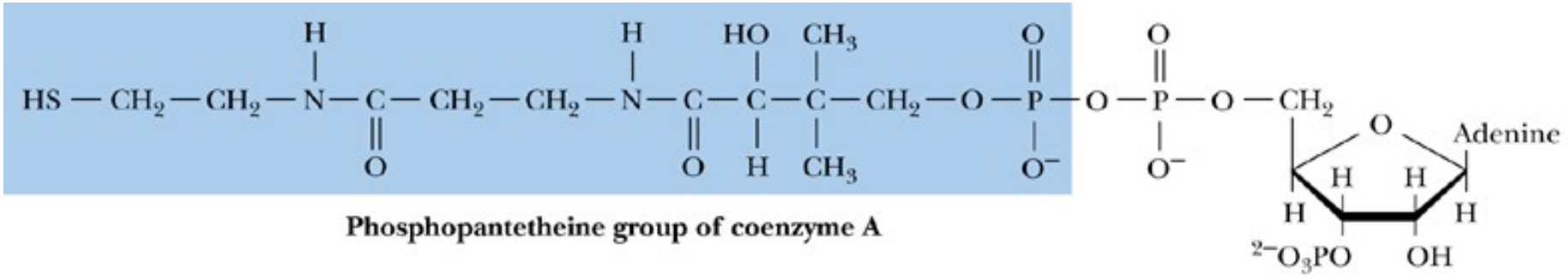
Proteins of the Fatty Acid Synthase Complex of *E. coli*

Protein	Role
Acyl carrier protein (ACP)	Carries acyl groups in thioester linkage
Acetyl-CoA–ACP transacetylase (AT)	Transfers acyl group from CoA to Cys residue of KS
β-Ketoacyl-ACP synthase (KS)	Condenses acyl and malonyl groups (there are at least three isozymes of KS)
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Enoyl-ACP reductase (ER)	Reduces double bond, forming saturated acyl-ACP

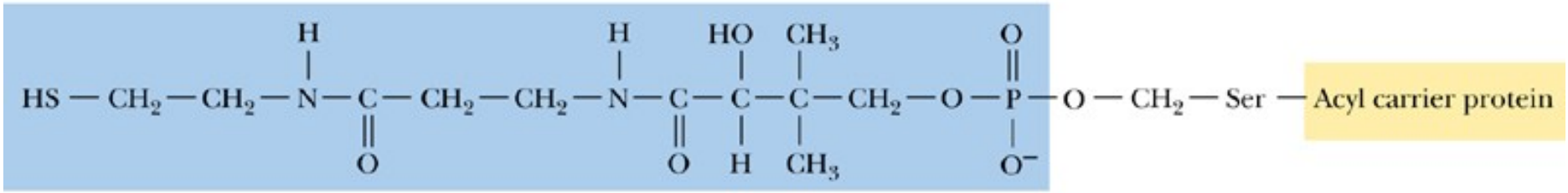



HS
HS
KS
MT
AT
ACP
KR
ER
HD

The acyl carrier protein (ACP) is very similar to CoA (thus can be regarded as “macro CoA”)

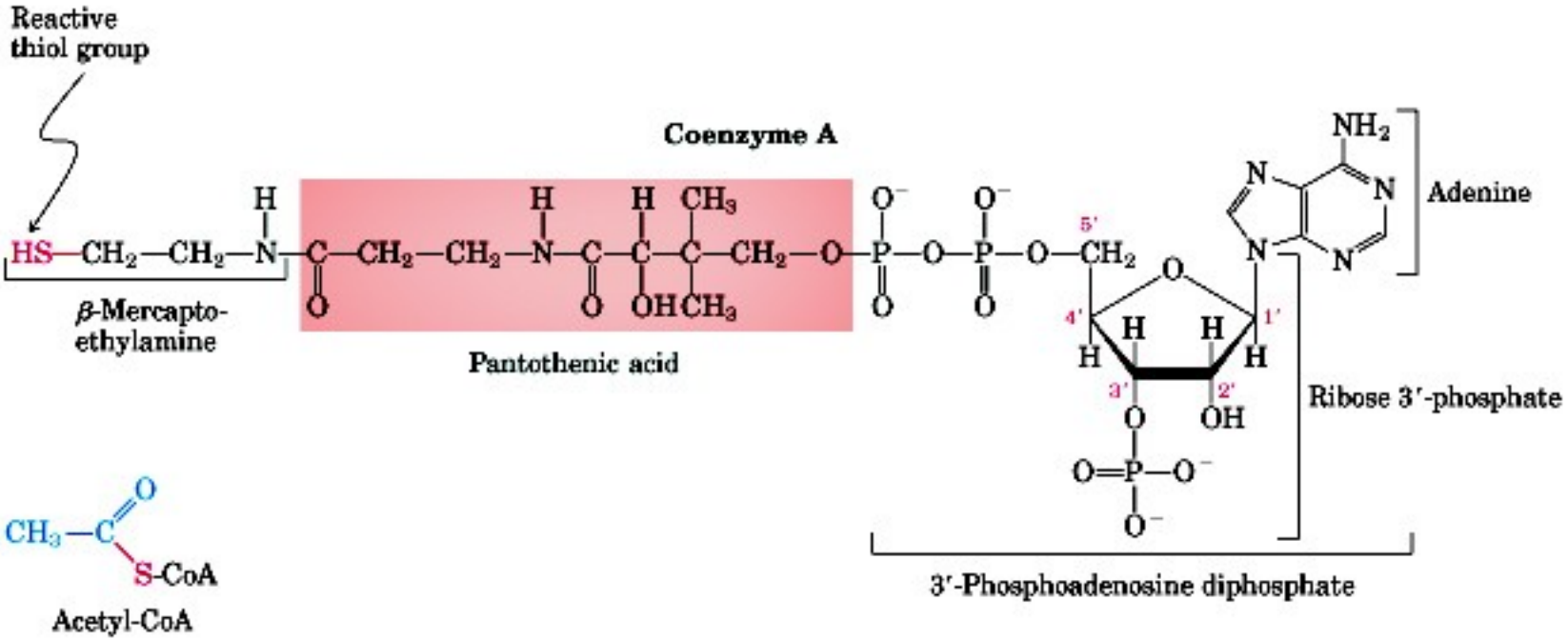


Phosphopantetheine group of coenzyme A

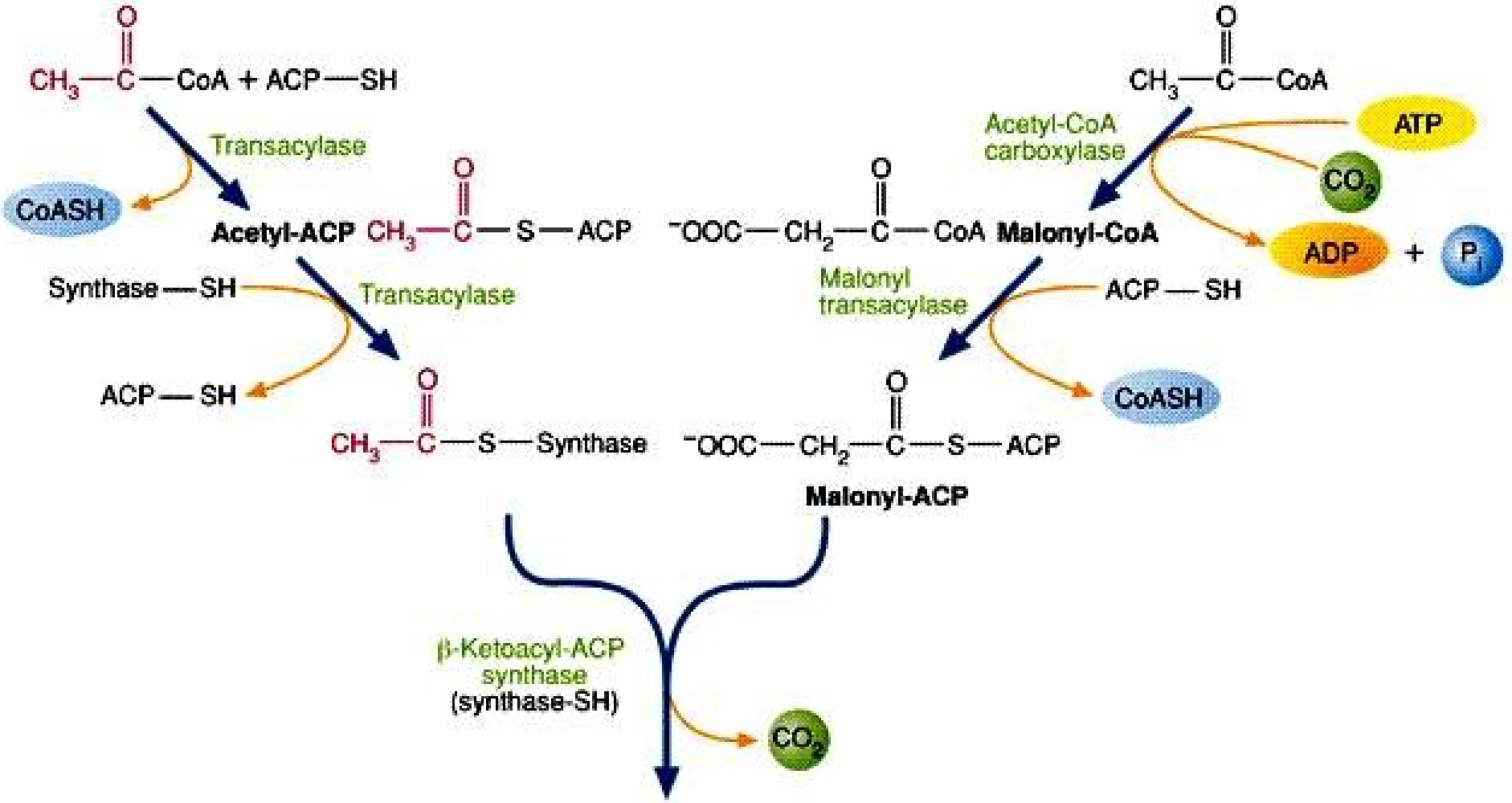


Phosphopantetheine prosthetic group of ACP

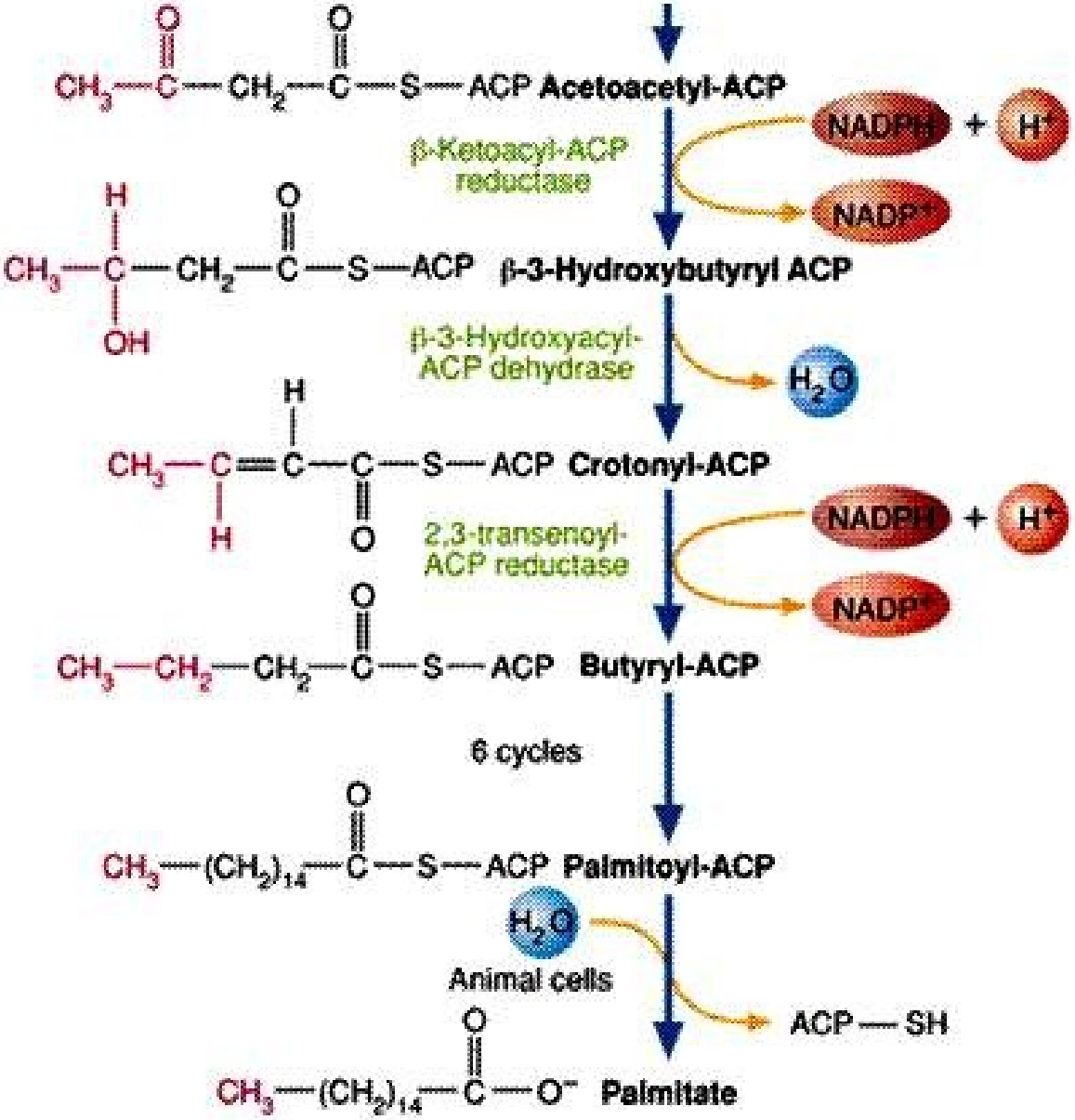
B5 泛酸（遍多酸） 自然界中广泛



Biochemistry by McKee/McKee, © 1996 McGraw-Hill Higher Education



Biochemistry by McKee/McKee, ©1996 McGraw-Hill Higher Education

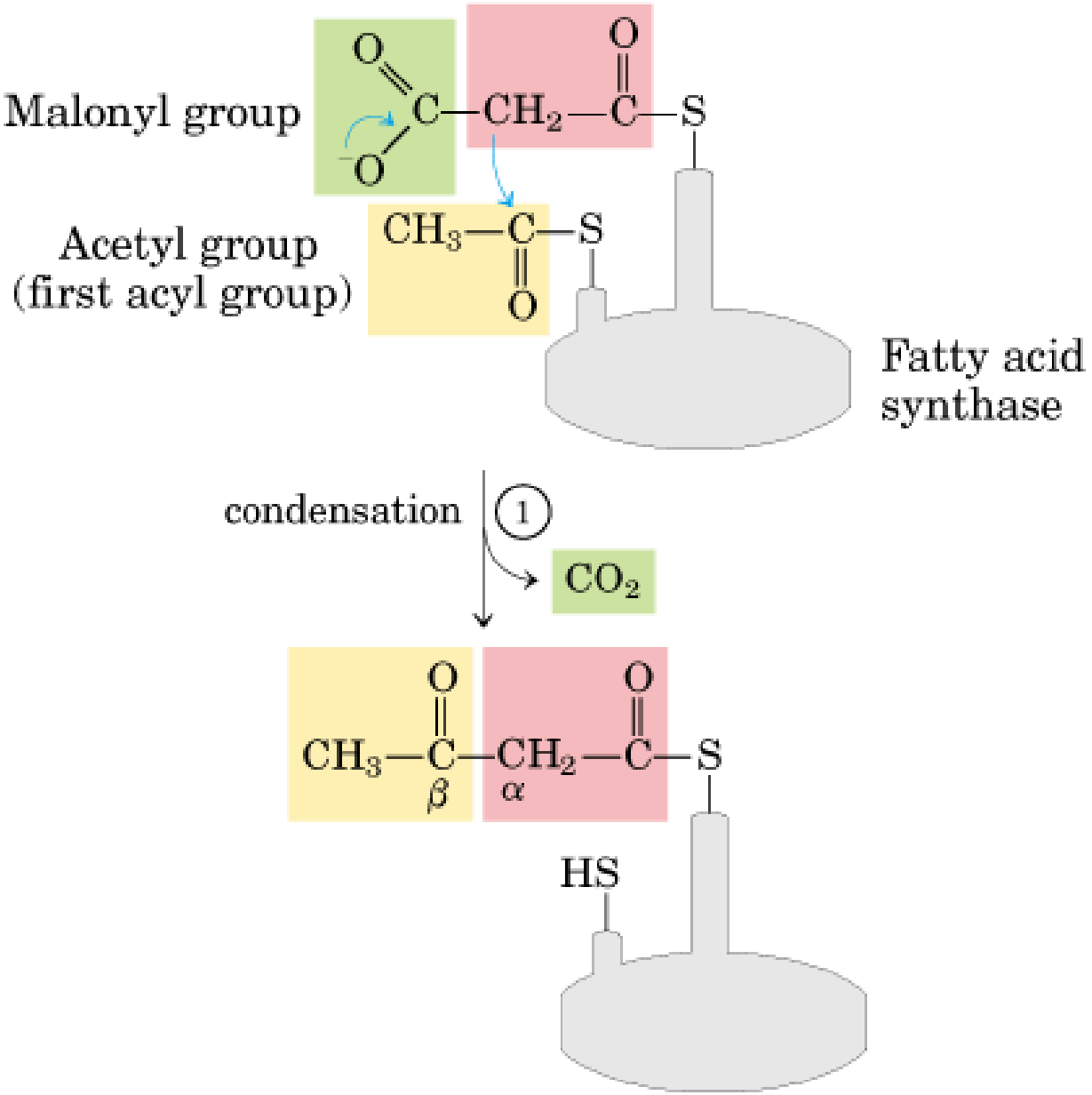


4. Fatty acids are synthesized by a repeating four-step reaction sequence

step 1

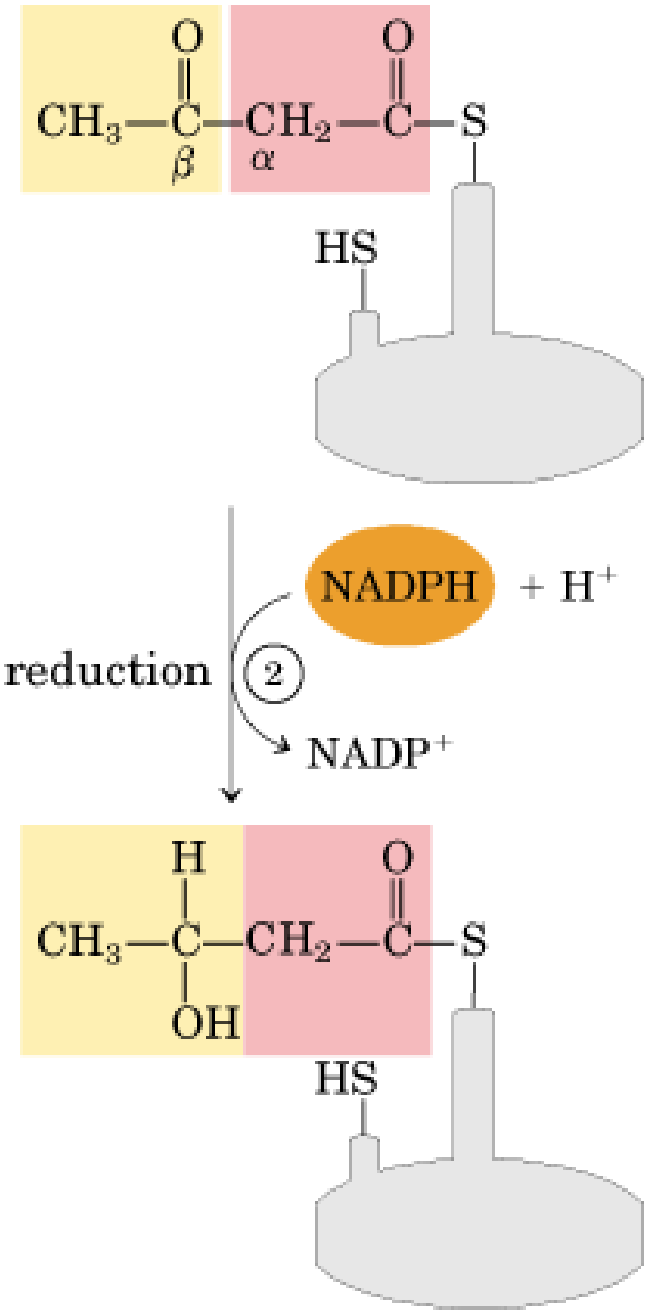
Condensation of acetyl-ACP and malonyl-ACP to form acetoacetyl-ACP, releasing free ACP and CO_2.




Malonyl group
Acetyl group
(first acyl group)
Fatty acid
synthase
condensation
1
CO_2
HS
CH_3—C—CH_2—C—S
β
α

Step 2

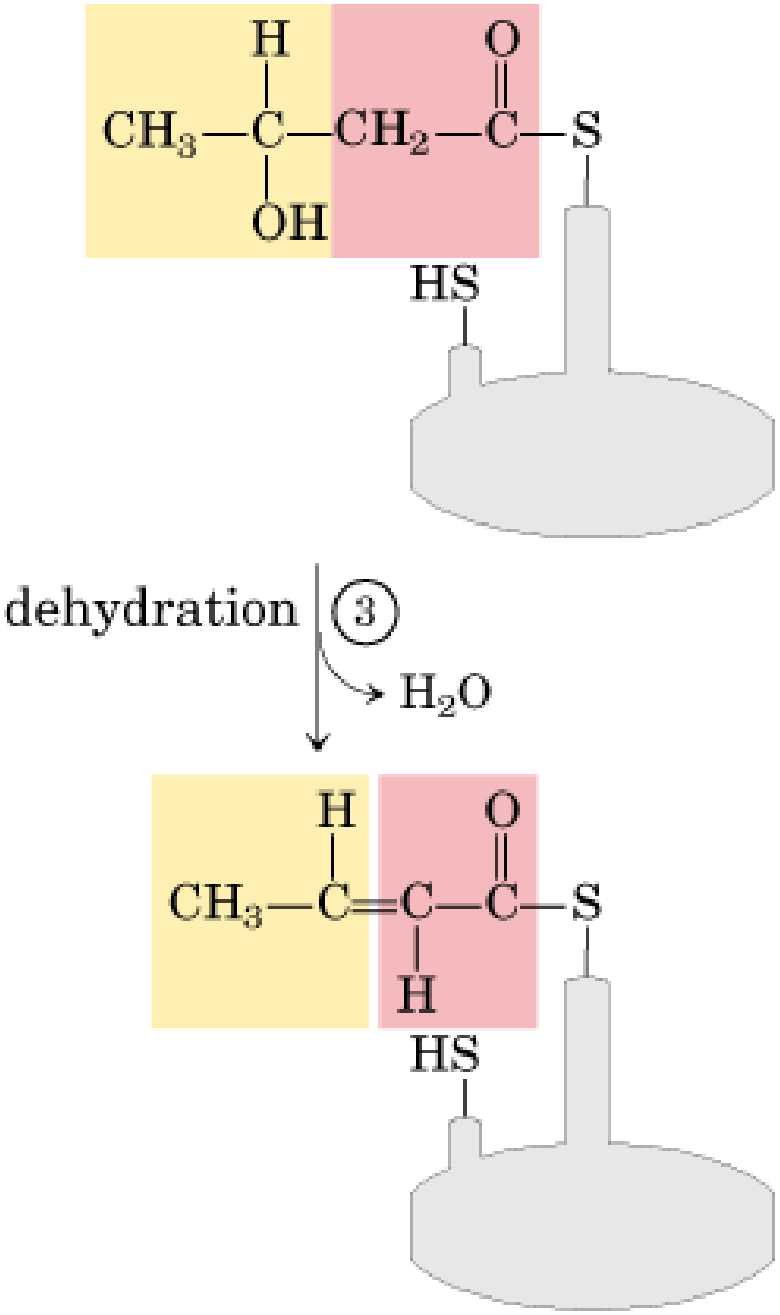



CH3—C—CH2—C—S
β
α
HS
NADPH
+ H+
reduction
2
NADP+
CH3—C—CH2—C—S
H
OH
HS

Step 3

dehydration of hydroxybutyryl-ACP to produce crotonyl-ACP.

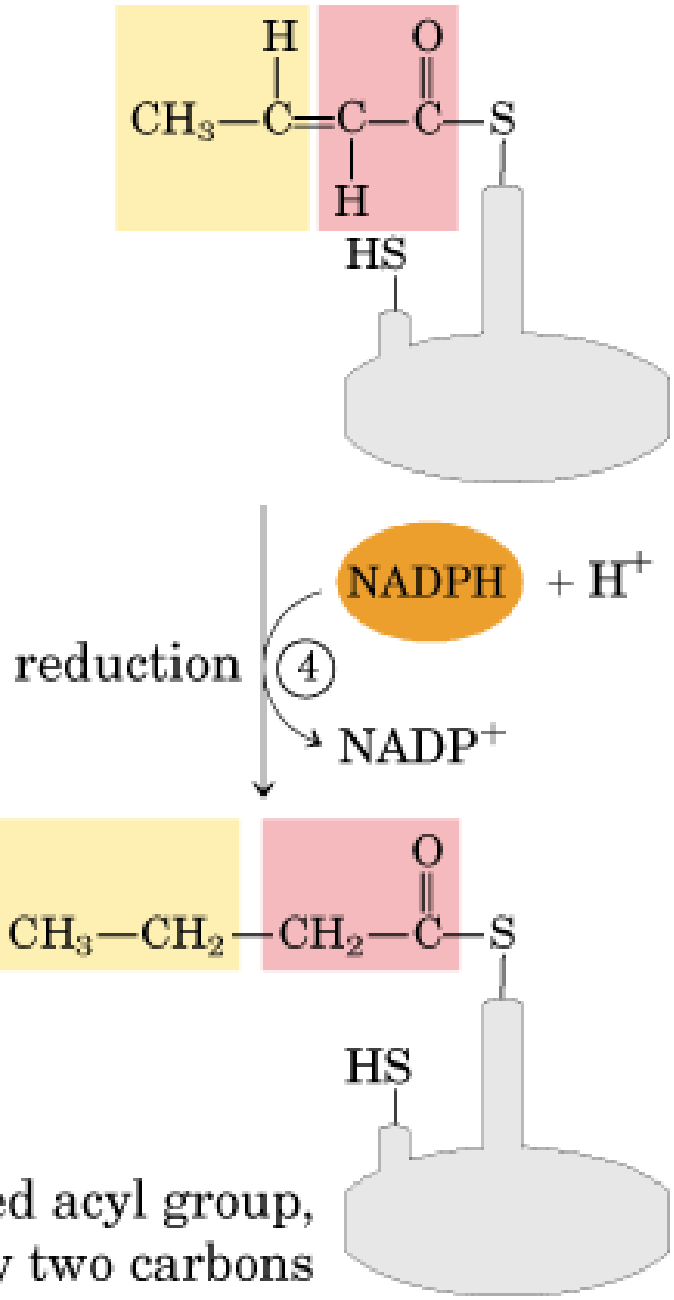



H
O
CH3—C—CH2—C—S
OH
HS
dehydration
3
H2O
H
O
CH3—C=C—C—S
H
HS

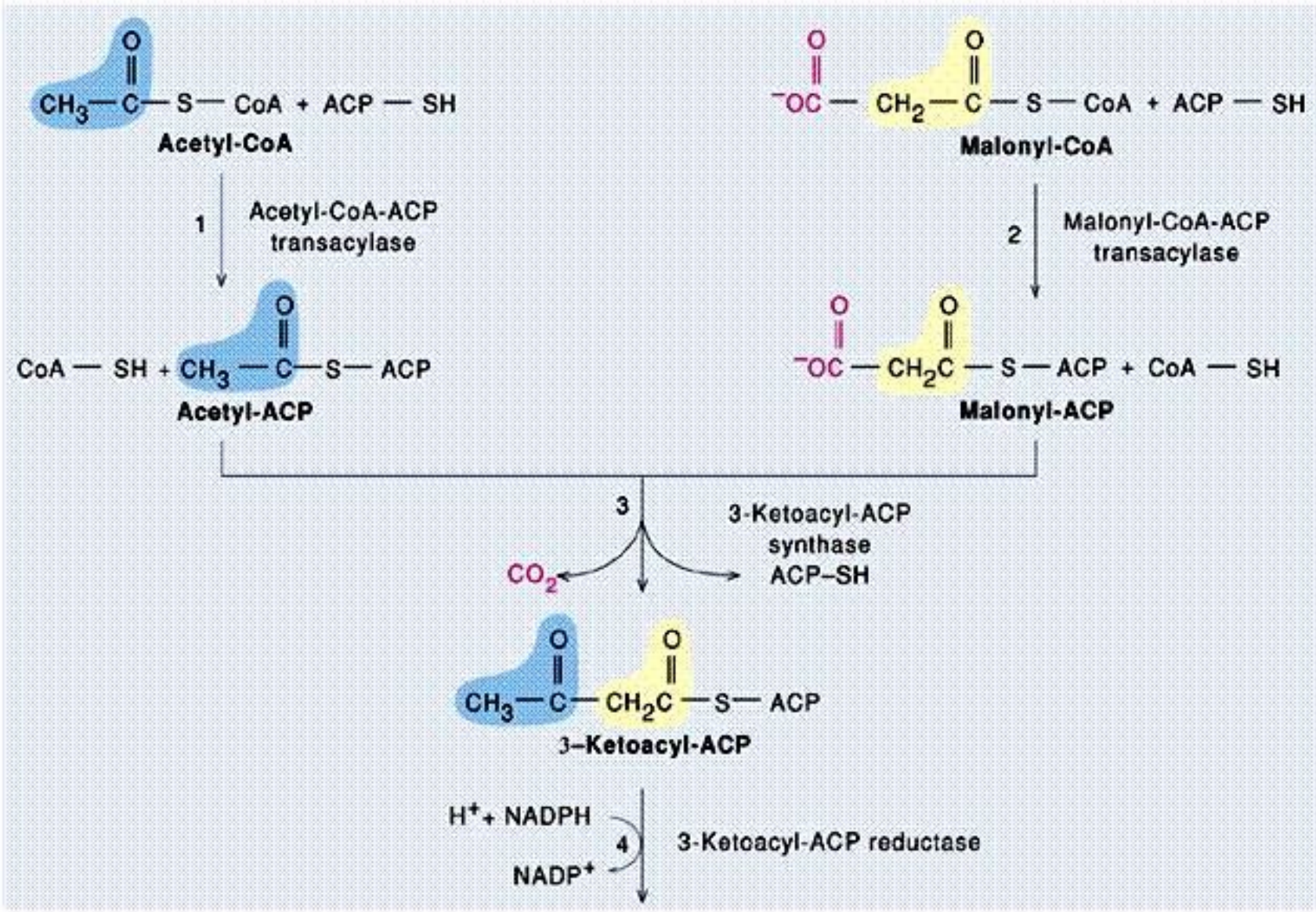
Step 4

Reduction of crotonyl-ACP by a second NADPH molecule to give butyryl-ACP .




H
O
CH3—C=C—C—S
H
HS
NADPH
+ H+
reduction
4
NADP+
O
CH3—CH2—CH2—C—S
HS
Saturated acyl group,
lengthened by two carbons

Principles of Biochemistry by Zubay/Parson/Vance, © 1995 McGraw-Hill Higher Education



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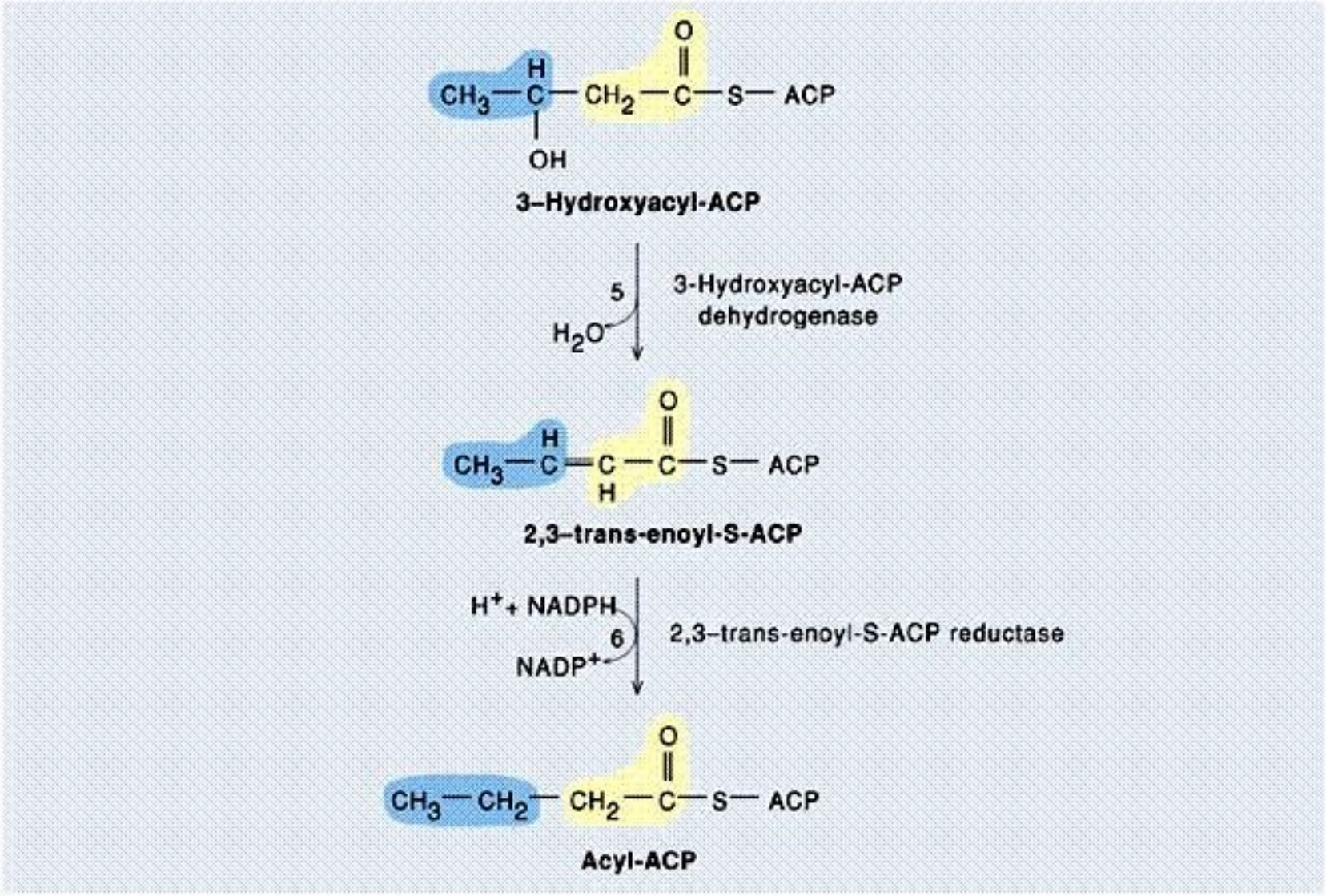
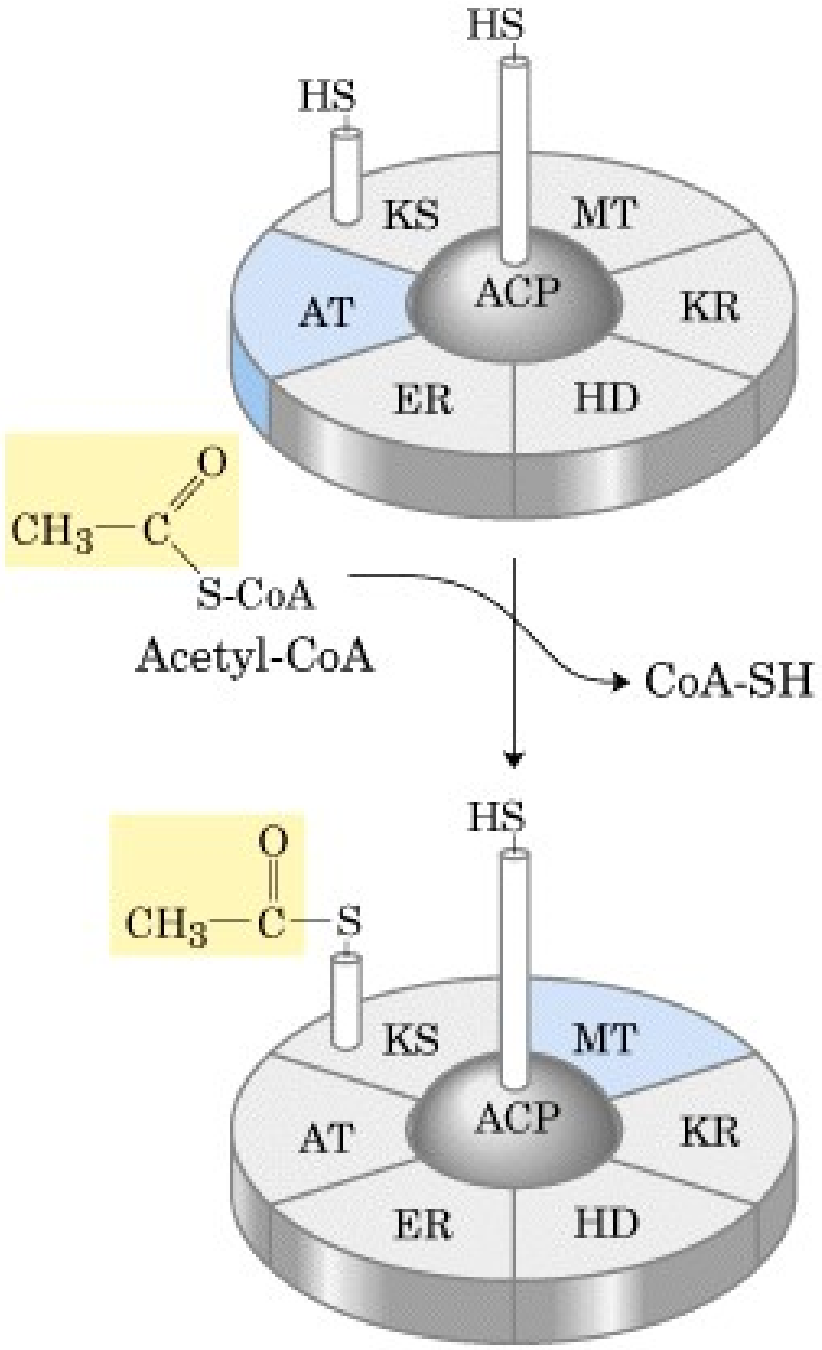


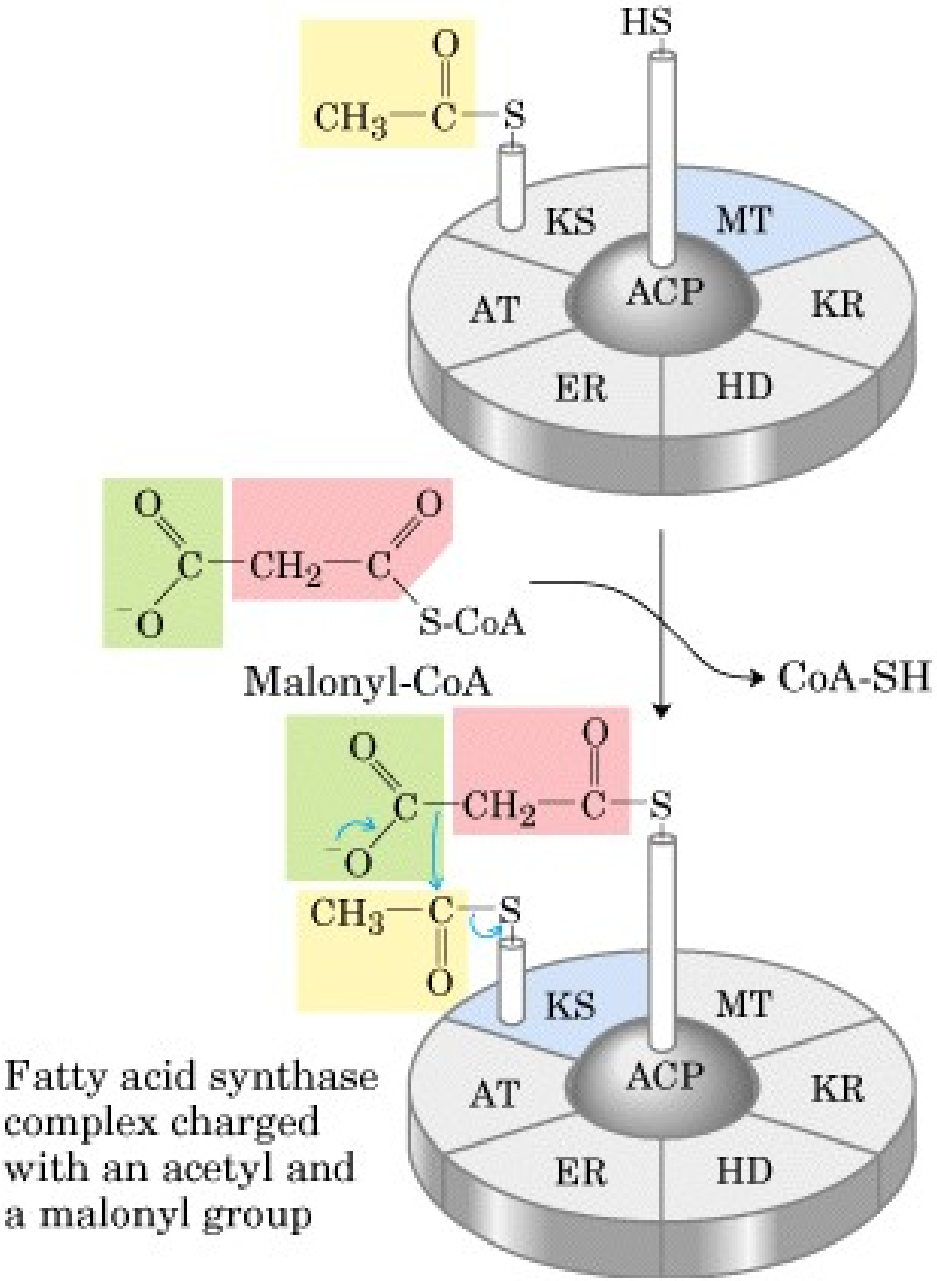
table 21–1

Proteins of the Fatty Acid Synthase Complex of *E. coli*

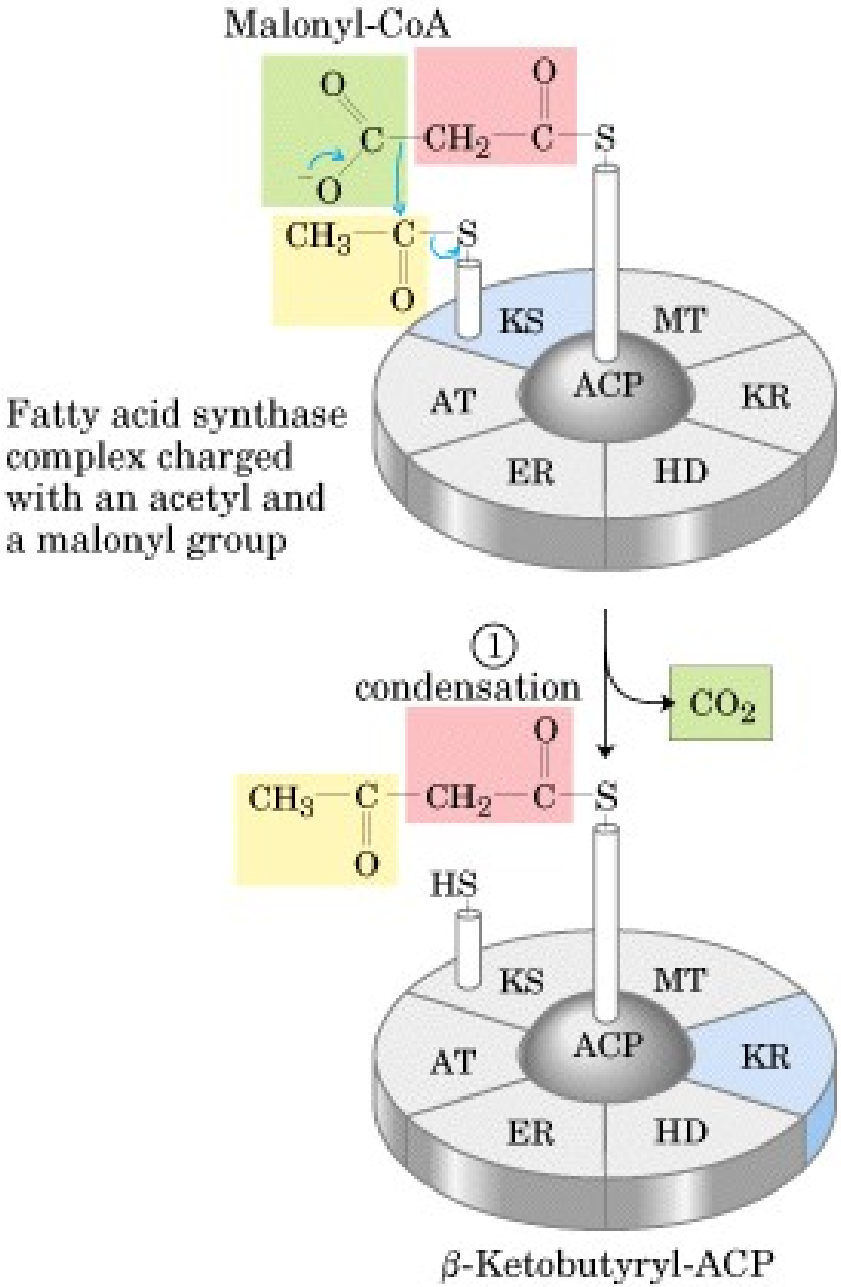
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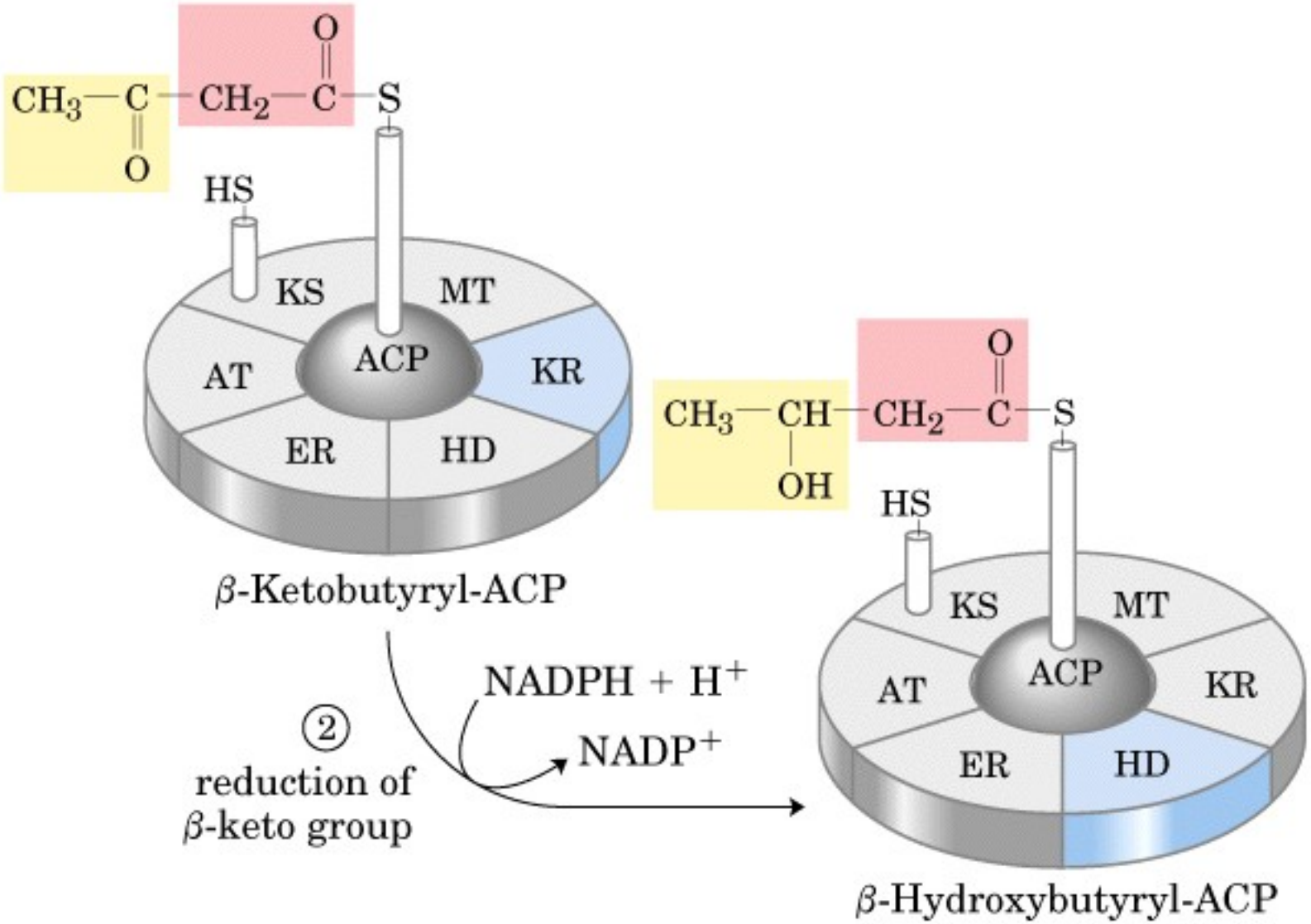


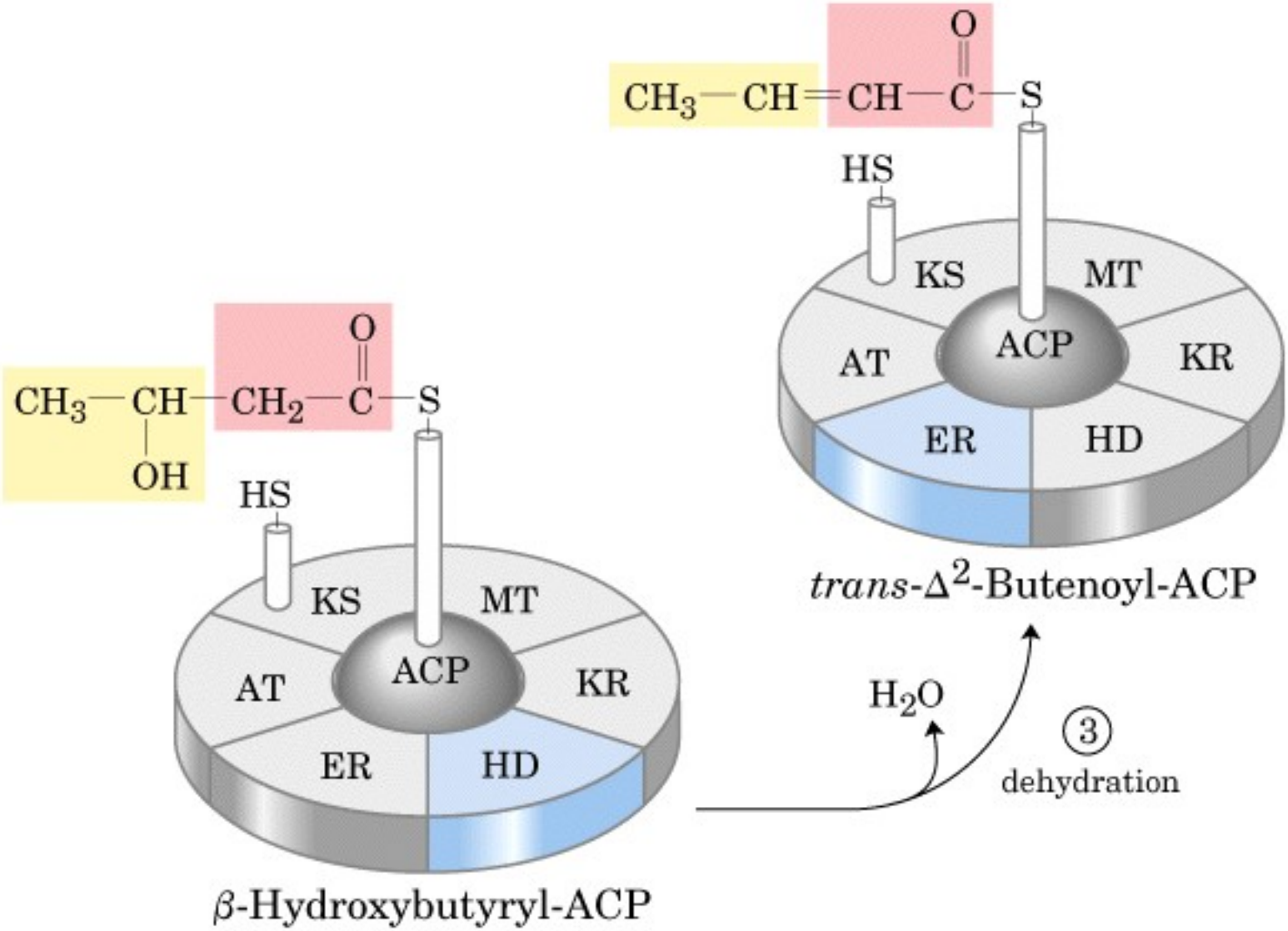

HS
HS
KS
MT
AT
ACP
KR
ER
HD
CH3—C—S-CoA
O
Acetyl-CoA
CoA-SH
HS
CH3—C—S
O
KS
MT
AT
ACP
KR
ER
HD

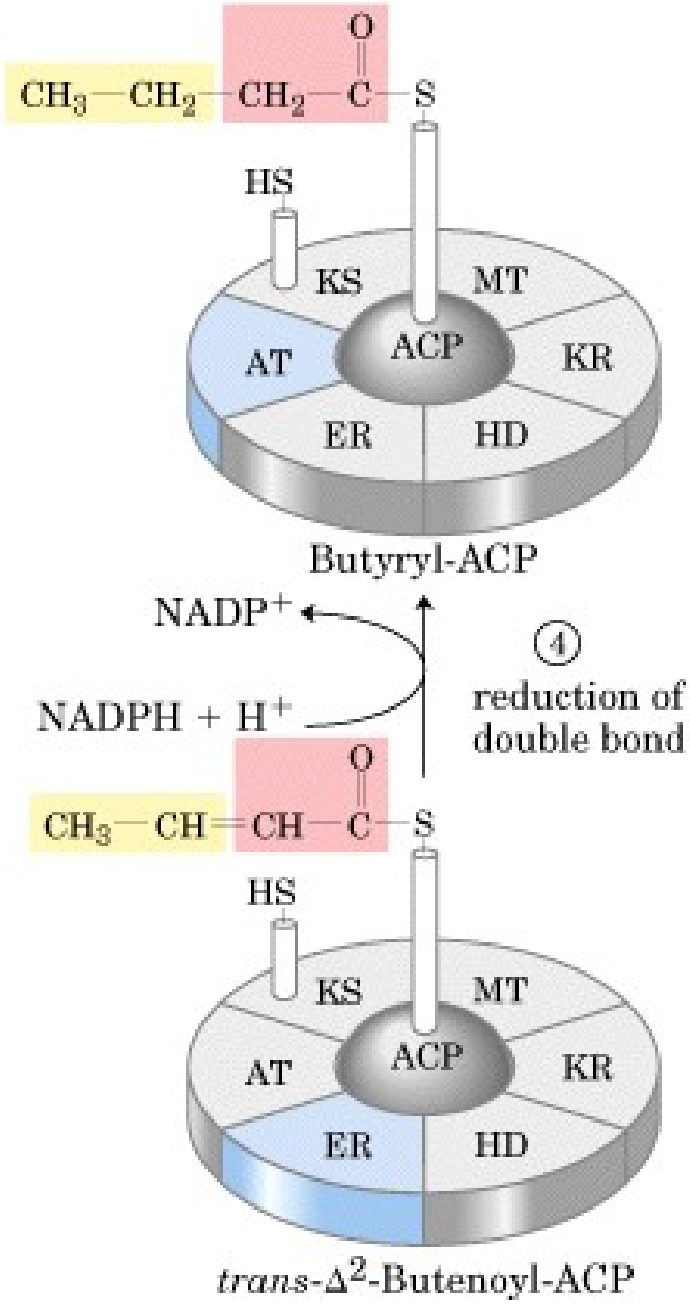


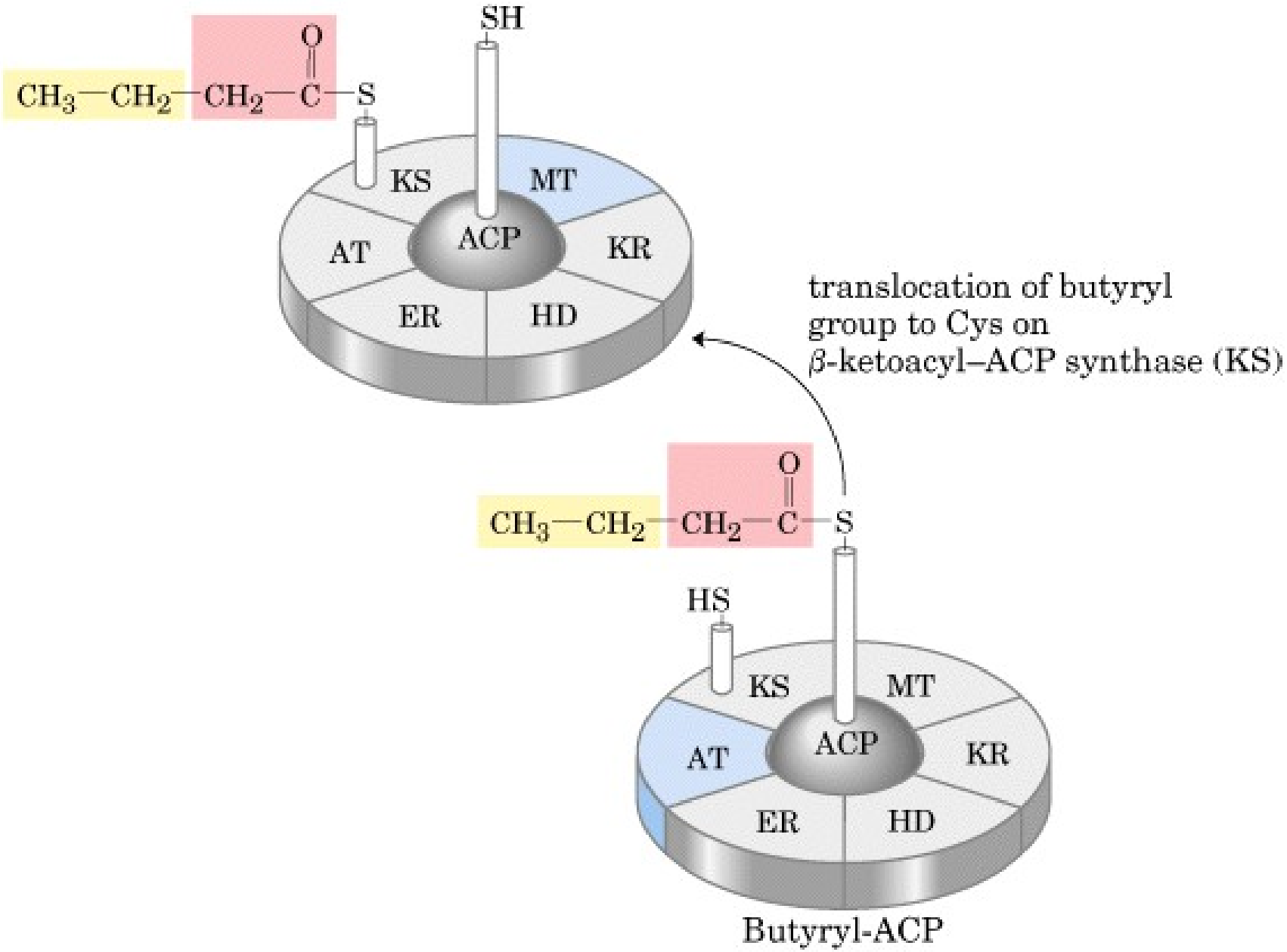
Fatty acid synthase complex charged with an acetyl and a malonyl group

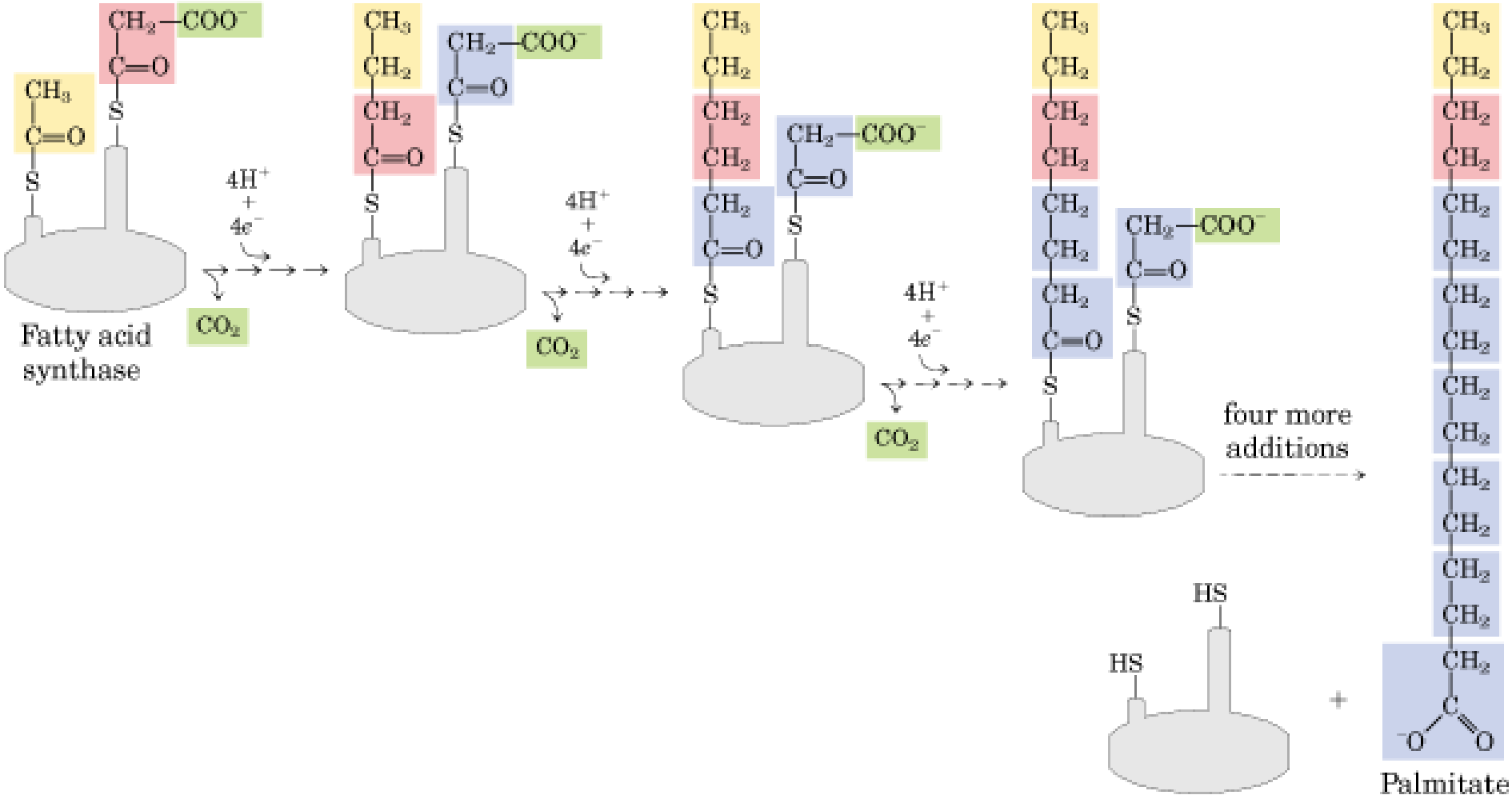

Malonyl-CoA
Fatty acid synthase complex charged with an acetyl and a malonyl group
KS
MT
AT
ACP
KR
ER
HD
1
condensation
CO_2
HS
β-Ketobutyryl-ACP


CH3—C—CH2—C—S
O
O
HS
KS
MT
AT
ACP
KR
ER
HD
β-Ketobutyryl-ACP
NADPH + H+
NADP+
② reduction of β-keto group
CH3—CH—CH2—C—S
OH
HS
KS
MT
AT
ACP
KR
ER
HD
β-Hydroxybutyryl-ACP


CH3—CH—CH2—C—S
O
OH
HS
KS
MT
AT
ACP
KR
ER
HD
β-Hydroxybutyryl-ACP
CH3—CH=CH—C—S
O
HS
KS
MT
AT
ACP
KR
ER
HD
trans-Δ2-Butenoyl-ACP
H2O
3
dehydration

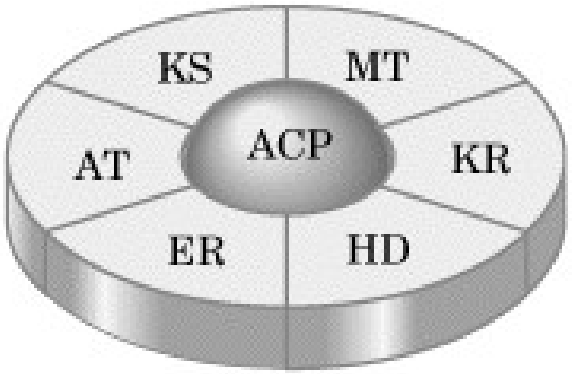

CH3—CH2—CH2—C—S
O
HS
KS
MT
AT
ACP
KR
ER
HD
Butyryl-ACP
NADP+
NADPH + H+
4
reduction of double bond
CH3—CH=CH—C—S
O
HS
KS
MT
AT
ACP
KR
ER
HD
trans-Δ2-Butenoyl-ACP


SH
O
CH3—CH2—CH2—C—S
KS
MT
AT
ACP
KR
ER
HD
translocation of butyryl group to Cys on β-ketoacyl–ACP synthase (KS)
O
CH3—CH2—CH2—C—S
HS
KS
MT
AT
ACP
KR
ER
HD
Butyryl-ACP


Fatty acid synthase
4H+ + 4e−
CO2
four more additions
HS
HS
Palmitate

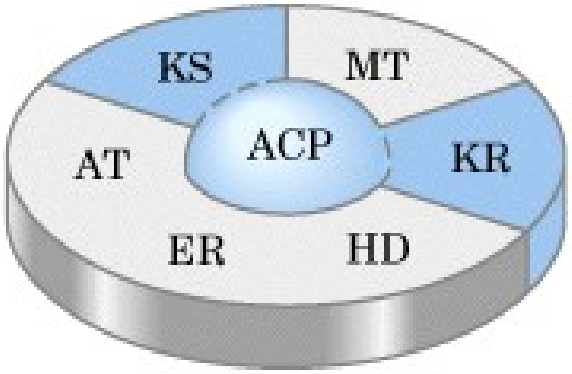
Bacteria, Plants

Seven activities in seven separate polypeptides



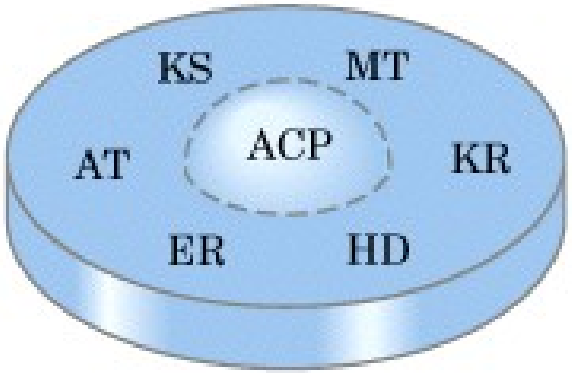
Yeast

Seven activities in two separate polypeptides



Vertebrates

Seven activities in one large polypeptide



Bioch

5. Palmitate can be further elongated and desaturated in smooth ER

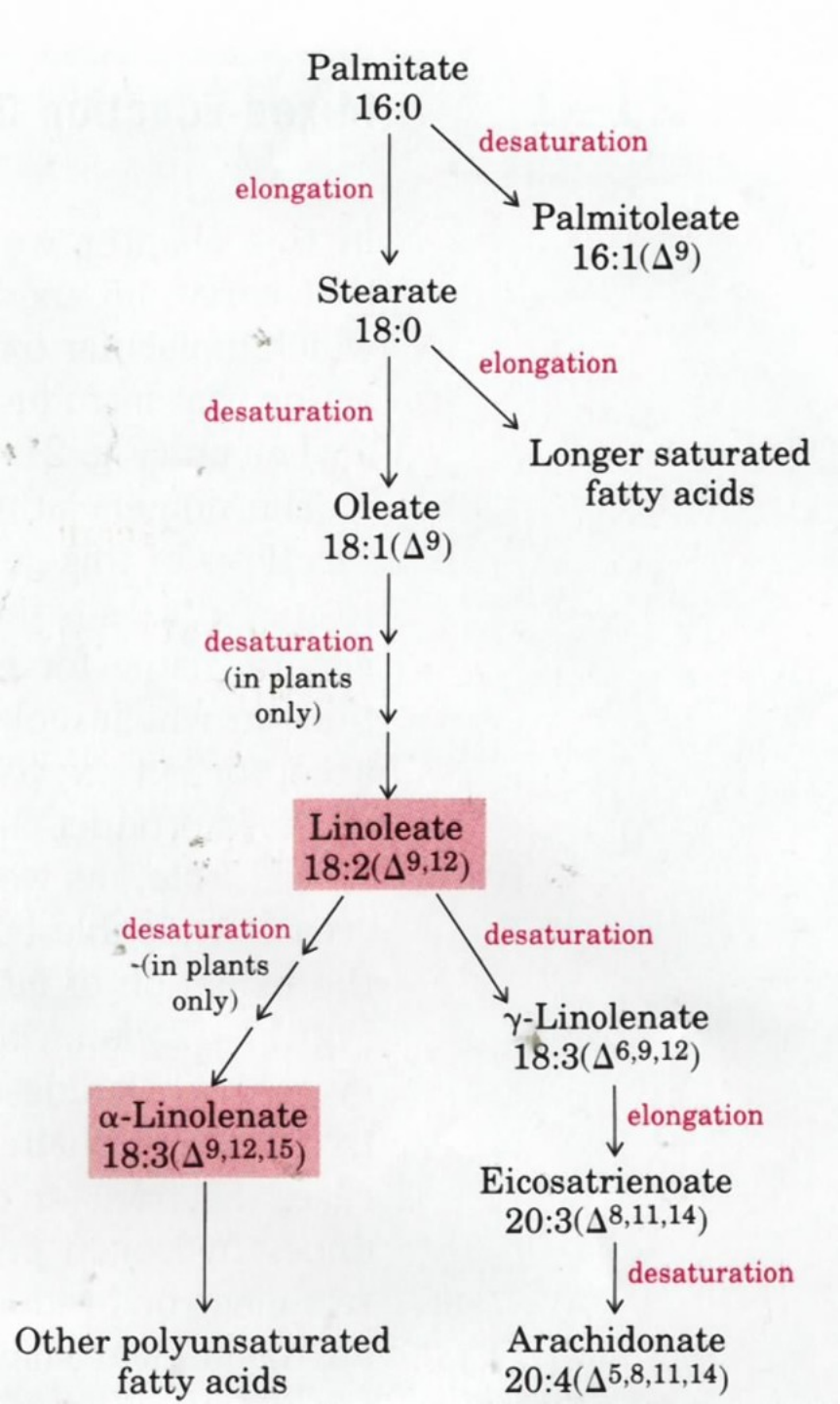
Palmitoyl-CoA can be further **elongated** by the fatty acid elongation system present mainly in the **smooth endoplasmic reticulum （ ER ）** , with two-carbon units also donated by malonyl-CoA. Stearoyl-CoA (硬脂酰） can be desaturated between C-9 and C-10 to produce oleate （油酸） , 18:1(Δ^9).



The double bonds are introduced by the catalysis of fatty acyl-CoA **desaturase**, where both the fatty acyl group and NADPH are oxidized by O_2.

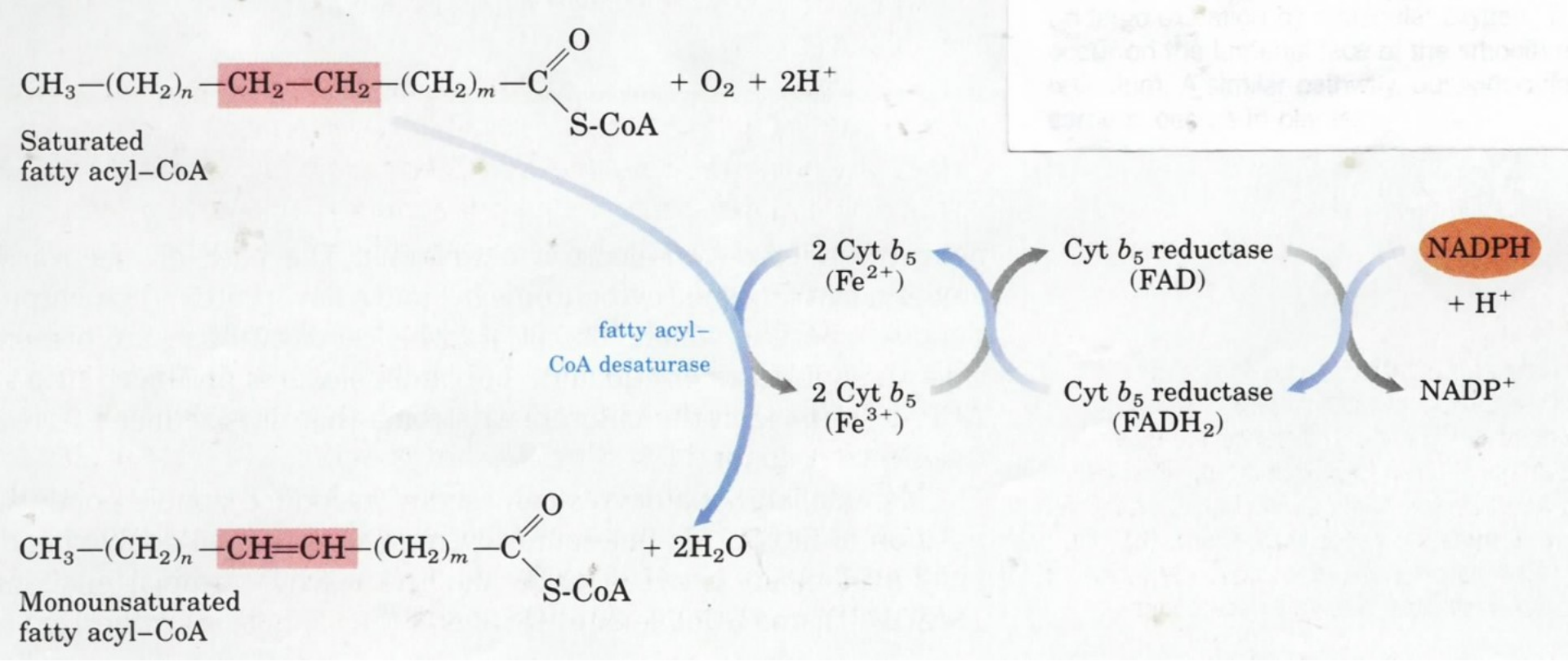
Linoleate (亚油酸) and linolenate （亚麻酸） cannot be synthesized by mammals and are therefore termed essential fatty acids as they have to be ingested in the diet.





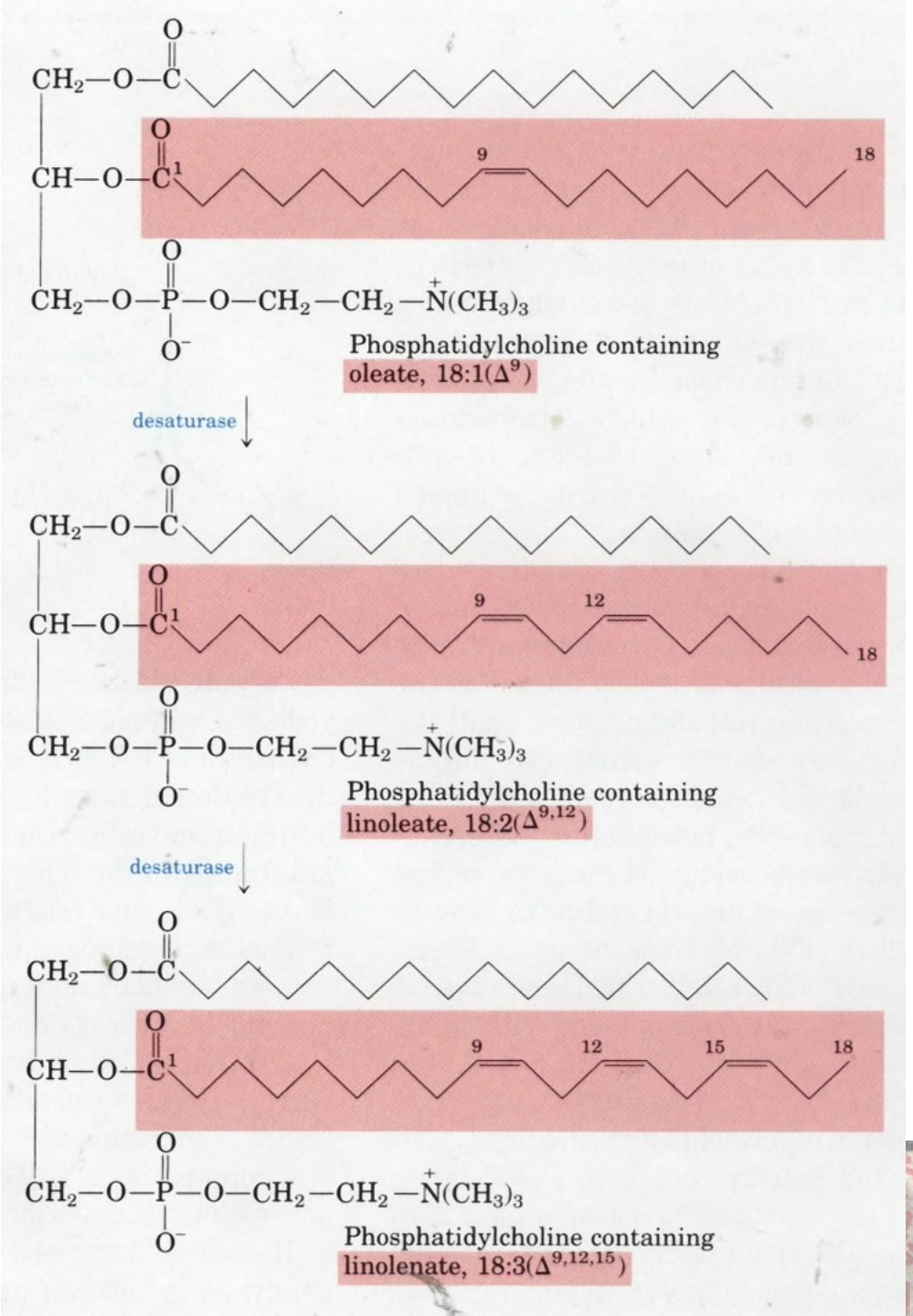
Palmitate is the Precursor for the biosynthesis of other fatty acids





Fatty acyl-CoA is desaturated (oxidized) by O_2 and NADPH.

Oleate can be desaturated on Phosphatidylcholine



6. Essential fatty acids

The polyunsaturated fatty acids **linoleate and linolenate** cannot be synthesized by mammals and are therefore termed essential fatty acids as they have to be ingested in the diet.



7. Newly synthesized fatty acids have mainly two alternative fates in cells

Fate I: be incorporated into triacylglycerols as a form to store metabolic energy in long terms.

Fate II: be incorporated into membrane phospholipids (during rapid growth).





The Nobel Prize in Physiology or Medicine 1964

"for their discoveries concerning the mechanism and regulation of the cholesterol and fatty acid metabolism"

Presentation Speech



Konrad Bloch

USA

Harvard University
Cambridge, MA, USA

1912 -

Biography



Feodor Lynen

Germany

Max-Planck-Institut f r Zellchemie
Munich, Germany

1911 - 1979

Biography





The Nobel Prize in Physiology or Medicine 1985

"for their discoveries concerning the regulation of cholesterol metabolism"

Press release



Michael S. Brown

USA

University of Texas Health
Science Center at Dallas
Dallas, TX, USA

1941 -

Biography



Joseph L. Goldstein

USA

University of Texas Health
Science Center at Dallas
Dallas, TX, USA

1940 -

Biography



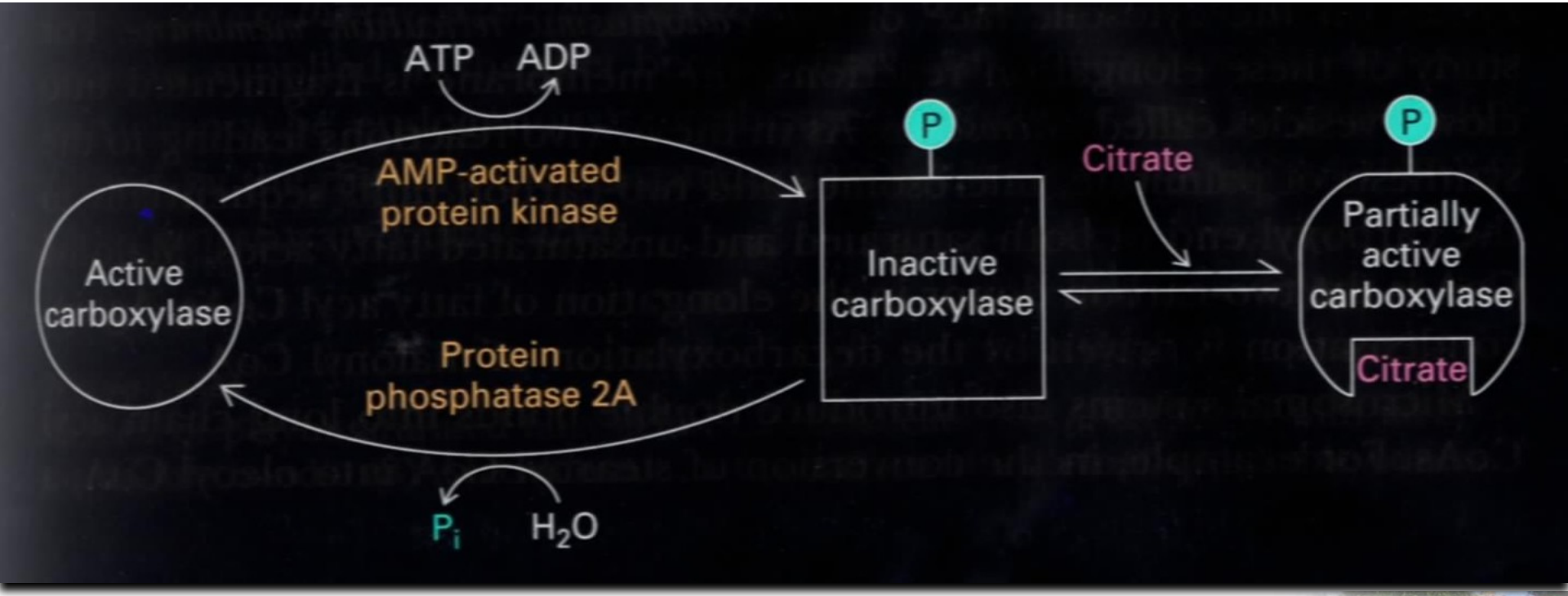


8. The rate of fatty acid biosynthesis is controlled by acetyl-CoA carboxylase

Acetyl-CoA carboxylase is regulated by allosteric effectors and reversible phosphorylation.



Citrate partially activate the phosphorylated acetyl-CoA carboxylase



- Acetyl-CoA carboxylase, catalyzing the committing and rate-limiting step of fatty acid synthesis, is allosterically **inhibited by palmitoyl-CoA and activated by citrate.**
- **Glucagon** and **epinephrine** inhibit protein phosphatase 2A, thus keeping acetyl CoA carboxylase in the inactive form and blocking fatty acid synthesis. **Insulin stimulates acetyl CoA carboxylase.**



4. Insulin stimulates conversion of dietary carbohydrates/proteins into fat

Diabetes patients due to lack of insulin would neither be able to use glucose properly, nor to synthesize fatty acids from carbohydrates and amino acids.

They show increased rates of fatty acid oxidation and ketone body formation, thus losing weight.



Summary

Fatty acid biosynthesis takes a different pathway from the reverse of its degradation and takes place in different cellular compartments.

The aceytl-CoA units are transported out of mitochondrial matrix as citrate.



Acetyl-CoA carboxylase catalyzes the rate-limiting step of fatty acid synthesis and is highly regulated by allosteric and covalent modifications.

Palmitate, the usual final product of fatty acid synthesis, can be further elongated and desaturated in sER.



Metabolism of Triacylglycerols K4

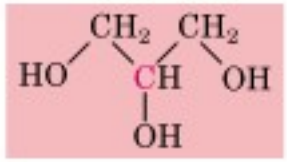


1. Newly synthesized fatty acids have mainly two alternative fates in cells

Fate I: be incorporated into triacylglycerols as a form to store metabolic energy in long terms.

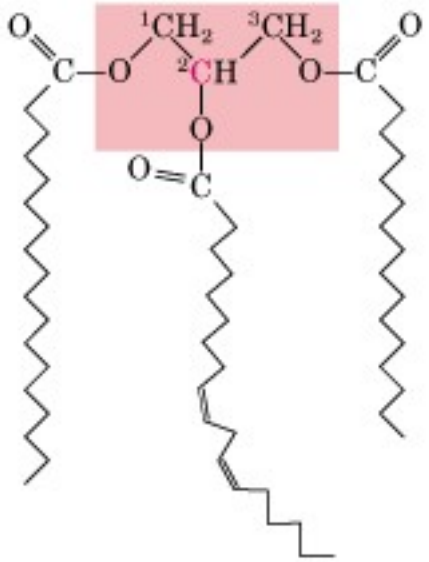
Fate II: be incorporated into membrane phospholipids (during rapid growth).



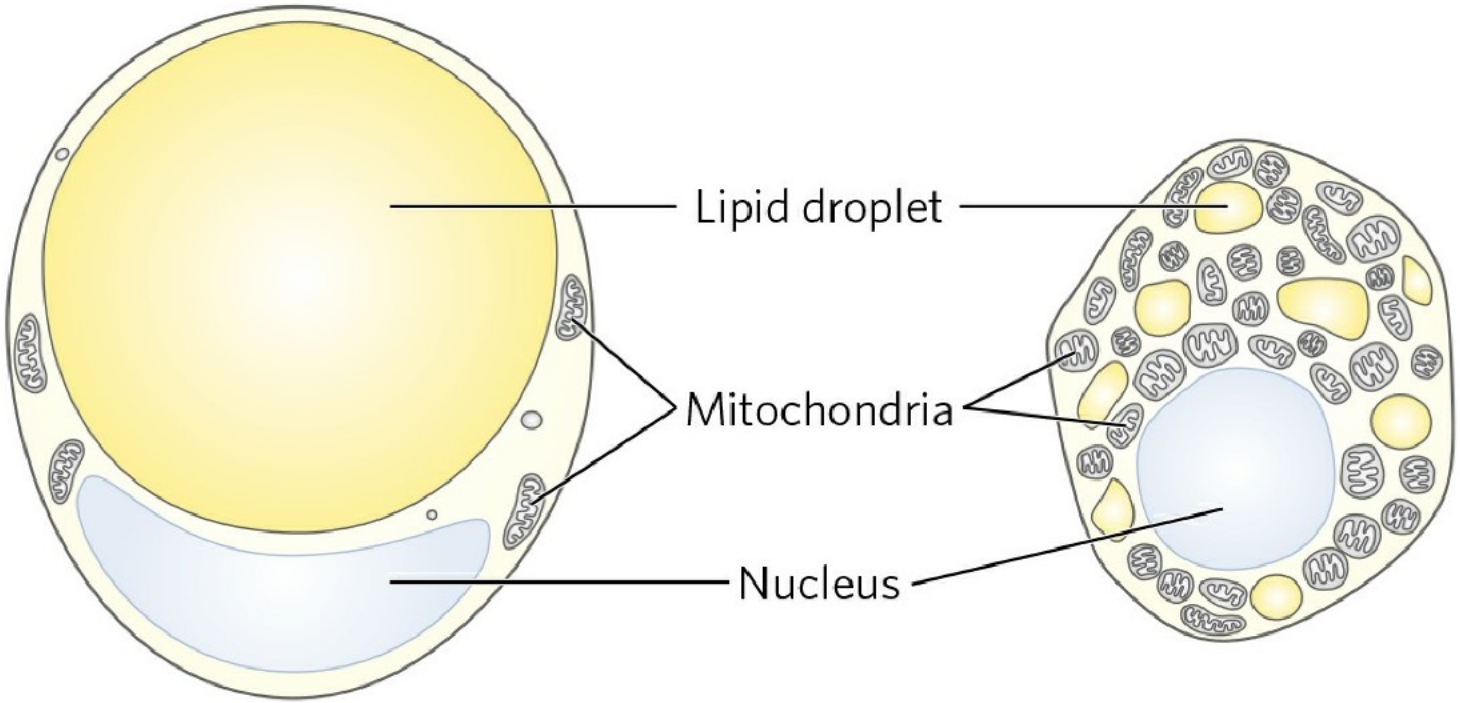


Glycerol



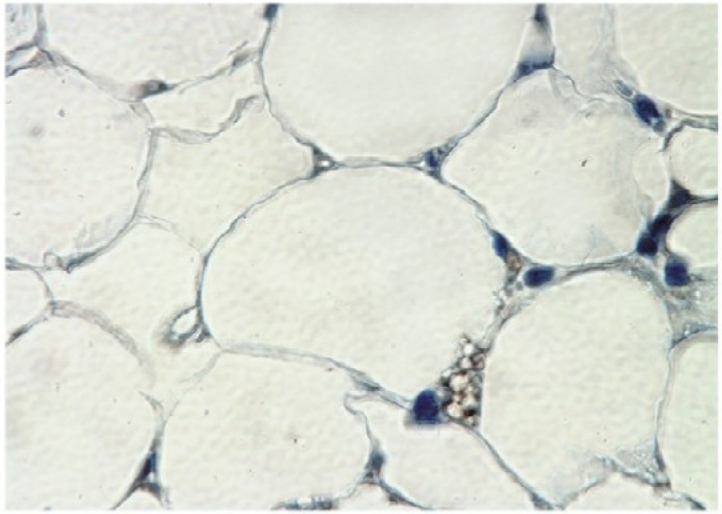


1-Stearoyl, 2-linoleoyl, 3-palmitoyl glycerol,
a mixed triacylglycerol

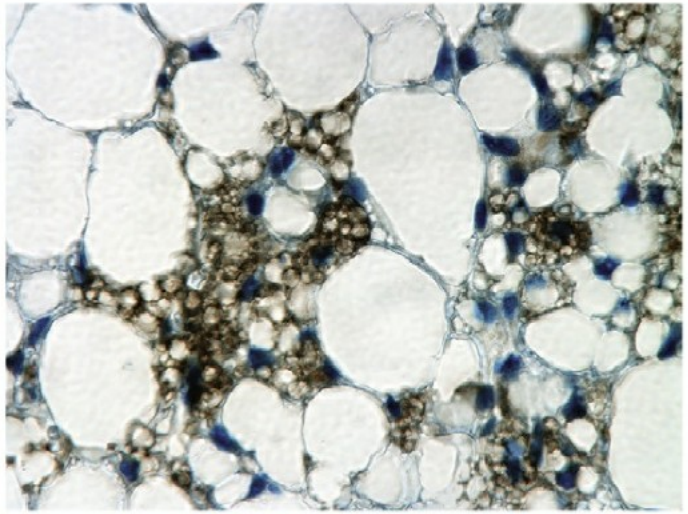


(a) White adipocyte

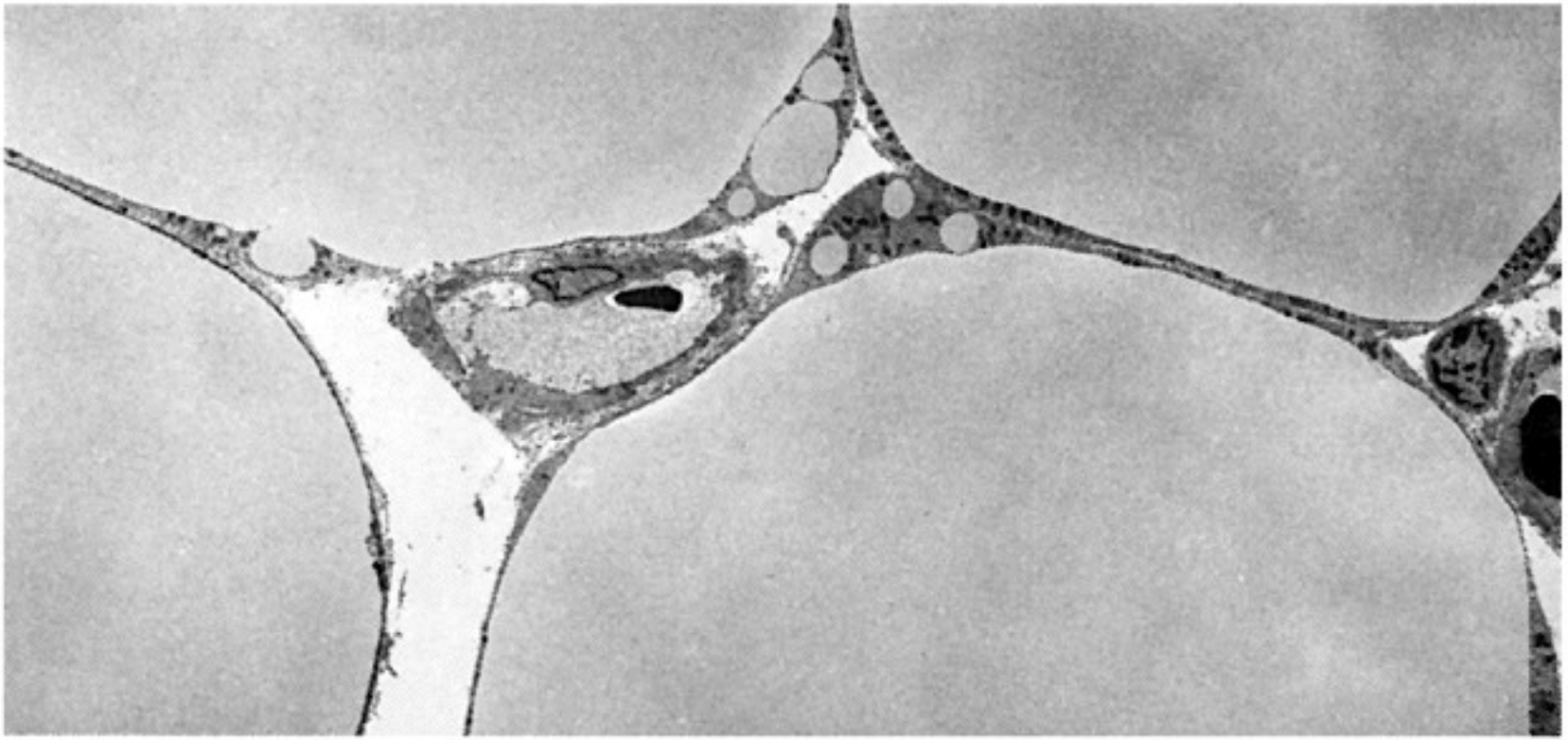
(b) Brown adipocyte



(c)

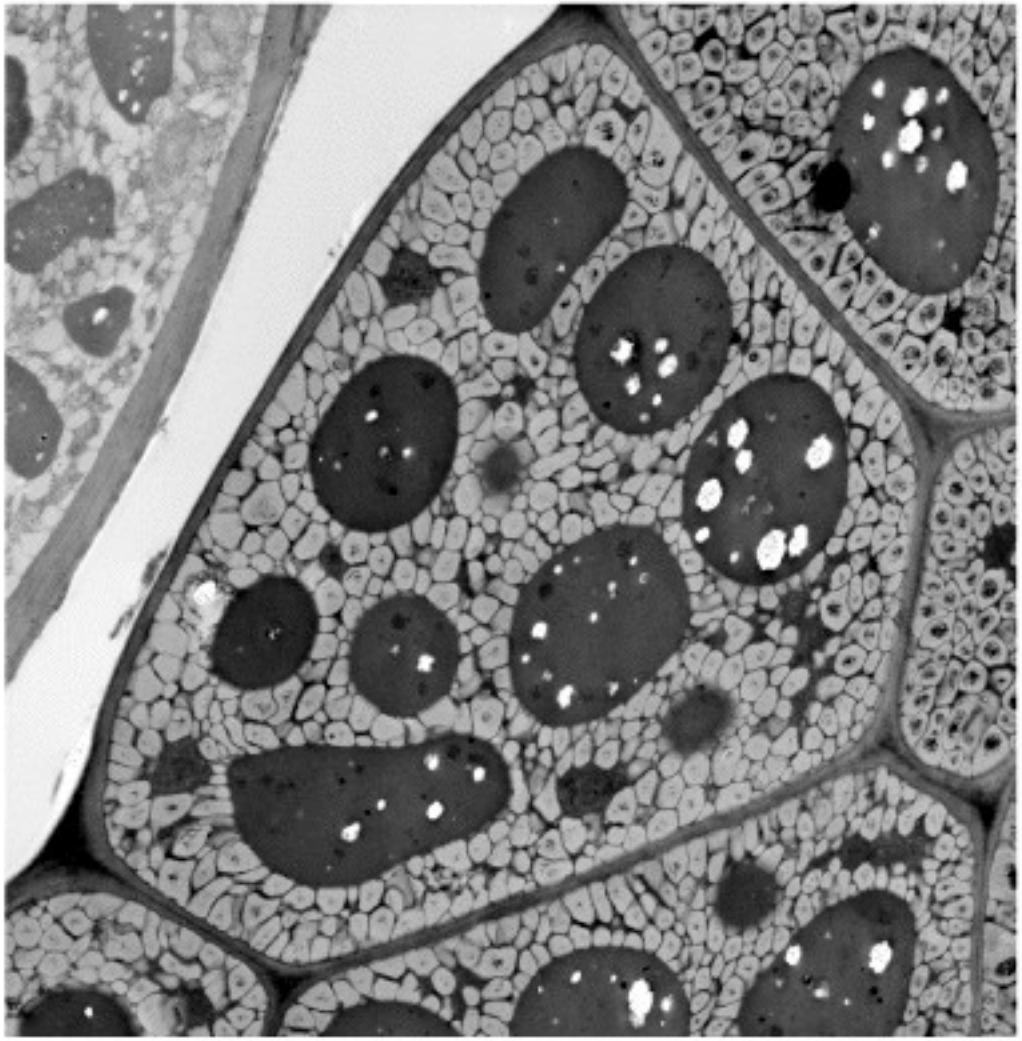


(d)



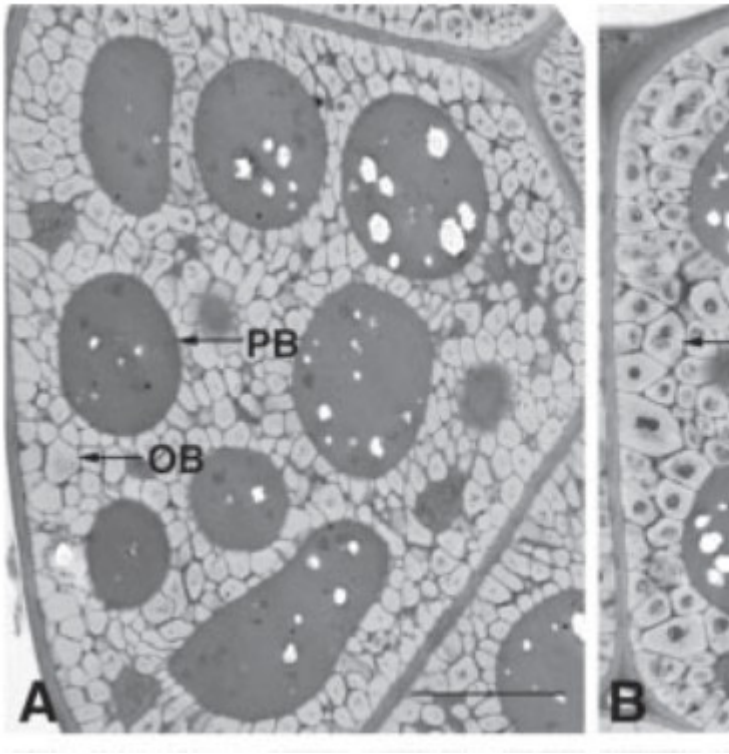
(a)

8 μm



(b)

3 μm





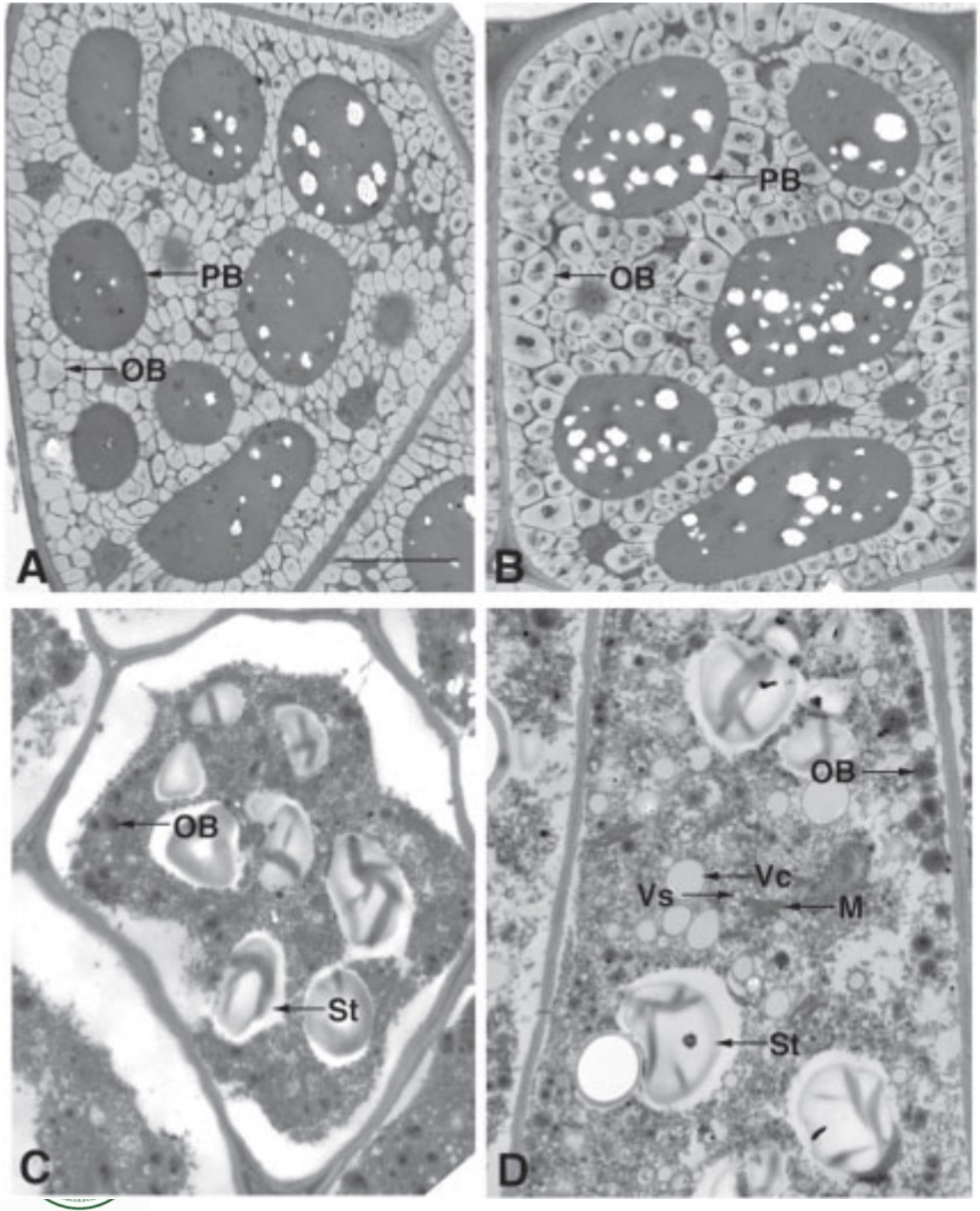
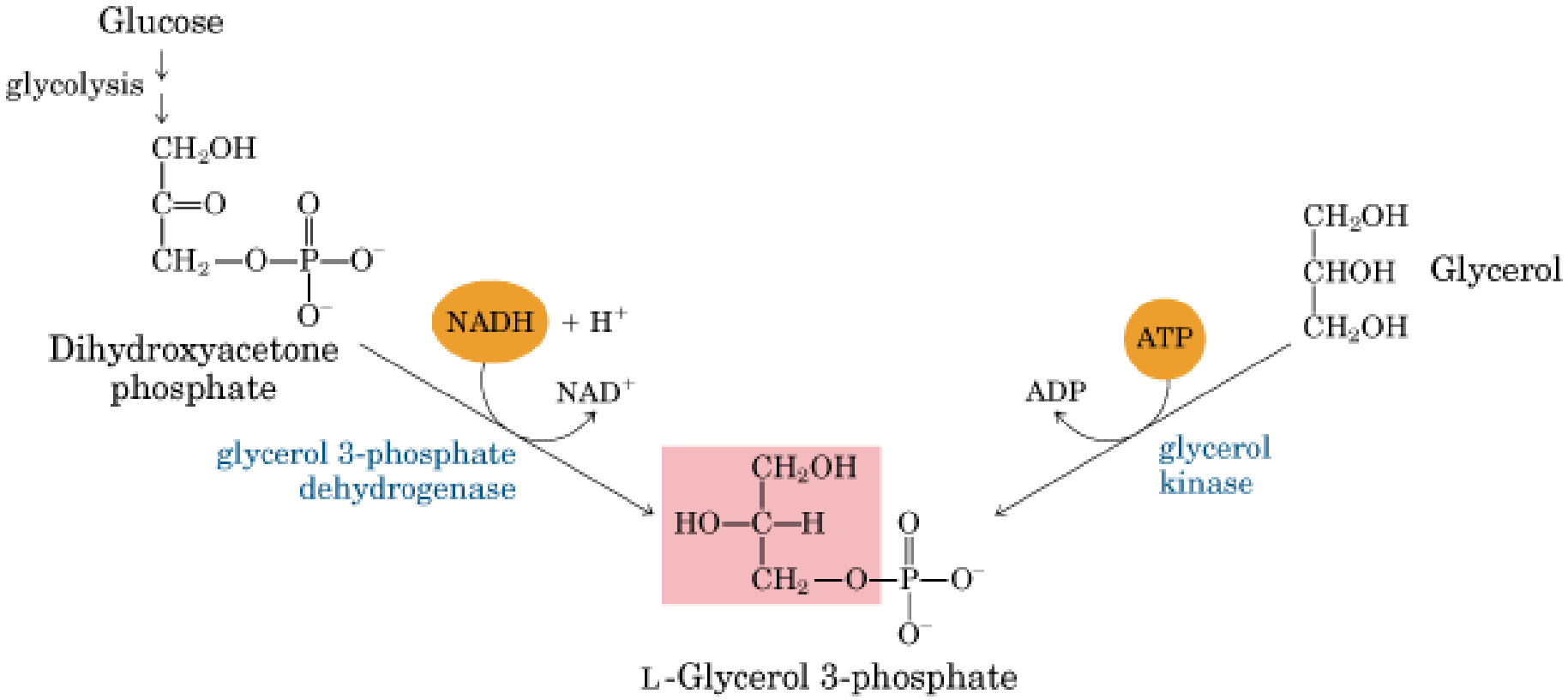


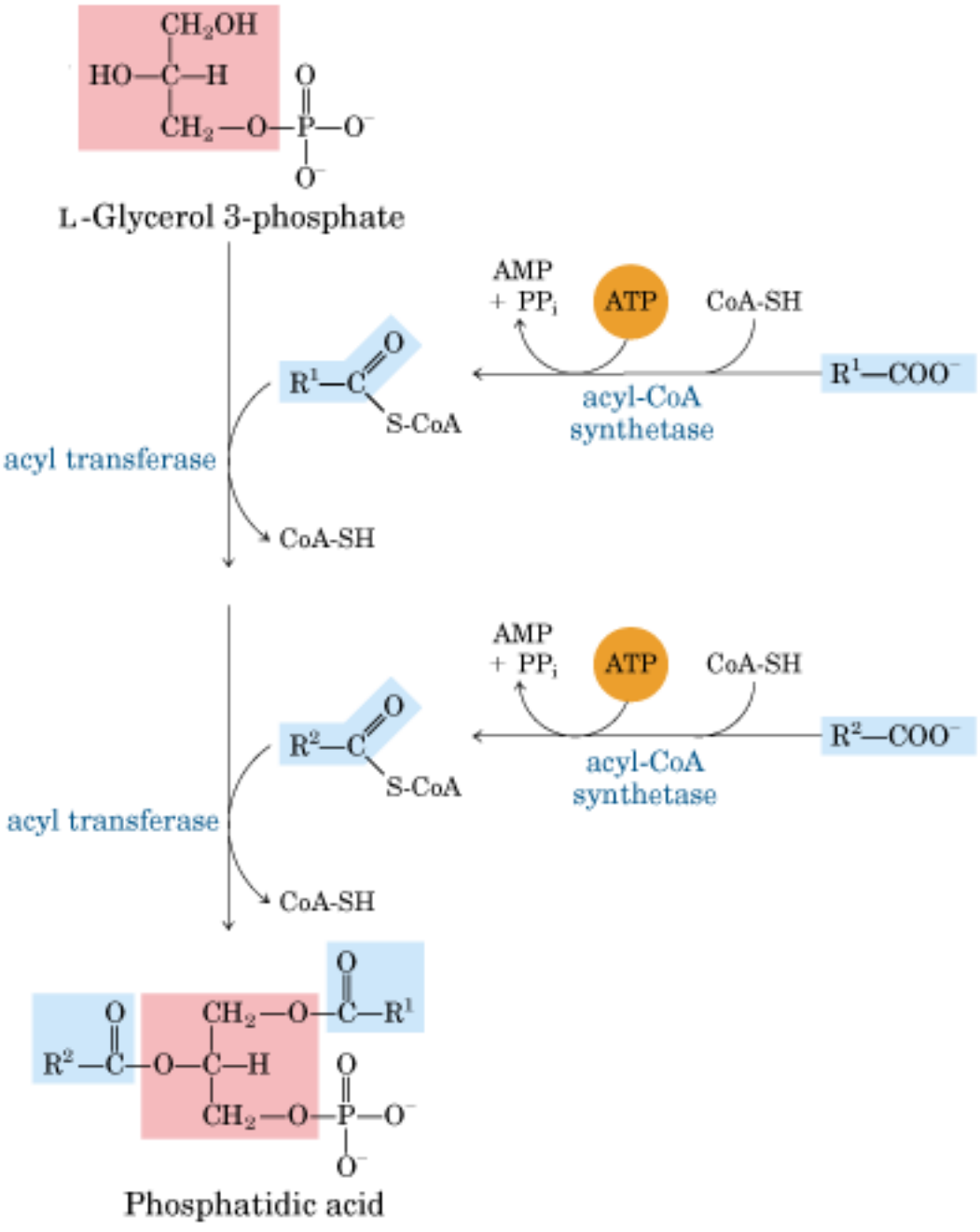
Fig. 1. Abnormal storage deposition and the shrunken phenotype of *sse1* seeds. Transmission electron micrograph (*21*) of a representative cell from the wild-type cotyledon (**A**) or hypocotyl (**B**) and the *sse1* cotyledon (**C**) or hypocotyl (**D**). Wild-type cells are filled with numerous oil bodies (OB) and a few large protein bodies (PB). *sse1* cells contain few oil bodies and additional structures such as starch granules (St), vacuoles (Vc), stacks of membranes (M), and vesicles (Vs). Wild-type C24 (**E**) and *sse1* (**F**) seeds. Bar, 3.1 μM (A to D).

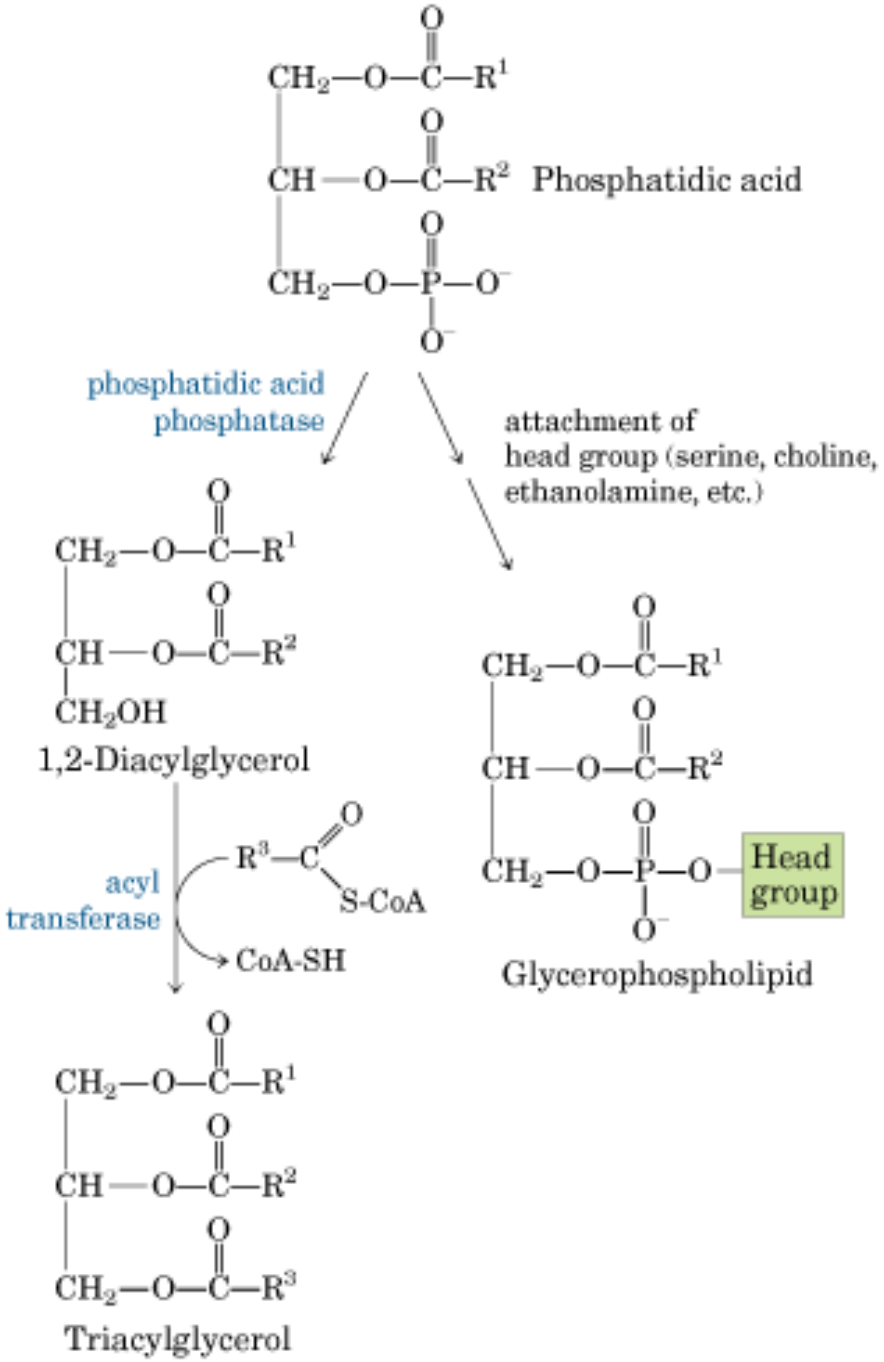
2. Triacylglycerols are synthesized from glycerol 3-phosphate and acyl CoAs

L-glycerol 3-phosphate is derived from either glycerol or dihydroxyacetone phosphate.
Acyl CoAs are added on to glycerol 3-phosphate to form phosphatidic acid.
A phosphatidic acid is converted to a triacylglycerol via a dephosphorylation reaction (catalyzed by **phosphatidic acid phosphatase**) and a acyl transferring reaction.

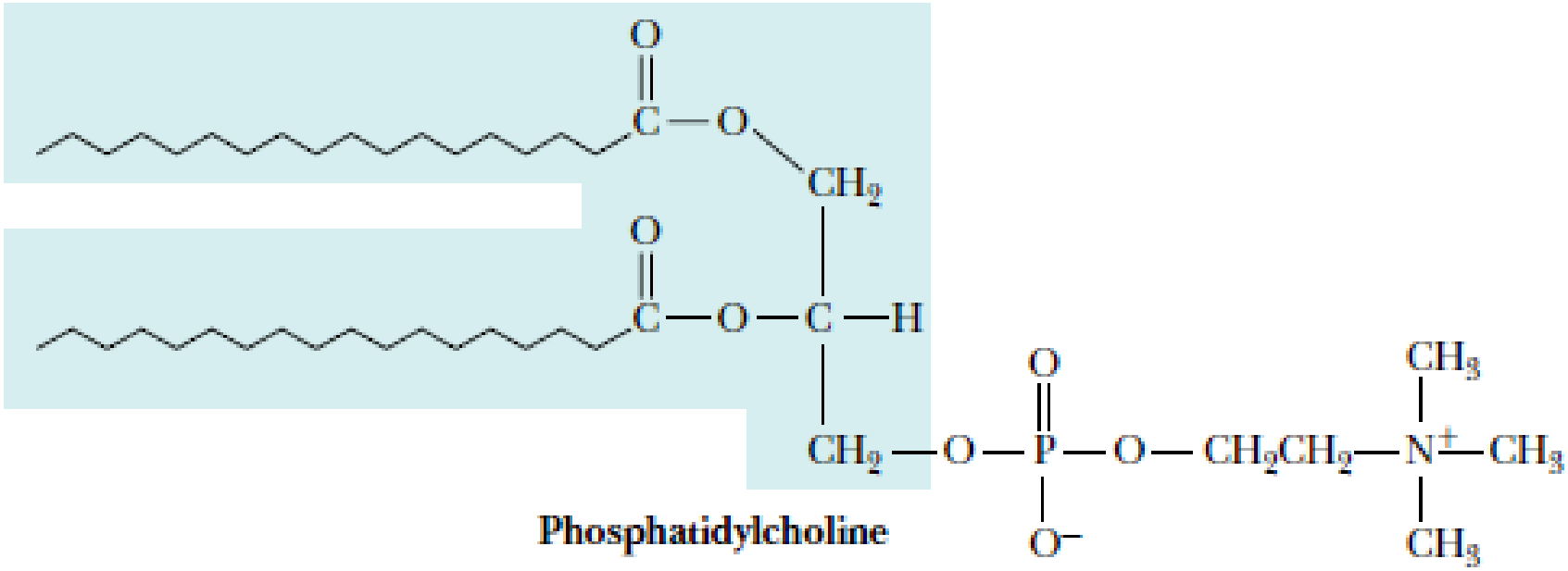



Glucose
glycolysis
Dihydroxyacetone phosphate
NADH + H+
NAD+
glycerol 3-phosphate dehydrogenase
ATP
ADP
glycerol kinase
Glycerol
L-Glycerol 3-phosphate

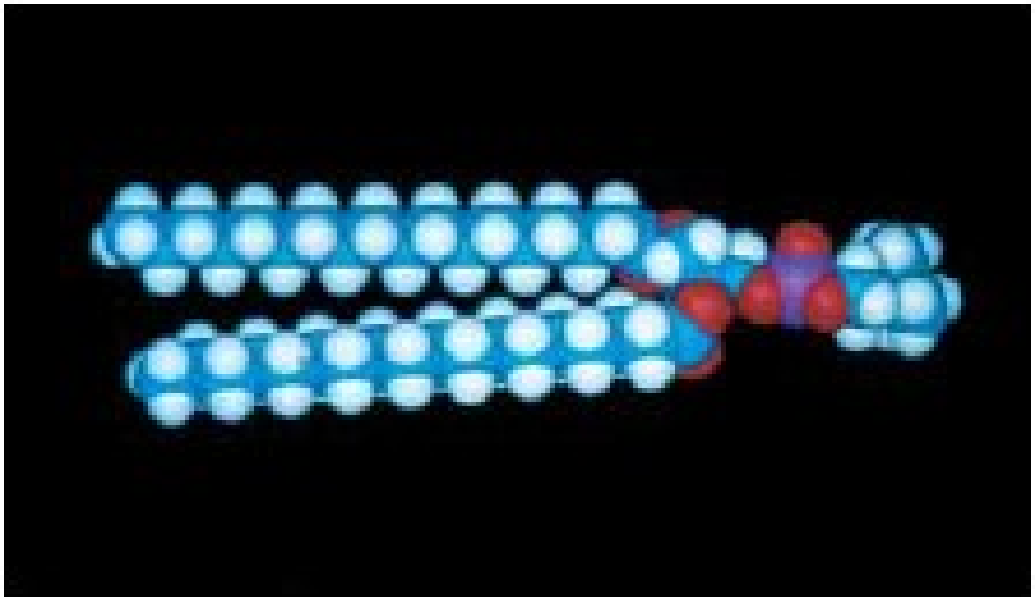

L-Glycerol 3-phosphate
AMP
+ PPi
ATP
CoA-SH
R1—COO⁻
acyl-CoA
synthetase
R1—C(=O)—S-CoA
acyl transferase
CoA-SH
AMP
+ PPi
ATP
CoA-SH
R2—COO⁻
acyl-CoA
synthetase
R2—C(=O)—S-CoA
acyl transferase
CoA-SH
Phosphatidic acid


Phosphatidic acid
phosphatidic acid phosphatase
attachment of head group (serine, choline, ethanolamine, etc.)
1,2-Diacylglycerol
acyl transferase
S-CoA
CoA-SH
Head group
Glycerophospholipid
Triacylglycerol

Biochemist



Phosphatidylcholine



3. Breakdown of triacylglycerols

The fatty acids in triacylglycerols are released from the glycerol backbone by the action of **lipases**.

The free fatty acids can then degraded by β–oxidatin to produce energy.

The glycerol is converted into dihydroxyacetone phosphate which enters glycolysis.



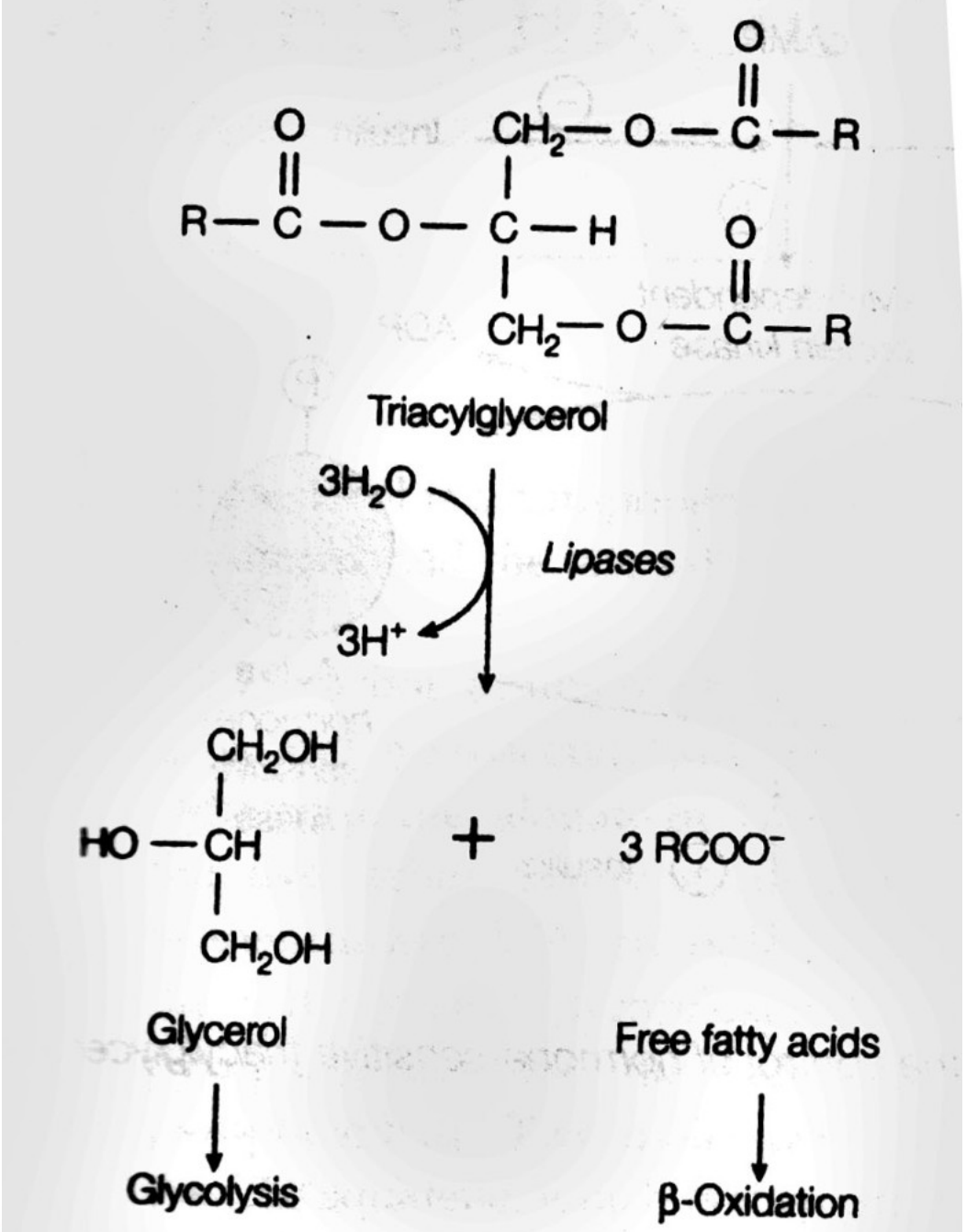
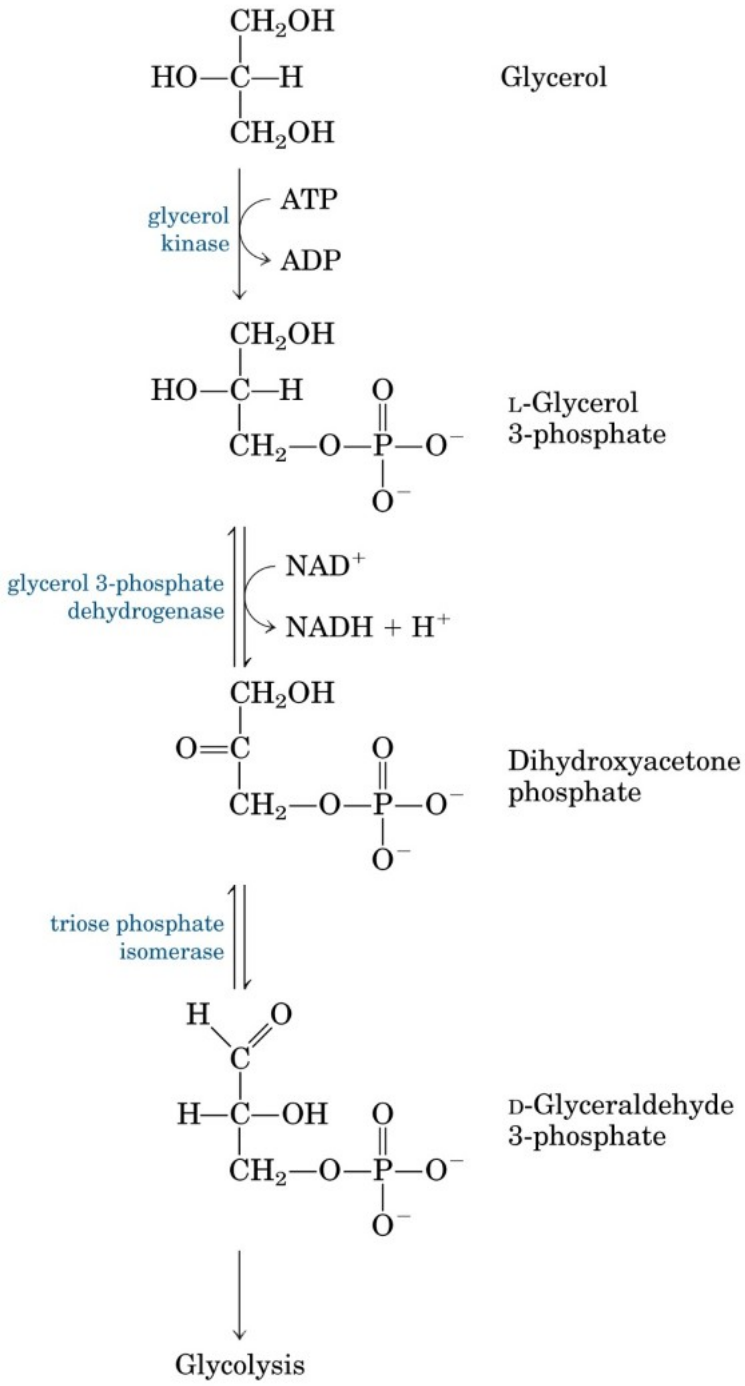
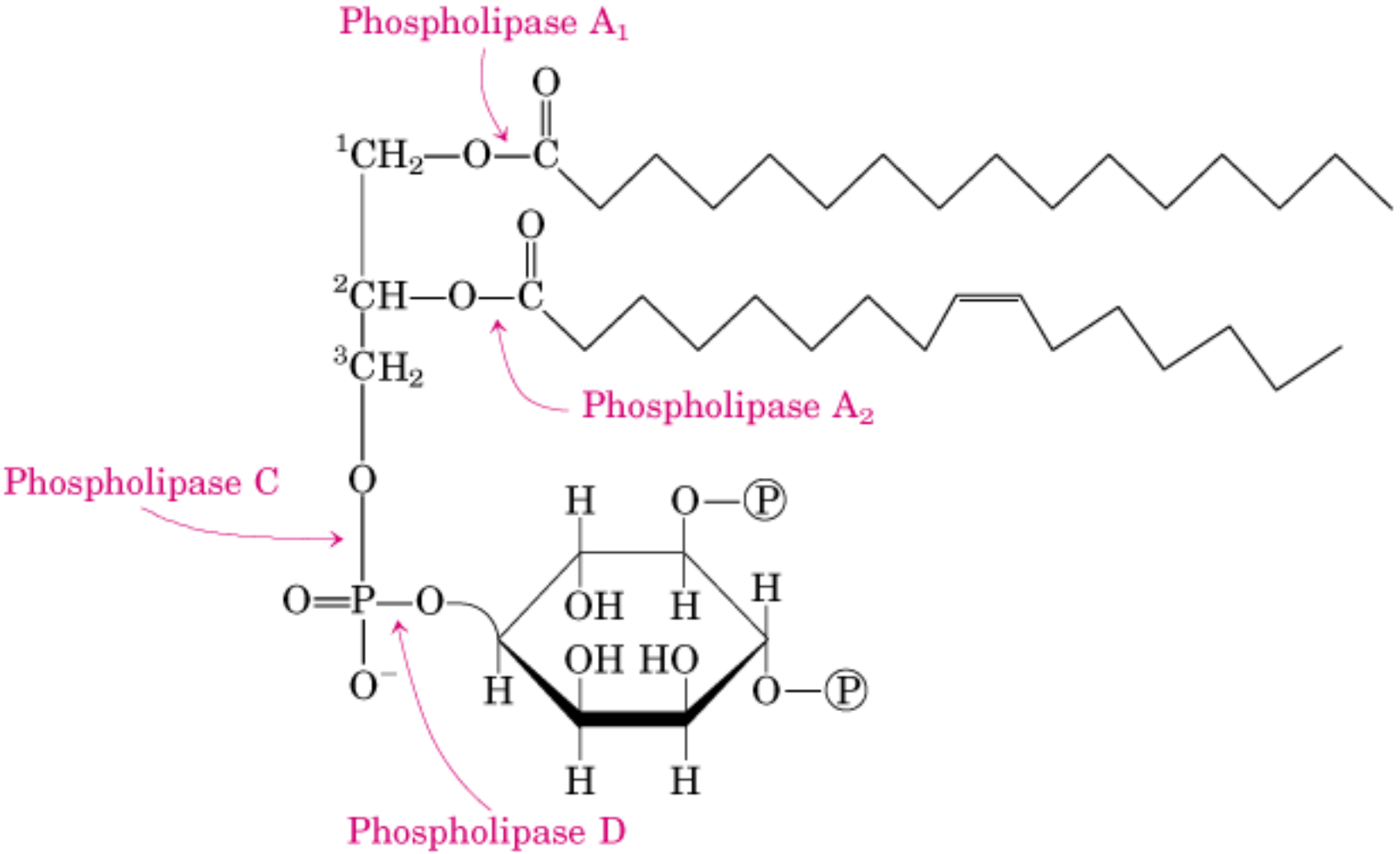


Fig. 3. *Breakdown of triacylglycerols.*


CH2OH
HO—C—H
CH2OH
Glycerol
glycerol kinase
ATP
ADP
CH2OH
HO—C—H
CH2—O—P—O⁻
O⁻
L-Glycerol 3-phosphate
glycerol 3-phosphate dehydrogenase
NAD+
NADH + H+
CH2OH
O=C
CH2—O—P—O⁻
O⁻
Dihydroxyacetone phosphate
triose phosphate isomerase
H
O
C
H—C—OH
CH2—O—P—O⁻
O⁻
D-Glyceraldehyde 3-phosphate
Glycolysis


Phospholipase A1
Phospholipase A2
Phospholipase C
Phospholipase D

4. Insulin stimulates conversion of dietary carbohydrates/proteins into fat

Diabetes patients due to lack of insulin would neither be able to use glucose properly, nor to synthesize fatty acids from carbohydrates and amino acids.

They show increased rates of fatty acid oxidation and ketone body formation, thus losing weight.



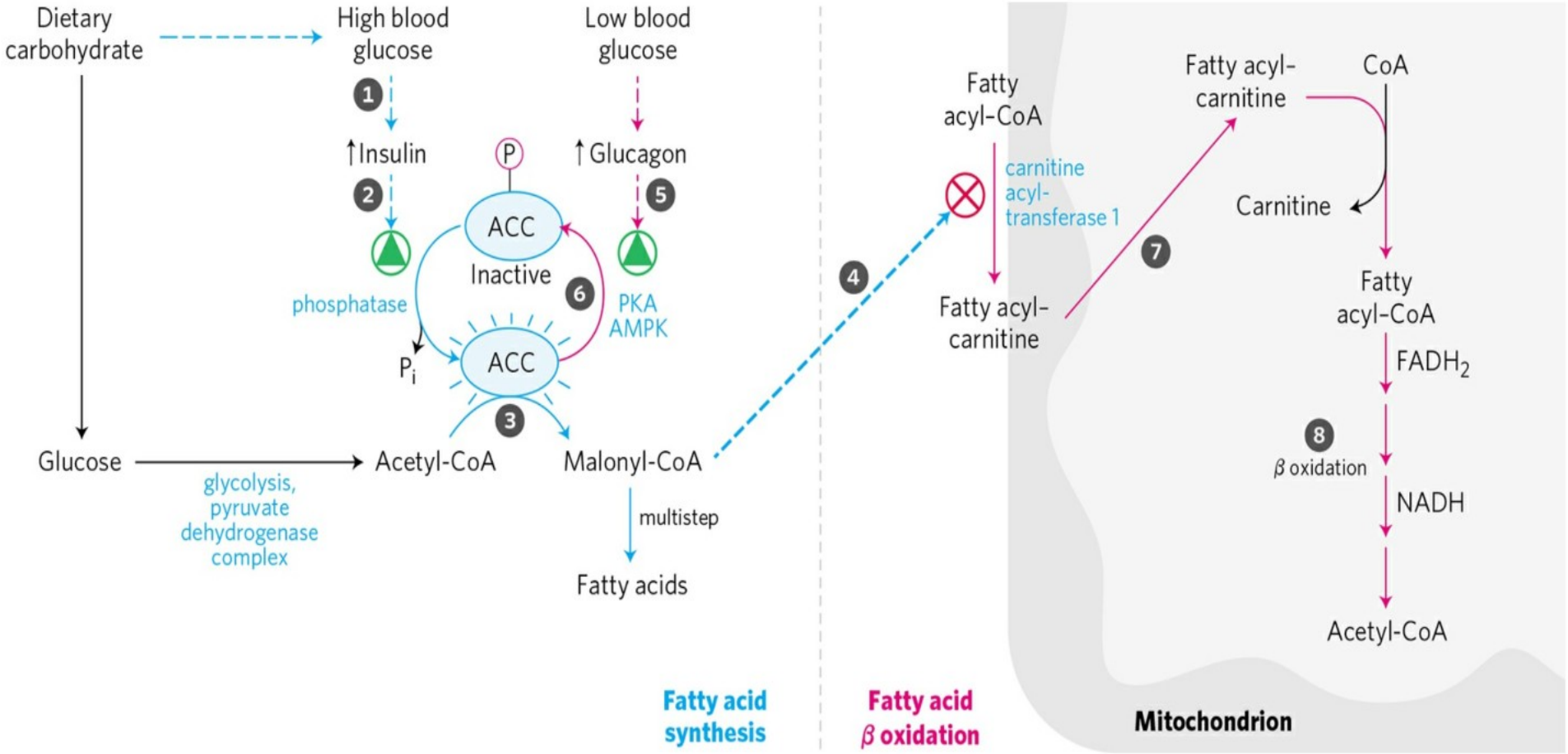
胰岛素对糖代谢的作用，主要是刺激糖原和脂肪酸的合成。

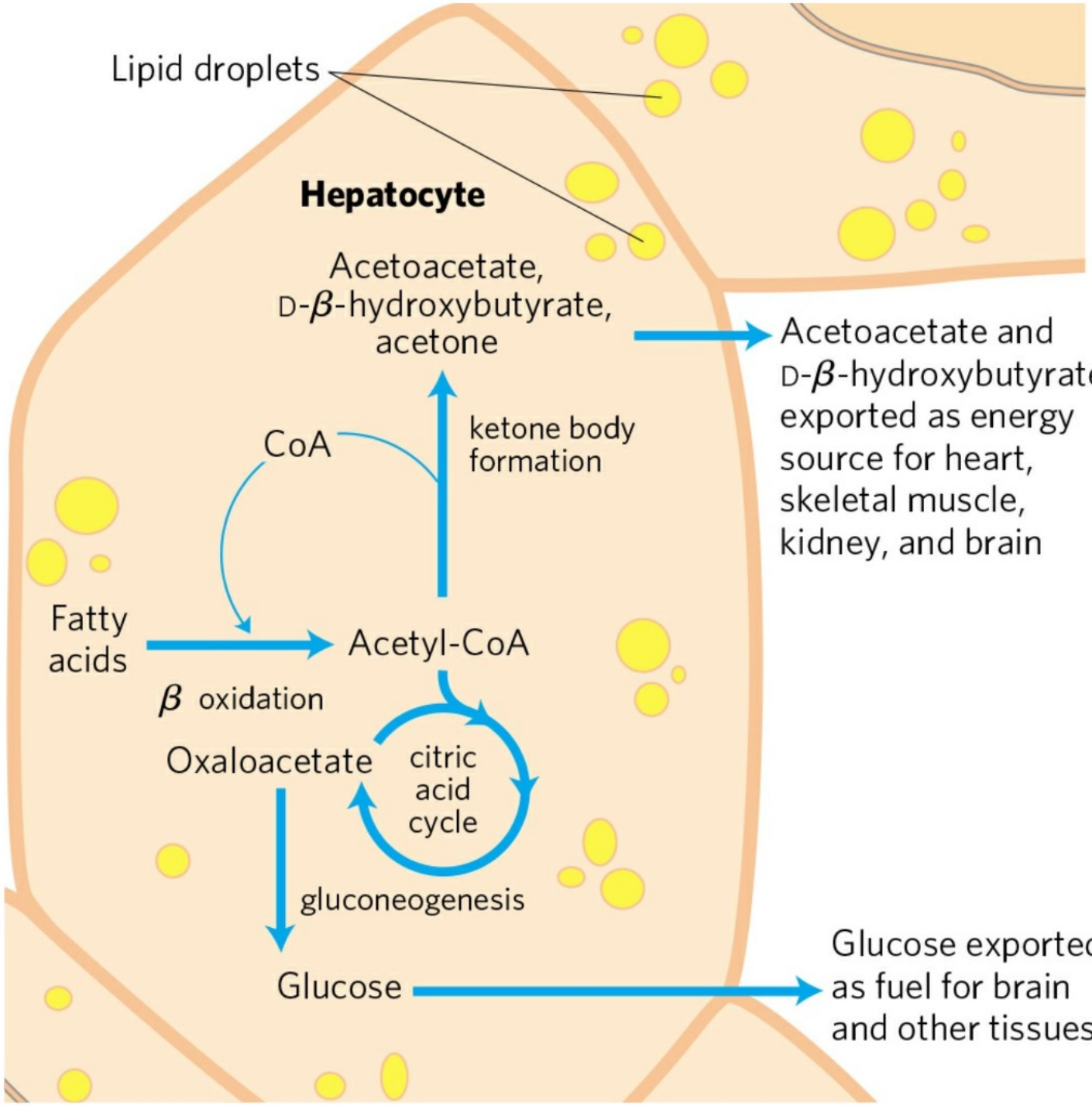
糖尿病的发生，直接原因是血中胰岛素含量不足。

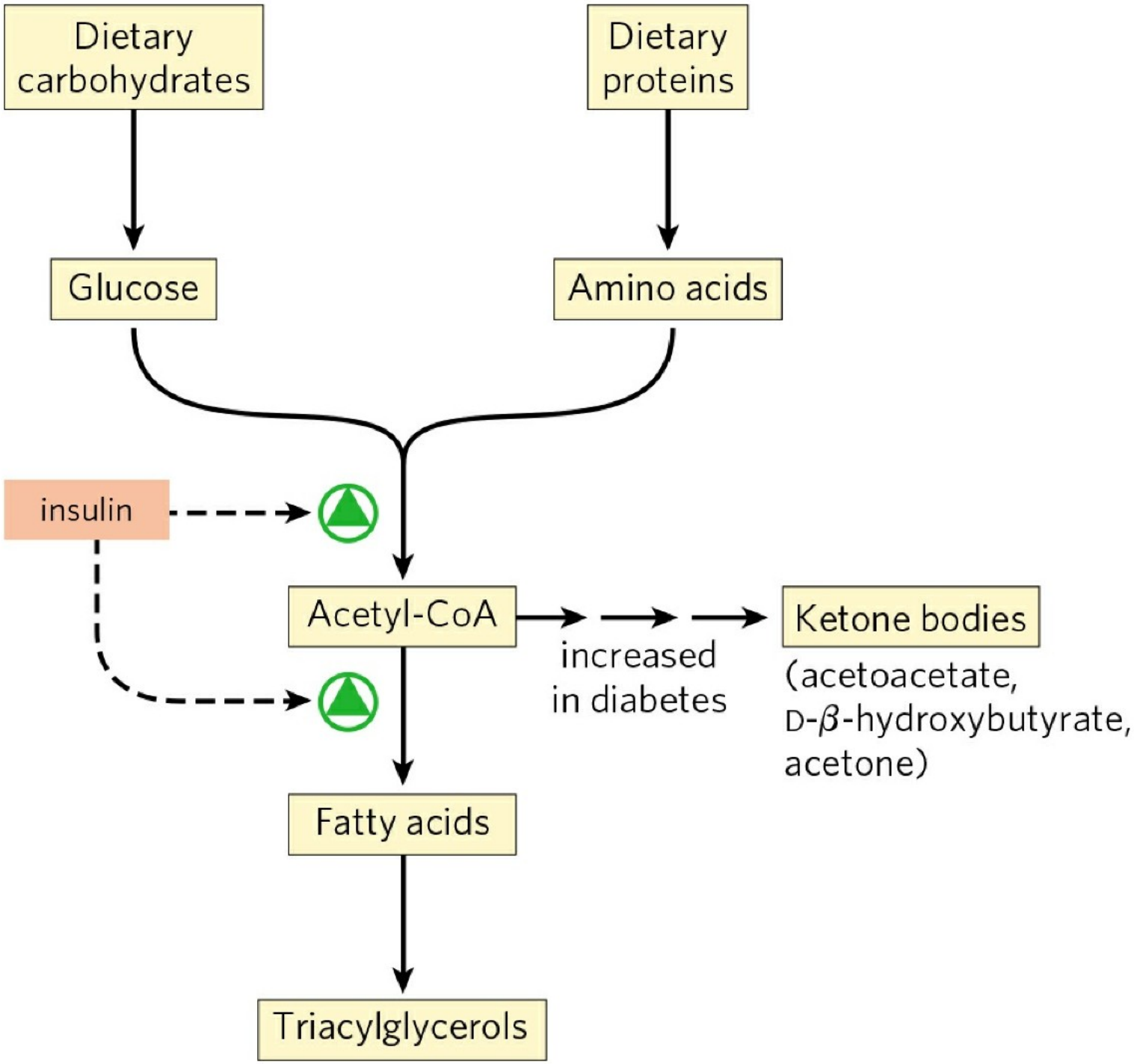
患者的肝脏往往产生过量的葡萄糖，由于缺少胰岛素，葡萄糖不能正常地进入细胞。又因胰高血糖素的浓度高于胰岛素的浓度，一方面导致肝脏中－ ΦΒΠ 的浓度下降，使糖酵解受到抑制，又刺激了葡萄糖异生，另一方面又加速了糖原的降解。这些过程都促使肝脏产生过量的葡萄糖进入血液。糖尿病患者由于糖代谢紊乱进而使脂类代谢和蛋白质代谢都受到损害。

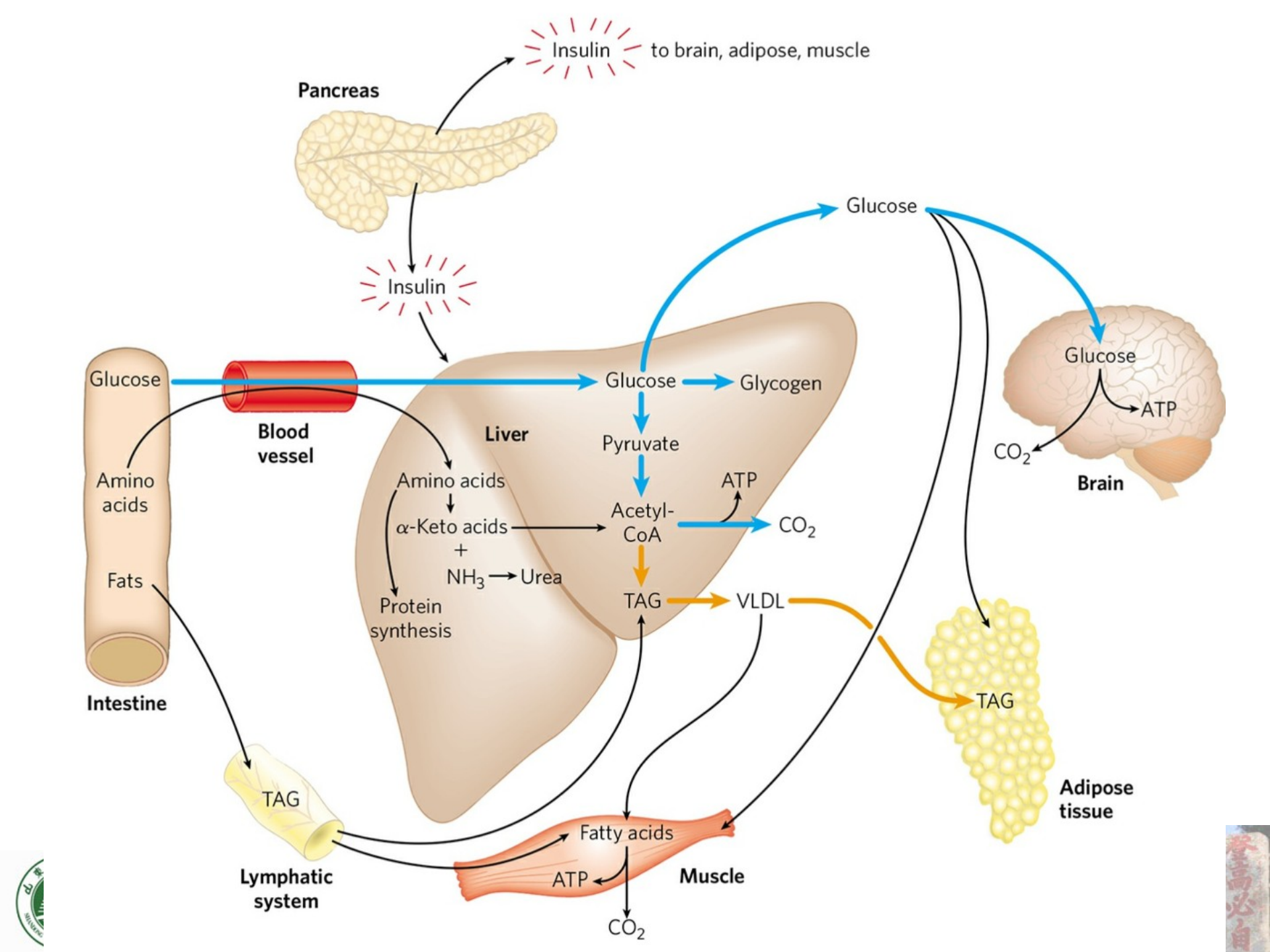
I 型—— ινσυλιν δεπενδεντ—— 儿童

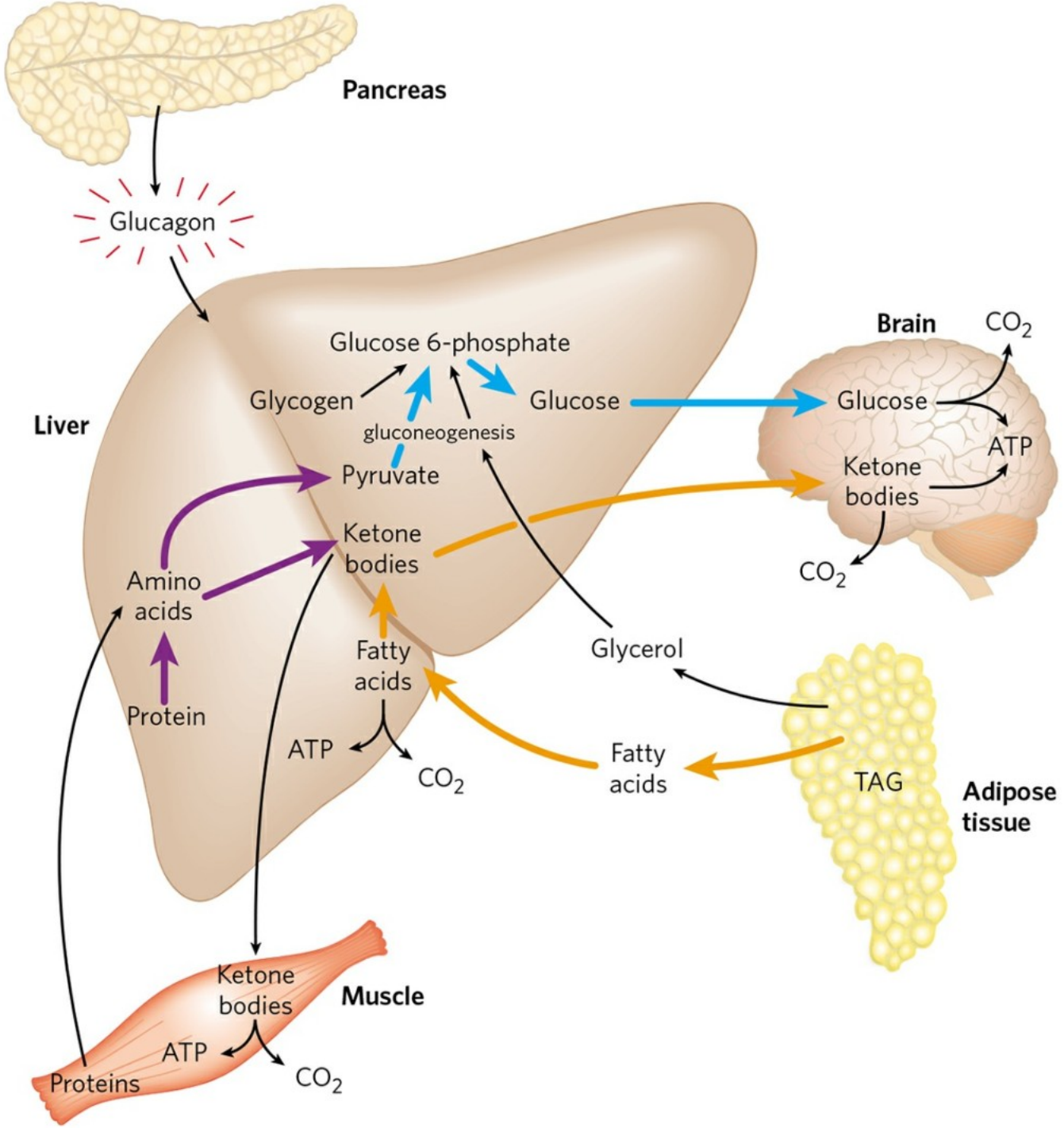
II 型—— νονινσυλιν–δεπενδεντ—— 成人


Dietary carbohydrate
High blood glucose
Low blood glucose
1
↑Insulin
↑Glucagon
2
5
P
ACC
Inactive
6
phosphatase
PKA
AMPK
P_i
ACC
3
Glucose
glycolysis, pyruvate dehydrogenase complex
Acetyl-CoA
Malonyl-CoA
multistep
Fatty acids
4
Fatty acyl-CoA
carnitine acyl-transferase 1
Fatty acyl-carnitine
7
Fatty acyl-carnitine
CoA
Carnitine
Fatty acyl-CoA
$FADH_2$
8
β oxidation
NADH
Acetyl-CoA
Fatty acid synthesis
Fatty acid β oxidation
Mitochondrion


Lipid droplets
Hepatocyte
Acetoacetate,
D-β-hydroxybutyrate,
acetone
Acetoacetate and
D-β-hydroxybutyrate
exported as energy
source for heart,
skeletal muscle,
kidney, and brain
CoA
ketone body
formation
Fatty
acids
Acetyl-CoA
β oxidation
Oxaloacetate
citric
acid
cycle
gluconeogenesis
Glucose
Glucose exported
as fuel for brain
and other tissues


Dietary carbohydrates
Dietary proteins
Glucose
Amino acids
insulin
Acetyl-CoA
increased in diabetes
Ketone bodies
(acetoacetate, D-β-hydroxybutyrate, acetone)
Fatty acids
Triacylglycerols


Insulin
to brain, adipose, muscle
Pancreas
Insulin
Glucose
Glucose
Glycogen
Glucose
ATP
CO_2
Brain
Glucose
Blood vessel
Liver
Pyruvate
Amino acids
Amino acids
α-Keto acids
+
NH_3 → Urea
Protein synthesis
Acetyl-CoA
ATP
CO_2
TAG
VLDL
Fats
Intestine
TAG
TAG
Adipose tissue
Lymphatic system
Fatty acids
ATP
Muscle
CO_2


Pancreas
Glucagon
Liver
Glucose 6-phosphate
Glycogen
Glucose
gluconeogenesis
Pyruvate
Ketone bodies
Amino acids
Protein
Fatty acids
ATP
CO_2
Glycerol
Fatty acids
TAG
Adipose tissue
Brain
CO_2
Glucose
ATP
Ketone bodies
CO_2
Ketone bodies
Muscle
ATP
CO_2
Proteins

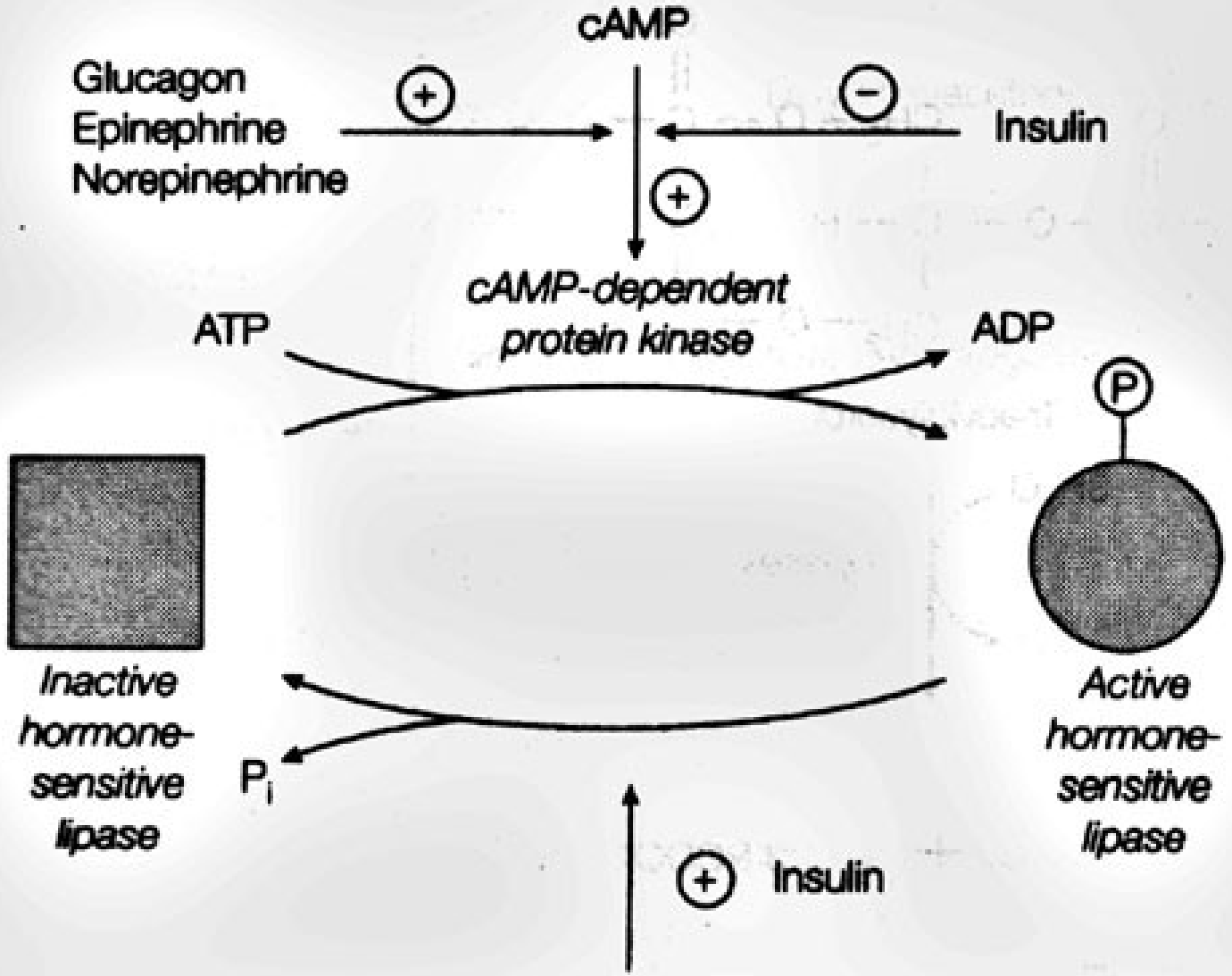


Fig. 5. Summary of the control of hormone-sensitive triacylglycerol li

END

Thank you

